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Massive MIMO Networks: Spectral, Energy, and Hardware Efficiency

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Massive MIMO Networks: Spectral, Energy, and Hardware Efficiency

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ABSTRACT

Massive multiple-input multiple-output (MIMO) is one of the most promising technologies for the next generation of wireless communication networks because it has the potential to provide game-changing improvements in spectral efficiency (SE) and energy efficiency (EE). This monograph summarizes many years of research insights in a clear and self-contained way and provides the reader with the necessary knowledge and mathematical tools to carry out independent research in this area. Starting from a rigorous definition of Massive MIMO, the monograph covers the important aspects of channel estimation, SE, EE, hardware efficiency (HE), and various practical deployment considerations.

From the beginning, a very general, yet tractable, canonical system model with spatial channel correlation is introduced. This model is used to realistically assess the SE and EE, and is later extended to also include the impact of hardware impairments. Owing to this rigorous modeling approach, a lot of classic "wisdom" about Massive MIMO, based on too simplistic system models, is shown to be questionable.

The monograph contains many numerical examples, which can be reproduced using Matlab code that is available online at https://dx.doi.org/10.1561/2000000093_supp.

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Practical Deployment Considerations

This section describes some important tradeoffs and considerations for the design, optimization, and deployment of Massive MIMO in practical networks. While the previous sections described the fundamental theory that is generally well understood, the topics covered by this section are still under development at the time of writing this monograph. Section 7.1 describes power allocation schemes for the maximization of network utility functions. Key topics in spatial resource allocation are outlined in Section 7.2, which include pilot assignment, scheduling, and load balancing. Channel modeling is considered in Section 7.3 with focus on the spatial characteristics that are resolved by large arrays. The configuration and deployment of antenna arrays are covered by Section 7.4. Massive MIMO technology for hotspots operating at mmWave frequencies is described in Section 7.5, while the role of Massive MIMO in heterogeneous networks is described in Section 7.6. We conclude with a case study in Section 7.7, taking some of the practical considerations into account, and a summary of key points in Section 7.8.

7.1 Power Allocation

Although the SE analysis in the previous sections applies to arbitrarily selected transmit powers, the numerical results were based on the assumption of equal transmit power per UE, in both UL and DL. This is generally not the optimal strategy if we want to maximize some *utility function*. Firstly, the sum SE can be increased by unequal power allocation, exploiting the different propagation conditions of the UEs. Secondly, the sum SE only measures the aggregated throughput of the network, while ignoring how fairly it is distributed among the UEs. This can lead to substantial unfairness. In addition to the sum SE utility, there are alternative utility functions that balance between aggregate throughput and fairness [46]. In this section, we describe several different network utility functions and provide power allocation schemes that maximize them. The channel hardening makes power allocation different in Massive MIMO than in single-antenna systems. There is no need to adapt the transmit powers to small-scale fading variations, but only to the large-scale fading characteristics. This unique characteristic of Massive MIMO makes advanced power allocation schemes, that previously were overly complex, practically feasible.

In a network with $\sum_{l=1}^{L} K_l$ UEs, there are equally many SE expressions to take into consideration—in both UL and DL. These SEs are all connected due to interference. This can, for example, be seen from the DL SE in Theorem 4.6 on p. 163, which takes the form

$$\underline{\mathsf{SE}}_{jk}^{\mathrm{DL}} = \frac{\tau_d}{\tau_c} \log_2 \left(1 + \frac{\rho_{jk} a_{jk}}{\sum\limits_{l=1}^{L} \sum\limits_{i=1}^{K_l} \rho_{li} b_{lijk} + \sigma_{\mathrm{DL}}^2} \right) \tag{7.1}$$

for UE k in cell j, where

$$a_{jk} = |\mathbb{E}\{\mathbf{w}_{jk}^{\mathrm{H}} \mathbf{h}_{jk}^{j}\}|^2 \tag{7.2}$$

$$b_{lijk} = \begin{cases} \mathbb{E}\{|\mathbf{w}_{li}^{\mathrm{H}} \mathbf{h}_{jk}^{l}|^2\} & (l,i) \neq (j,k) \\ \mathbb{E}\{|\mathbf{w}_{jk}^{\mathrm{H}} \mathbf{h}_{jk}^{j}|^2\} - |\mathbb{E}\{\mathbf{w}_{jk}^{\mathrm{H}} \mathbf{h}_{jk}^{j}\}|^2 & (l,i) = (j,k). \end{cases} \tag{7.3}$$

Notice that $\underline{\mathsf{SE}}_{jk}^{\mathrm{DL}}$ in (7.1) is an increasing function of ρ_{jk}, which is the DL transmit power allocated to this UE, while it is a decreasing

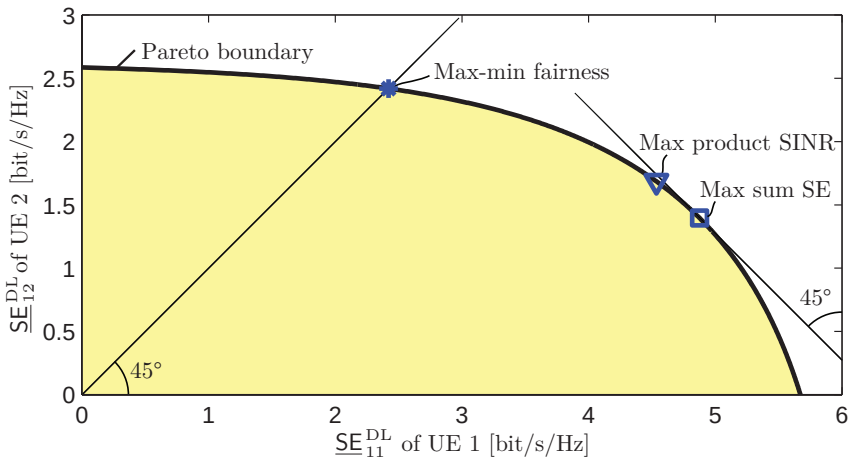


Figure 7.1: Example of an SE region (shaded) with different combinations $(\underline{\mathsf{SE}}_{11}^{\mathrm{DL}}, \underline{\mathsf{SE}}_{12}^{\mathrm{DL}})$ of SEs that can be achieved by different power allocations. The three operating points that maximize the utility functions in (7.4) are indicated.

function of the transmit power ρ_{li} of any other UE. Hence, there is a conflicting relation between the SEs of two UEs, not only due to mutual interference but also because each BS has a limited power budget to allocate among its UEs.

This conflicting relation can be illustrated by an *SE region* with $\sum_{l=1}^{L} K_l$ dimensions, which contains all simultaneously achievable SE combinations. An example is provided in Figure 7.1 for the DL with $L = 1$ and $K_1 = 2$ UEs. The shaded region contains all $(\underline{\mathsf{SE}}_{11}^{\mathrm{DL}}, \underline{\mathsf{SE}}_{12}^{\mathrm{DL}})$-points that can be achieved by different allocations of power between the UEs. In this example, we assumed that UE 1 has a better channel than UE 2, which results in an SE region that contains larger SE values for UE 1. Any point in the interior of the SE region is strictly suboptimal because it is possible to jointly increase $\underline{\mathsf{SE}}_{11}^{\mathrm{DL}}$ and $\underline{\mathsf{SE}}_{12}^{\mathrm{DL}}$ by changing the power allocation. Hence, an efficient network must operate on the outer boundary of the SE region, which is referred to as the *Pareto boundary*. For points on the Pareto boundary, we cannot increase the SE of a UE without decreasing the SE of another UE. The shape of the Pareto boundary describes the conflicting relation [47].

There are infinitely many points on the Pareto boundary, so which point should we choose? No objective answer exists to this question because the typical goal of every UE is to maximize its own SE [47], which would in principle require to switch off all other UEs. As a network designer, we need to find a subjective balance between the individual goals of the UEs. A structured way to do this is by defining a network utility function, $U(\mathsf{SE}_{11}, \ldots, \mathsf{SE}_{LK_L})$, that takes the SEs of all UEs as inputs and gives a scalar that measures the utility as output [250]; that is, the larger the better. To keep it general, SE_{jk} for UE k in cell j can either denote the UL SE $\underline{\mathsf{SE}}_{jk}^{\mathrm{UL}}$ or the DL SE $\underline{\mathsf{SE}}_{jk}^{\mathrm{DL}}$. The utility function is to be maximized and should be selected so that the input is more preferred the larger the output is. Some prominent examples of utility functions are:

$$U(\mathsf{SE}_{11}, \ldots, \mathsf{SE}_{LK_L}) = \begin{cases} \sum_{j=1}^{L} \sum_{k=1}^{K_j} \mathsf{SE}_{jk} & \text{Max sum SE} \\ \min_{j,k} \mathsf{SE}_{jk} & \text{Max-min fairness} \\ \prod_{j=1}^{L} \prod_{k=1}^{K_j} \mathsf{SINR}_{jk} & \text{Max product SINR} \end{cases} \tag{7.4}$$

where SINR_{jk} denotes the effective SINR of SE_{jk}. Notice that "max" indicates that we want to maximize these utilities, while the actual utility functions are the sum SE, minimum SE, and product of the SINRs, respectively.

By definition, the max sum SE utility leads to the highest aggregate SE, but without any fairness guarantees—some UEs with bad channel conditions might get zero SE when maximizing this utility. By normalizing the sum SE by the number of UEs, we can also interpret this utility as the arithmetic mean SE. The other extreme is max-min fairness, which provides complete fairness by only counting the SE achieved by the weakest UE in the network. One can easily convince oneself that maximizing this utility function results in the same SE for everyone, thus a UE has no benefit of having a good channel condition. There is a variety of tradeoffs between these extremes, represented either by weighing the SEs of the UEs differently in (7.4) or by using alternative functions; for example, the geometric mean or the harmonic mean of the SEs [46, 176, 221]. The book [210] advocates a heuristic approach for maximizing the minimum SE locally in each cell, while allowing for

different SE levels in different cells. This is made possible by neglecting coherent interference and assuming that every BS and at least one UE per cell transmit with full power.

The max product SINR utility is also defined in (7.4). To understand the motivation behind maximizing the product of the effective SINRs, let us consider the DL and notice that

$$\begin{aligned}\sum_{j=1}^{L}\sum_{k=1}^{K_j}\underline{\mathsf{SE}}_{jk}^{\mathrm{DL}} &= \sum_{j=1}^{L}\sum_{k=1}^{K_j}\frac{\tau_d}{\tau_c}\log_2\left(1+\underline{\mathsf{SINR}}_{jk}^{\mathrm{DL}}\right) \\ &\geq \sum_{j=1}^{L}\sum_{k=1}^{K_j}\frac{\tau_d}{\tau_c}\log_2\left(\underline{\mathsf{SINR}}_{jk}^{\mathrm{DL}}\right) = \frac{\tau_d}{\tau_c}\log_2\left(\prod_{j=1}^{L}\prod_{k=1}^{K_j}\underline{\mathsf{SINR}}_{jk}^{\mathrm{DL}}\right)\end{aligned} \tag{7.5}$$

which shows that this utility seeks to maximize a lower bound on the sum SE where the "1+" term is neglected in every logarithm. This has little effect on UEs that support high SINRs, but underestimates the SEs of the weakest UEs. Hence, a maximization of the product of the SINRs leads to higher SEs for the weakest UEs, as compared to maximizing the sum SE. The max product SINR utility also guarantees that every UE gets a non-zero SE, thus this utility function provides more fairness than the sum SE utility. Recall from Theorem 4.13 on p. 192 that, with M-MMSE precoding, the SINR of every UE increases without bound when the number of antennas grows large. The "1+" in the logarithms is then negligible and maximizing the product SINR or the sum SE is asymptotically the same thing. When there are very many antennas, equal power allocation will be nearly optimal for both utilities, since the SNR variations created by power allocation have little impact on the SEs when everyone has very high SINRs.

The points in the SE region that maximize the three utility functions in (7.4) are exemplified in Figure 7.1. The max-min fairness point is the intersection between the Pareto boundary and a line from the origin with 45° slope. This is the line that contains all points where the UEs have equal SE. The max sum SE point lies on another line with 45° slope, which is defined by the equation $\underline{\mathsf{SE}}_{11}^{\mathrm{DL}} + \underline{\mathsf{SE}}_{12}^{\mathrm{DL}} = v$, where v represents the maximum value of the sum SE. Every point on this line would give this sum SE, but the max sum SE point is the one that

intersects with the SE region. Note that this point gives substantially higher SE to UE 1, compared with max-min fairness, at the expense of decreasing the SE of UE 2. Finally, the max product SINR point gives slightly higher SE to the weaker UE 2, but in this example, the point is still very close to the line that maximizes the sum SE.

7.1.1 Downlink Power Allocation

We will now use the utility framework developed above to optimize the DL power allocation, for the case when the hardening bound in Theorem 4.6 on p. 163 is used. In practice, the transmit power of a BS is limited by regulations and hardware constraints, which we model by a maximum total transmit power $P_{\max}^{\mathrm{DL}} \geq 0$ per BS. The resulting utility maximization problem is

$$\begin{aligned} \underset{\rho_{11} \geq 0, \ldots, \rho_{LK_L} \geq 0}{\text{maximize}} \quad & U(\underline{\mathsf{SE}}_{11}^{\mathrm{DL}}, \ldots, \underline{\mathsf{SE}}_{LK_L}^{\mathrm{DL}}) \\ \text{subject to} \quad & \sum_{k=1}^{K_j} \rho_{jk} \leq P_{\max}^{\mathrm{DL}}, \quad j = 1, \ldots, L \end{aligned} \tag{7.6}$$

where the DL transmit powers $\rho_{11}, \ldots, \rho_{LK_L}$ are the $\sum_{l=1}^{L} K_l$ optimization variables. The actual power transmitted by BS j is $\sum_{k=1}^{K_j} \rho_{jk}$. We can use this problem formulation along with any precoding scheme and we keep the precoding vectors fixed while optimizing the transmit powers. The type of power allocation problems in (7.6) has been studied for decades [81, 341, 46] and the computational complexity of finding the solution strongly depends on the choice of the utility function U. The max sum SE problem is non-convex and hard to solve to global optimality [201], but there exist successive approximation algorithms that find locally optimal solutions in polynomial time [341] and global optimization algorithms that find the global optimum with a complexity that grows exponentially with the number of UEs [269]. These methods can be used for benchmarking, but are of limited practical use. See Appendix B.6 on p. 422 for details on practical optimization algorithms, as well as definitions of linear, convex, and geometric programs (the three problem categories that will be of importance in this section).

The solution to the power allocation problem in (7.6) depends on a_{jk} in (7.2) and b_{lijk} in (7.3), which are the average channel gains and average interference gains, respectively. The average is computed with respect to the small-scale fading realizations so that the optimized power allocation is only a function of the channel statistics and the choice of precoding. The same power allocation solution can be used for many coherence blocks, in both the frequency and time domains (cf. Figure 2.1). It is mainly the macroscopic mobility of UEs that determines the time interval over which the channel statistics are static; see Remark 2.3 on p. 72. This is a key feature of Massive MIMO as compared to systems with few antennas, in which there is no channel hardening and hence a need to adapt the power allocation to the substantial power variations that occur between coherence blocks in both time and frequency. The bottom line is that Massive MIMO systems can afford more complex power allocation schemes since the same solution can be used for many coherence blocks.

When the max-min fairness or max product SINR utility functions are used, the globally optimal solution to (7.6) can be obtained efficiently, as shown by the following theorems.

Theorem 7.1. The max-min fairness problem, for given values of a_{jk} and b_{lijk}, can be expressed as

$$\underset{\rho_{11}\geq 0,\ldots,\rho_{LK_L}\geq 0,\gamma\geq 0}{\text{maximize}} \quad \gamma \tag{7.7}$$

$$\text{subject to} \quad \frac{\rho_{jk}a_{jk}}{\sum\limits_{l=1}^{L}\sum\limits_{i=1}^{K_l}\rho_{li}b_{lijk}+\sigma^2_{\mathrm{DL}}} \geq \gamma, \ j=1,\ldots,L, \ k=1,\ldots,K_j$$

$$\sum_{k=1}^{K_j}\rho_{jk} \leq P^{\mathrm{DL}}_{\max}, \quad j=1,\ldots,L$$

and the globally optimal solution is obtained by Algorithm 1, to a tolerance $\epsilon > 0$.

Proof. The first step is to notice that maximization of $\min_{j,k}\underline{\mathsf{SE}}^{\mathrm{DL}}_{jk}$ is equivalent to maximization of $\min_{j,k}\underline{\mathsf{SINR}}^{\mathrm{DL}}_{jk}$. Using the latter utility function, we obtain (7.7) by writing (7.6) on epigraph form [67]; that

is, to introduce the auxiliary variable γ that must satisfy the constraint $\underline{\mathsf{SINR}}_{jk}^{\mathrm{DL}} \geq \gamma$ for all j, k, and maximize that variable instead. Moreover, we notice that the constraints in (7.7) are linear in the power parameters. The SINR constraints can also be written in the linear form $\rho_{jk} a_{jk} \geq \gamma(\sum_{l=1}^{L} \sum_{i=1}^{K_l} \rho_{li} b_{lijk} + \sigma_{\mathrm{DL}}^2)$ for any fixed value of γ. Hence, we can solve (7.7) as a linear feasibility problem when γ is fixed. If we solve this subproblem and the SINR constraints $\underline{\mathsf{SINR}}_{jk}^{\mathrm{DL}} \geq \gamma$ are satisfied for a given value of γ, we need to increase the transmit powers as γ increases to keep the constraints satisfied. We can thus make a line search over γ to find the largest value for which all power constraints are satisfied when solving the feasibility problem. Algorithm 1 finds that value, to a tolerance ϵ, by bisection over the search range between $\gamma = 0$ and $\gamma = \min_{j,k} P_{\max}^{\mathrm{DL}} a_{jk} / \sigma_{\mathrm{DL}}^2$, where the latter is the effective SINR of the weakest UE when neglecting all interference. □

This theorem shows that the max-min fairness problem can be solved to global optimality by Algorithm 1, which solves a sequence of linear feasibility problems. Each subproblem is similar to the classical power optimization works [64, 370, 360] that find the minimum power that satisfies given SINR constraints. Appendix B.6 on p. 422 describes how such subproblems are solved by general-purpose algorithms. The outer while-loop in the algorithm performs a bisection search for the optimal SINR value, which means that the search space for the max-min SINR value is halved in every iteration. The convergence is thus very fast.

Theorem 7.2. The max product SINR problem, for given values of a_{jk} and b_{lijk}, can be expressed as

$$\begin{aligned} \underset{\rho_{11} \geq 0, \ldots, \rho_{LK_L} \geq 0, c_{11} \geq 0, c_{LK_L} \geq 0}{\text{maximize}} \quad & \prod_{j=1}^{L} \prod_{k=1}^{K_j} c_{jk} \\ \text{subject to} \quad & \sum_{l=1}^{L} \sum_{i=1}^{K_l} \frac{c_{jk} \rho_{li} b_{lijk}}{\rho_{jk} a_{jk}} + \frac{c_{jk} \sigma_{\mathrm{DL}}^2}{\rho_{jk} a_{jk}} \leq 1, \\ & \qquad j = 1, \ldots, L, k = 1, \ldots, K_j \\ & \sum_{k=1}^{K_j} \rho_{jk} \leq P_{\max}^{\mathrm{DL}}, \quad j = 1, \ldots, L \end{aligned} \tag{7.8}$$

Algorithm 1: Bisection algorithm for solving the max-min fairness problem in (7.7).

Input : $\{a_{jk}\}$, $\{b_{lijk}\}$, $P_{\max}^{\mathrm{DL}}$, ϵ
Output: γ^{lower}, $\{\rho_{jk}^{\mathrm{opt}}\}$

`/* Initialization */`

$\gamma^{\mathrm{lower}} \leftarrow 0$

$\gamma^{\mathrm{upper}} \leftarrow \min_{j,k} P_{\max}^{\mathrm{DL}} a_{jk}/\sigma_{\mathrm{DL}}^2$

$\rho_{jk}^{\mathrm{opt}} \leftarrow 0$ for all j, k

`/* Bisection over potential SINR values */`

do

$\gamma^{\mathrm{candidate}} \leftarrow \frac{\gamma^{\mathrm{lower}}+\gamma^{\mathrm{upper}}}{2}$

Solve the linear feasibility problem

find $\rho_{11} \geq 0, \ldots, \rho_{LK_L} \geq 0$

$$\text{subject to} \quad \gamma^{\mathrm{candidate}} \left(\sum_{l=1}^{L} \sum_{i=1}^{K_l} \rho_{li} b_{lijk} + \sigma_{\mathrm{DL}}^2 \right) - \rho_{jk} a_{jk} \leq 0,$$

$$j = 1, \ldots, L, \quad k = 1, \ldots, K_j$$

$$\sum_{k=1}^{K_j} \rho_{jk} \leq P_{\max}^{\mathrm{DL}}, \quad j = 1, \ldots, L$$

if *feasible* **then**

$\gamma^{\mathrm{lower}} \leftarrow \gamma^{\mathrm{candidate}}$

$\rho_{jk}^{\mathrm{opt}} \leftarrow \rho_{jk}$ for all j, k, based on the solution to the feasibility problem

else

$\gamma^{\mathrm{upper}} \leftarrow \gamma^{\mathrm{candidate}}$

while $\gamma^{\mathrm{upper}} - \gamma^{\mathrm{lower}} > \epsilon$

which is a geometric program.

Proof. Inserting the max product SINR utility function into (7.6) yields

$$\begin{aligned} \underset{\rho_{11}\geq 0,\ldots,\rho_{LK_L}\geq 0}{\text{maximize}} \quad & \prod_{j=1}^{L}\prod_{k=1}^{K_j} \frac{\rho_{jk}a_{jk}}{\sum\limits_{l=1}^{L}\sum\limits_{i=1}^{K_l}\rho_{li}b_{lijk}+\sigma^2_{\mathrm{DL}}} \\ \text{subject to} \quad & \sum_{k=1}^{K_j}\rho_{jk} \leq P^{\mathrm{DL}}_{\max}, \quad j=1,\ldots,L \end{aligned} \tag{7.9}$$

from which (7.8) follows by introducing the auxiliary variables c_{jk} such that

$$c_{jk}\left(\sum_{l=1}^{L}\sum_{i=1}^{K_l}\rho_{li}b_{lijk}+\sigma^2_{\mathrm{DL}}\right) \leq \rho_{jk}a_{jk} \tag{7.10}$$

and by maximizing $\prod_{j=1}^{L}\prod_{k=1}^{K_j} c_{jk}$ instead. Finally, we notice that the objective function and the constraint functions in (7.8) are posynomials, thus it is a geometric program. □

This theorem shows that the max product SINR problem can also be solved efficiently, but this time as a geometric program. The notion of a geometric program is defined in Appendix B.6 on p. 422. Geometric programs can also be transformed into convex programs by a change of variable; see [66, 81] or Appendix B.6 for details.

The optimal power allocation solutions in Theorems 7.1 and 7.2 are straightforward to obtain using either general-purpose solvers of linear/geometric programs or dedicated implementations. Alternatively, distributed algorithms can be developed by using decomposition methods [250]. The considered utility maximization problems have a total power constraint per BS, but other types of constraints can be also included; for example, to limit the minimum or maximum power allocated per UE or to limit the average interference that is caused to a particular UE in the network. The algorithmic solutions provided by Theorems 7.1 and 7.2 can still be applied, as long as the additional constraints are linear or geometric, respectively.

Remark 7.1 (Per-antenna power constraints)**.** One type of additional constraint that cannot be treated in the framework described above is per-antenna (peak) power constraints, which are required in practice to handle the limited dynamic range of the individual power amplifiers. If all BS antennas exhibit roughly the same large-scale fading value to a UE (in terms of the diagonal elements of the spatial correlation matrix being roughly the same), the precoding will divide the transmit power almost equally over the antennas. The small-scale fading creates variations within a coherence block, but these are averaged out since the signals from many different coherence blocks separated in the frequency domain are transmitted simultaneously. The fact that the signals to different UEs are spatially multiplexed also contribute to this effect. In these cases, the per-antenna constraints can be reasonably neglected. Otherwise, we need to find an alternative solution. One approach is to take the solution from one of the optimization problems above and then heuristically reduce the total transmit power of each BS so that all per-antenna constraints are satisfied. It is also possible to optimize the precoding vectors jointly with the power allocation, as done in [346, 367, 46], but then the number of optimization variables is proportional to the number of antennas and, thus, very large in Massive MIMO.

Example of DL SE with Different Utility Functions

To illustrate the effect of different power allocation schemes on the individual SEs of the UEs, we continue the running example that was defined in Section 4.1.3 on p. 134. We consider $M = 100$, $K = 10$, $f = 2$, and spatially correlated channels based on the Gaussian local scattering model with ASD $\sigma_\varphi = 10°$. Moreover, we assume that the total DL power per BS is $P_{\max}^{\mathrm{DL}} = 30\,\mathrm{dBm}$. Except for pilots, all samples in each coherence block are used for DL data transmission.

Figure 7.2 shows CDF curves of the UEs' individual DL SEs, where the randomness is due to the UE locations and shadow fading realizations. We consider MR, RZF, and M-MMSE precoding and compare three power allocation schemes: *i*) Equal power allocation of 20 dBm per UE; *ii*) Max-min fairness; *iii*) Max product SINR. The latter two are implemented using Theorems 7.1 and 7.2.

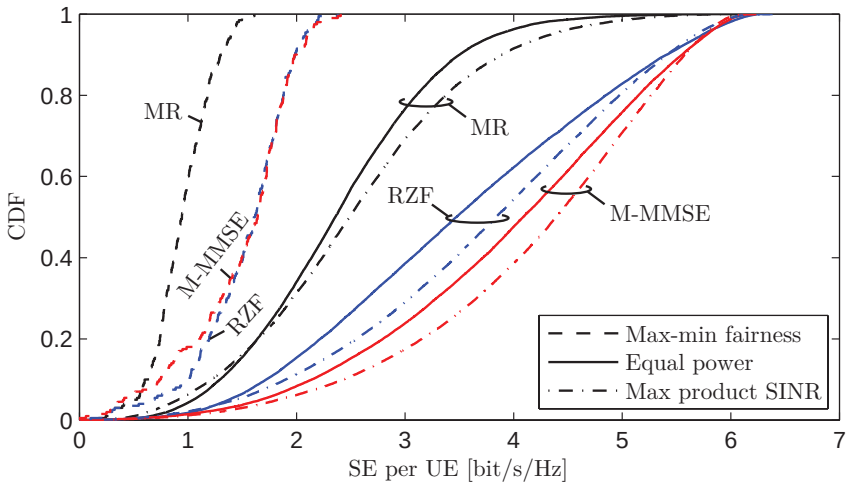


Figure 7.2: CDF of the DL SE per UE for the running example with $M = 100$, $K = 10$, $f = 2$, and using the Gaussian local scattering model with ASD $\sigma_\varphi = 10°$. We compare three power allocation schemes applied with MR, RZF, or M-MMSE precoding.

The general observation from Figure 7.2 is that, irrespective of the precoding scheme, the CDF curve with max product SINR is to the right of the equal power allocation curve, which in turn is to the right of the max-min fairness curve. This basically means that every UE will, statistically speaking, achieve better performance with max product SINR power allocation than with the other schemes.

If we look at the tails of the CDF curves, we notice that this general observation is not fully accurate. The UEs with the 10% strongest channels could achieve slightly higher SE using equal power allocation. Similarly, a small fraction of the UEs is better off with max-min fairness. For a UE at a given location, the simulation data shows that there is around 5% chance that this UE will achieve its highest SE with max-min fairness. However, for a UE that is placed at a random location and that requires a certain SE level, the chance that it will be supported is always larger with max product SINR power allocation than with max-min fairness—because the CDF curves with max product SINR in Figure 7.2 are to the right of the max-min fairness curves, except at SE levels very close to zero. Table 7.1 shows the SE level that is guaranteed to 95% of

Scheme	Max-min fairness	Equal power	Max product SINR
M-MMSE	0.40	1.67	**1.83**
RZF	0.67	1.38	**1.44**
MR	0.49	**1.05**	0.91

Table 7.1: SE in bit/s/Hz that can be guaranteed to 95% of the UEs, at random locations, with $M = 100$, $K = 10$, and $f = 2$. The results correspond to the 0.05 percentile in Figure 7.2. The largest value for each precoding scheme is in bold face.

the UEs and we note that max-min fairness provides the smallest SEs. This non-intuitive behavior is explained by the fact that the max-min fairness allocation makes every UE operate at the SE achieved by the weakest UE in the entire network. Statistically speaking, everyone gets a higher performance with max product SINR power allocation.

The three precoding schemes behave as expected, with M-MMSE giving the highest SEs and MR giving the lowest SEs. However, RZF actually works slightly better than M-MMSE with max-min fairness power allocation, as can be seen from both the CDF curves and Table 7.1. This is due to the equal power allocation assumption made for the channel estimation in this example. Since the precoding vectors depend on the UL powers, the result is that M-MMSE overemphasizes on interference suppression in the DL. This changes if we also optimize the UL powers, as described in Section 7.1.2 below.

In summary, DL power allocation can greatly affect the SE distribution among the UEs. The max product SINR utility provides a good balance between sum SE and fairness. In this simulation, it provides around 2.5× higher sum SE than max-min fairness with almost negligible SE losses for the statistically weakest UEs. We will compare these power allocation schemes again in the case study of Section 7.7. The results are similar, but one key difference is that the max-min fairness utility provides some benefits for the weakest UEs.

7.1.2 Uplink Power Control

It is more complicated to optimize the transmit powers in the UL than in the DL because the UL powers affect not only the UL data

transmission but also the quality of the channel estimates and indirectly the combining vectors. This makes the power control in practical systems (that operate with imperfect CSI) different from the classic power control algorithms studied in [369, 360, 81], which assume that the channels are perfectly known. The SE expressions for the UL in Theorem 4.1 on p. 123 and Theorem 4.4 on p. 149 appear to be too complicated for finding the optimal transmit powers by convex optimization. However, in the special case of i.i.d. Rayleigh fading channels, one can obtain closed-form SE expressions with MR and ZF that can be utilized for tractable power optimization [178, 359, 80].

We consider the general case with arbitrary spatial correlation matrices and combining schemes and assume that each UE has a maximum UL transmit power of $P_{\max}^{\mathrm{UL}} > 0$. To pave the way for the use of low-resolution ADCs at the BSs (see Section 6.4.1 on p. 289), it is important to limit the power differences between the intra-cell signals that are received at the BS. Recall that $\beta_{jk}^j = \frac{1}{M_j}\mathrm{tr}(\mathbf{R}_{jk}^j)$ denotes the average channel gain from UE k in cell j to any antenna at BS j. In a cellular network, there are commonly channel gain differences of up to 50 dB between the UEs in a cell, but when using low-resolution ADCs it is beneficial to substantially reduce these differences to avoid that weak signals drown in the quantization distortion from stronger signals.

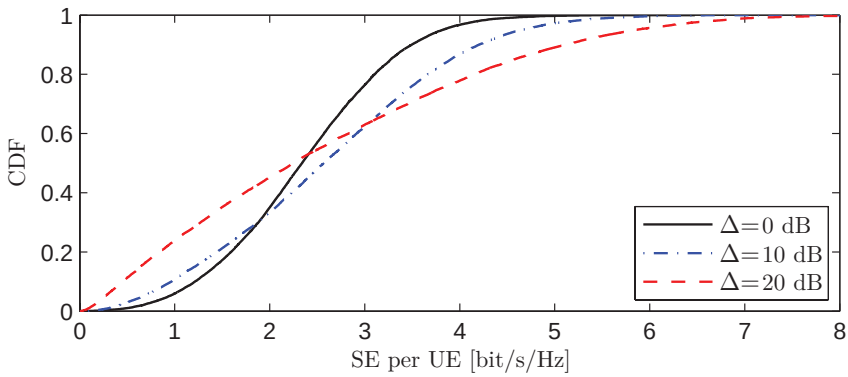
In LTE systems, this issue is managed by letting UEs at the cell edge transmit at maximum power and by gradually reducing the power for UEs that are located closer to the BS [288]. Inspired by this principle, we define a maximum received power ratio $\Delta \geq 0$ dB and consider the heuristic power control policy

$$p_{jk} = \begin{cases} P_{\max}^{\mathrm{UL}} & \Delta > \frac{\beta_{jk}^j}{\beta_{j,\min}^j} \\ P_{\max}^{\mathrm{UL}} \Delta \frac{\beta_{j,\min}^j}{\beta_{jk}^j} & \Delta \leq \frac{\beta_{jk}^j}{\beta_{j,\min}^j} \end{cases} \tag{7.11}$$

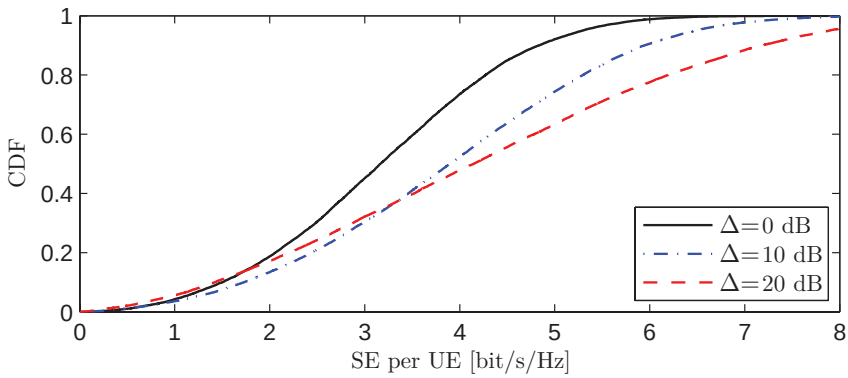
for $k = 1, \ldots, K_j$ in cell j, where $\beta_{j,\min}^j = \min_{i=1,\ldots,K_j} \beta_{ji}^j$ is the average channel gain of the weakest UE in cell j. We refer to (7.11) as power control, rather than power allocation, since no power can be reallocated between the UEs in the UL.

With the power control policy in (7.11), the UE with the smallest channel gain in the cell will use the full transmit power $P_{\max}^{\mathrm{UL}}$. The UEs with slightly larger channel gains that satisfy $\beta_{jk}^{j} < \beta_{j,\min}^{j}\Delta$ will also use maximum power, while UEs with better channel conditions will use less power so that the received signal power becomes at most Δ times larger than the received signal of the weakest UE. This heuristic policy is applied independently in every cell and does not take the spatial channel characteristics of the UEs into account, thus there is certainly room for improvements. The next example shows how the choice of Δ affects the SE of the UEs.

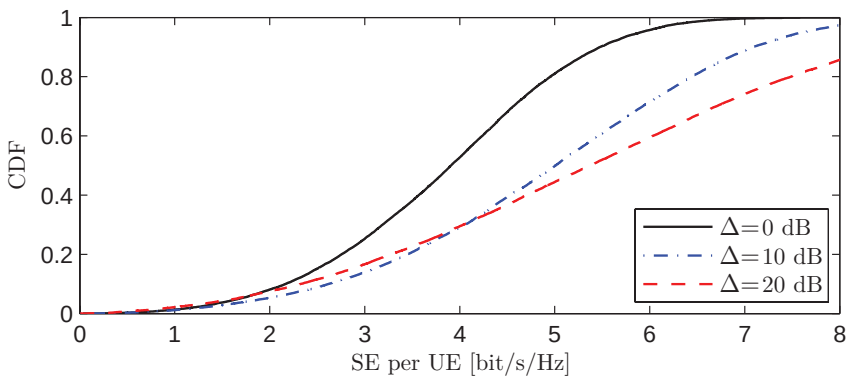
Remark 7.2 (Unequal pilot and data powers)**.** We have assumed that the UL transmit power used for pilot and data are equal, since having large differences in power between consecutive UL samples is generally not feasible due to hardware constraints, but some modulation schemes allow for it. By relaxing this constraint, we can assign unequal powers for UL data and pilot transmission. This gives greater flexibility to the power control. It can be used to set a fixed pilot power level and only optimize the data power, which leads to optimization problems with a similar structure as the DL utility maximization problem in (7.6). This approach has been adopted in [193, 210], among others, and the resulting power control solutions build on the heritage from utility maximization with perfect CSI [81]. Alternatively, the pilot and data powers can be treated as separate variables and optimized jointly, as carried out in [135, 247, 80] for some different utility functions. In particular, under the assumption of i.i.d. Rayleigh fading and MR or ZF combining, the utility maximization may lead to geometric programs that can be solved to global optimality. The gain from having an unequal pilot and data powers are large for UEs with weak channel conditions, which can then allocate more power for channel estimation and, thus, enable better receive combining (e.g., a larger array gain and better interference suppression).



(a) MR combining with the heuristic power control policy in (7.11).



(b) RZF combining with the heuristic power control policy in (7.11).



(c) M-MMSE combining with the heuristic power control policy in (7.11).

Figure 7.3: CDF of the UL SE per UE with $M = 100$, $K = 10$, $f = 2$, and using the local scattering model with Gaussian angular distribution and ASD $\sigma_\varphi = 10^\circ$.

Example of UL SE with Heuristic Power Control

We will now exemplify the SE distribution achieved by the heuristic power control scheme in (7.11) with MR, RZF, and M-MMSE combining. We continue the running example that was defined in Section 4.1.3 on p. 134. We consider $M = 100$, $K = 10$, $f = 2$, and the Gaussian local scattering model with ASD $\sigma_\varphi = 10°$. The maximum UL transmit power is $P_{\max}^{\mathrm{UL}} = 20\,\mathrm{dBm}$ per UE. Except for pilots, all samples in each coherence block are used for UL data transmission.

Figure 7.3 shows CDF curves of the UEs' individual SE, where the randomness is due to the UE locations and shadow fading realizations. For each scheme, we consider $\Delta \in \{0, 10, 20\}\,\mathrm{dB}$.

A first observation is that there are large variations in SE, even for the case of $\Delta = 0\,\mathrm{dB}$ in which all UEs in a cell adapt their transmit power to reach the same received signal power. There are several reasons for this. First, it is the UE with the worst channel condition in a cell that determines the SE level of all UEs in the cell and its location and shadow fading realization vary substantially. Second, the interference from other cells varies as well, depending on their UE distribution and power control. Third, the spatial channel correlation creates further variations, since the UEs with the smallest channel gain may have similar spatial characteristics as an interfering UE and the UE with the best channel gain may have spatial characteristics that are different from all interfering UEs. When running the same simulation with uncorrelated fading, the CDF curves are compressed around their mean values, in the sense that the lower and upper tails are smaller.

The choice of receive combining scheme impacts the shape of the CDF curves. With MR, the weakest UEs greatly benefits from using a small Δ, since the intra-cell interference from UEs with good channel conditions would be large unless the power control reduces their power. UEs with good channel conditions prefer a larger Δ, since they can then obtain a higher SE. With RZF or M-MMSE, the intra-cell interference is suppressed to such an extent that the UEs with good channels can transmit their signals to get a larger received signal power, while the detrimental effect on the weakest UEs is almost negligible. This shows that, by having a larger received power, the increased quality of the

channel estimate improves the receive combining to such an extent that it almost entirely counteracts the increased interference level that it creates.

7.2 Spatial Resource Allocation

There can be thousands of UEs residing in each cell of a cellular network, but only a small fraction of them are in general active in a given coherence block. The intermittent activity is both created by the end user and by the bursty nature of packet-based data traffic. While the number of active UEs can change rapidly, the number of pilot sequences τ_p is essentially fixed in practice—one BS cannot decide to reduce the number of pilots if another BS needs all the pilots. The number can change over the course of the day, to adapt to the substantial long-term variations in average traffic load that occur in practice [26], but it is not practical to adapt τ_p to short-term traffic variations.

In this section, we outline some key considerations in the allocation of spatial resources, which includes the pilot assignment, the interplay between spatial multiplexing and time-frequency scheduling, and the impact of traffic load variations on the sum SE. The number of pilot sequences and how these are allocated to the UEs will play a key role.

7.2.1 Pilot Assignment

In cellular networks, every UE that intends to connect to a BS must go through a network entry procedure. This refers to all the functions that a UE goes through in order to establish a communication link with the BS for data transmission and reception. In the LTE standard, this procedure is called random access (RA) and relies on a contention-based protocol. The development of RA procedures for Massive MIMO systems is still in its infancy and is therefore not treated in this monograph, but a few details are given in Remark 7.4 later. Next, we assume that the RA procedure has been successfully completed, and we will only deal with the pilot assignment problem. Once a UE is connected to a cell, it is assigned to one of the pilot sequences that are available in that cell. The pilot assignment implicitly determines which other

UEs in the network will cause pilot contamination to the UE. The assignment thereby impacts the SE that the UE will achieve. Joint optimization of the pilot assignment across cells is a way to empower the weakest UEs [192, 363, 378], which are the ones most susceptible to the reduced estimation quality and the coherent interference caused by pilot contamination. The pilot assignment is a combinatorial problem in which BS j should select K_j pilots from the set of τ_p pilots and assign them properly to its K_j UEs. The computational complexity grows very rapidly with the number of UEs and is therefore of limited practical use.

There are two categories of heuristic solutions in the literature. The first category contains greedy algorithms [192, 363, 378] that assign and reassign the pilots to the UEs in an iterative manner to improve a utility function. For example, if a UE at the cell edge is greatly affected by pilot contamination, it can switch pilot with a UE in the cell-center, motivated by the fact that UEs with strong channel conditions are less affected by pilot contamination. The second category consists of predetermined pilot reuse patterns [154, 358, 49], where the τ_p pilots are divided into f groups with τ_p/f pilots each. The integer f is called the pilot reuse factor. Each cell is associated with one of these disjunct pilot groups, according to a predetermined pattern. This approach was taken in the running example, for which Figure 4.4b shows three different pilot reuse patterns in a symmetric network. An example with $f = 3$ in an asymmetric cellular network is provided in Figure 7.4, where each color/pattern represents one pilot group. The classical four-color theorem proves that $f = 4$ is sufficient to make sure that every cell belongs to another cell group than its immediate neighbors [130]. Note that the use of pilot reuse patterns resembles the frequency reuse patterns in GSM, but in contrast to legacy systems the reuse patterns are only applied to the pilot transmission while all cells send payload data in parallel over the full bandwidth (i.e., universal frequency reuse). The reuse pattern should be designed to automatically give a large spatial separation between UEs that use the same pilot, to alleviate the need for further coordination between cells. Note that the two categories described above can also be used jointly, by having pilot reuse patterns

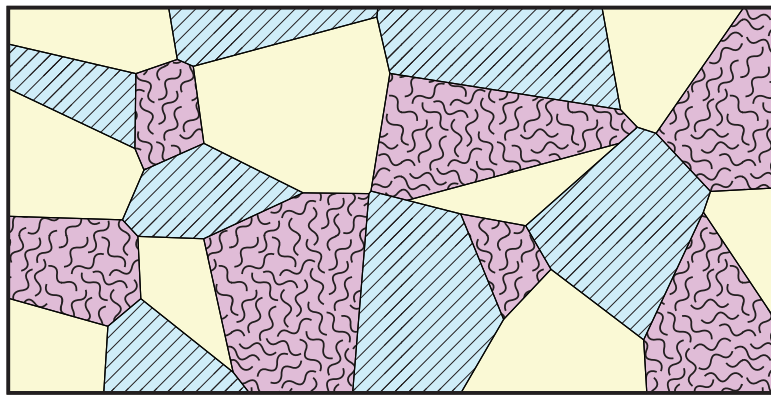


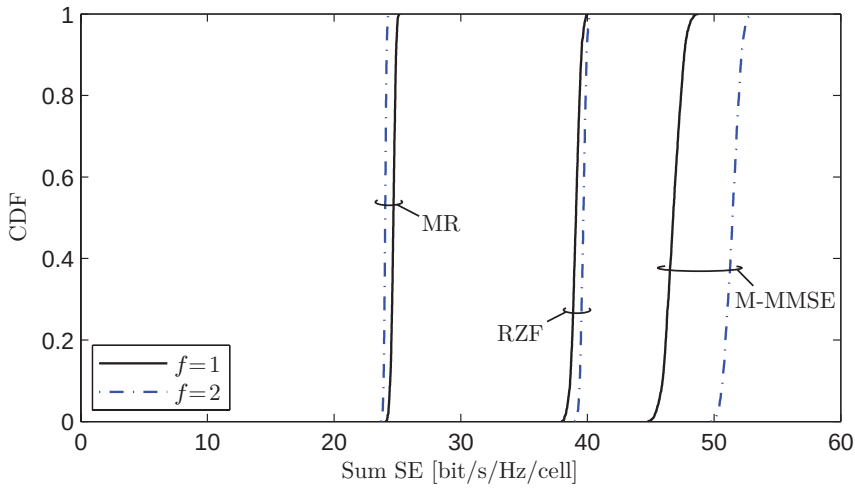
Figure 7.4: Illustration of the division of an asymmetric cellular network into $f = 3$ disjoint pilot groups, where f is called the pilot reuse factor. Each group is indicated with a distinct color and pattern.

and then apply a greedy algorithm for pilot assignment within each pilot group.

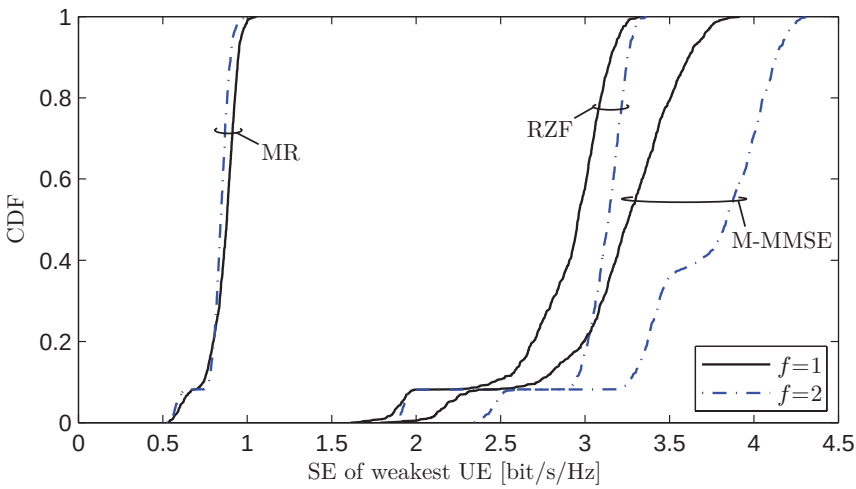
A pilot assignment that is good for a UE in the UL, in terms of giving little coherent interference, is not necessarily good for the UE in the DL. For example, the spatial channel correlation can be very different depending on who is transmitting, as illustrated in Figure 4.16. Channel gain differences can also create asymmetry since the UL interference is sent from UEs in other cells and the DL interference is sent from BSs in other cells.

To investigate the impact of pilot assignment on the SEs, we revisit the running example defined in Section 4.1.3 on p. 134. We consider one snapshot of the network with $K = 10$ UEs at fixed locations in each cell. We focus on the UL with $M = 100$ antennas, 20 dBm transmit power per UE, and the Gaussian local scattering model with ASD $\sigma_\varphi = 10°$. Except for the pilots, all samples in the coherence blocks are used for UL data transmission.

In order to show the impact that different types of pilot assignment can have on the SE, we consider the pilot reuse factors $f = 1$ and $f = 2$, as illustrated in Figure 4.4b. We uniformly randomize the pilot assignment in every cell and pilot group, and we plot CDF curves of the resulting SE variations. Figure 7.5a shows the sum SE, averaged



(a) Impact of pilot assignment on the average sum SE.



(b) Impact of pilot assignment on the SE of the weakest UE in a random cell.

Figure 7.5: CDFs of the average UL sum SE among the cells and the weakest UE's SE in an arbitrary cell for different random pilot assignments. The running example is considered for different pilot reuse factors f, $M = 100$, $K = 10$, and the Gaussian local scattering model with ASD $\sigma_\varphi = 10°$.

over the cells, and Figure 7.5b shows the SE of the UE with the weakest channel (smallest average channel gain) in an arbitrarily selected cell. Results are provided for MR, RZF, and M-MMSE combining.

The sum SE in Figure 7.5a is greatly affected by the choice of combining scheme and slightly affected by the choice of pilot reuse factor. More importantly, all the CDF curves are almost vertical. The largest value is only 3%–9% higher than the smallest value, with the largest gain achieved with M-MMSE. Hence, there is little to gain in sum SE from optimizing the pilot assignment (e.g., using a greedy algorithm).

If we look instead at the SE in Figure 7.5b of the UE with the weakest channel in a randomly selected cell, we observe that it fluctuates substantially. The curves are not smooth since the pilot assignment problem is combinatorial. In particular, we notice that there is one interfering UE that provides particularly high pilot contamination, which results in the discontinuity at 0.1 probability (which is the probability that two particular UEs use the same pilot when assigning 10 pilots uniformly at random). The absolute SE variations are small with MR, since its performance is mainly limited by non-coherent intra-cell interference, while the weakest UE has much to gain from optimizing the pilot assignment when using RZF or M-MMSE combining.

In summary, the sum SE can be optimized by selecting an appropriate pilot reuse factor. The exact pilot assignment within a cell has little impact on the sum SE, but can greatly affect how the sum SE is divided among the UEs. A reasonable goal for the pilot assignment is to improve the fairness, by making sure that UEs with bad channel conditions are assigned pilots that give little pilot contamination.

Remark 7.3 (Joint spatial-division and multiplexing)**.** An interesting transmission scheme that allows pilot reuse within a cell is joint spatial-division and multiplexing (JSDM) [7, 235]. The key idea behind JSDM is to capitalize on spatial channel correlation of large antenna arrays. This allows us to partition a cell into geographical regions which are characterized by spatial correlation matrices that are almost spatially orthogonal (see Definition 4.2 on p. 187). Suppose we can partition the K UEs in a cell into G groups consisting of K_g UEs, $g = 1, \ldots, G$, respectively, and denote by $\mathbf{h}_{gk} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{R}_{gk})$ the channel of the kth

UE in the gth group. The UE grouping is done in such a way that the correlation matrices of the UEs in each group are very similar, in the sense that $\frac{1}{\mathrm{tr}(\mathbf{R}_{gk})}\mathbf{R}_{gk} \approx \frac{1}{\mathrm{tr}(\mathbf{R}_{gl})}\mathbf{R}_{gl}$ for all g, k, l, but almost orthogonal to the correlation matrices of the UEs in other groups: $\mathrm{tr}(\mathbf{R}_{gk}\mathbf{R}_{hl}) \approx 0$ for all k, l and $g \neq h$. As there is little interference between the groups by design, we can reuse the same pilot sequences for each group without causing prohibitive amounts of pilot contamination. Due to the intra-cell pilot reuse, a larger number of UEs can be served per cell for a given τ_p. Although JSDM is a theoretically very appealing concept, it is at the time of writing this monograph unclear if practical channels have the necessary orthogonality properties.

Remark 7.4 (Random access in Massive MIMO)**.** The RA is a contention-based procedure by which a UE can connect to a cell in the network, either as an initial connection to the network or to switch cell. This procedure is used in LTE and, in its basic form, operates as follows: Each accessing UE acquires basic synchronization from the BS (e.g., determining LTE parameters, frequency synchronization, and frame timing) and makes use of the so-called random access channel (RACH)[1] to transmit a randomly selected pilot-like sequence. Since the accessing UEs are not coordinated in selecting their sequences, collisions may occur. After detecting the selected sequences and trying to identify the UEs that are using a given sequence (collision resolution), the BS broadcasts a response message, informing the identified UEs that the RA procedure has been successful while providing physical parameters (e.g., timing adjustments). The detected UEs send connection requests to further specify the resources needed for data transmission. On the other hand, the undetected UEs repeat the RA procedure after a random waiting time, until successful notification. As exemplified by this monograph, the benefits of Massive MIMO in terms of SE, EE and HE are nowadays well understood. On the other hand, the potential impact that Massive MIMO has on the network access functionalities has received less attention so far. Recent attempts in this direction can be found in [76, 41, 286, 305]. In [41], the authors exploit the

[1]The RACH is typically composed of a specified set of consecutive OFDM symbols and adjacent subcarriers.

channel hardening and spatial resolution of Massive MIMO to resolve collisions distributively. This approach can be used as an add-on to conventional resolution mechanisms. A coded RA protocol leveraging the channel hardening properties of Massive MIMO is presented in [305]. The large number of antennas were used in [286] to obtain accurate estimates of the timing misalignments of the accessing UEs, which were used to develop an RA procedure that resolves collisions with high probability.

7.2.2 Scheduling

While conventional BSs manage intra-cell interference by scheduling the active UEs in different coherence blocks, Massive MIMOs greatly alleviates this issue since the interference is mitigated spatially, by receive combining and transmit precoding. There is no fundamental upper limit on how many UEs can be served per cell in a given coherence block—it is even practically feasible to serve more UEs than there are BS antennas[2] and to assign a pilot sequence to multiple UEs within a cell (cf. Remark 7.3). As we increase the number of UEs, for a fixed number of BS antennas, the sum SE first increases, then reaches a maximum, and finally starts to decay again. As long as the increased multiplexing gain (i.e., the number of SEs that are summed up) outweighs the larger pilot overhead and extra interference (i.e., the reduction of the individual SEs), the sum SE increases. It was proved in [49] that one should not use more than half the samples in a coherence block for pilots, as $M \to \infty$, but in practice, the number of pilots can be optimized based on the anticipated number of UEs.

These behaviors are illustrated in Figure 7.6. The horizontal axis represents the number of single-antenna UEs or, equivalently, the number of data streams that are transmitted or received by the BS. The vertical axis illustrates the sum SE in a scenario that can either represent the UL or DL. Three operating regimes are identified in the figure. The first is the *multiplexing regime* in which the sum SE grows almost linearly with the number of served UEs, because the increased multiplexing gain

[2] All receive combining and transmit precoding schemes described in Section 4, except ZF, can be applied with an arbitrary number of UEs.

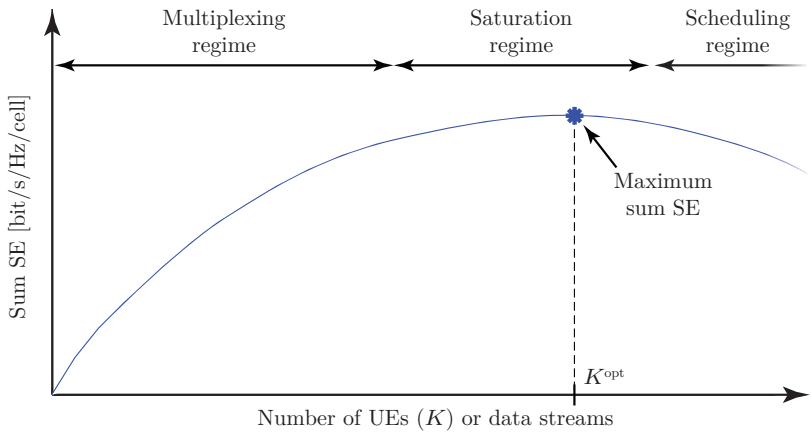


Figure 7.6: Illustration of how the number of UEs affect the sum SE, of either the UL or DL. There is a maximum at some K^{opt}. In order to maximize the area throughput, the network should serve all K UEs over the entire bandwidth when $K \leq K^{\text{opt}}$, while scheduling should be used to bring down the effective number of UEs to K^{opt} when $K > K^{\text{opt}}$.

dominates over the extra interference. The growth slows down in the *saturation regime*, where we eventually reach a number of UEs that provides the maximum sum SE. Beyond this point, for every additional UE, the sum SE decreases. A well-dimensioned network operates in the saturation regime and occasionally in the multiplexing regime (when the data traffic is low). The scheduling of UEs is then trivial because all time-frequency resources are allocated to all active UEs in an effort to maximize the throughput of the network.

The regime beyond the maximum point is called the *scheduling regime*. As the sum SE decreases, it does not pay off to serve all of them in parallel by spatial multiplexing. Time-frequency scheduling can then be used to reduce the effective number of UEs per coherence block to operate close to the maximum sum SE point. Conventional cellular networks with 1–8 antennas reach the scheduling regime as soon as there are more than a few active UEs in the cell, while the next example illustrates that Massive MIMO can handle many tens of UEs before reaching the scheduling regime.

Due to the natural traffic variations over the day, from rush hours in the mornings and evenings to low-traffic hours in the late nights, cellular networks are often dimensioned for the highest anticipated data traffic load. Consequently, during parts of the day, there are not sufficiently many active UEs to reach the maximum sum SE by sending one data stream per UE. It is then important to recall that we have focused on the single-stream transmission to single-antenna UEs in this monograph, while multiantenna UEs have only been briefly discussed in Remark 1.4 on p. 50. The key benefit of having multiple UE antennas is that we can spatially multiplex several data streams to/from the UE, for example, by treating each antenna as a separate UE in the signal processing. In a practical scenario where some UEs are equipped with multiple antennas, it is the scheduling algorithm that determines how many streams should be assigned to each UE. Recall that the horizontal axis in Figure 7.6 represents the number of data streams that are simultaneously spatially multiplexed by the BS, which not necessarily equals the number of UEs. When there are few active UEs in a cell—fewer than what is needed to reach the saturation regime by single-stream transmission—we can increase the sum SE by sending multiple data streams to some of the UEs. In contrast, when there are sufficiently many UEs to reach the saturation regime by sending only one data stream per UE, this is the preferred choice [194, 52]. In principle, we could instead schedule a subset of the UEs and send multiple streams to each of them. But since the channels to the different antennas of a UE can be strongly spatially correlated, the sum SE is typically larger when we schedule a larger number of UEs and send only one stream to each of them.

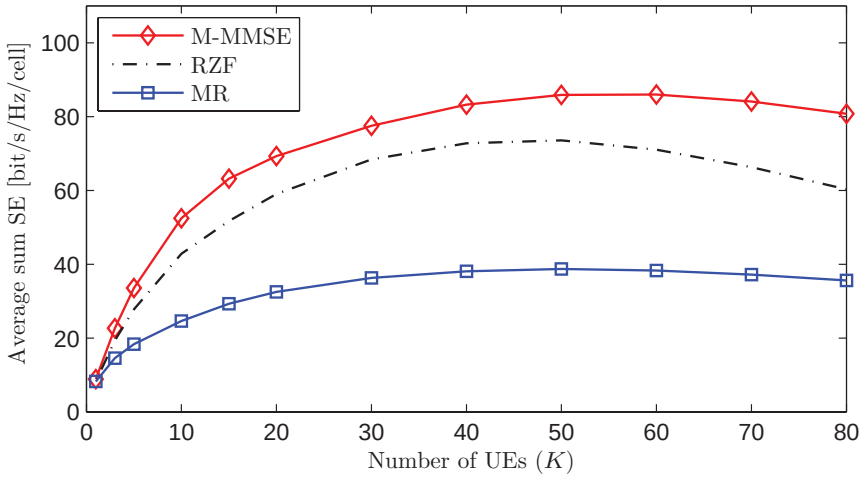
We will now exemplify the multiplexing, saturation, and scheduling regimes by revisiting the running example that was defined in Section 4.1.3 on p. 134. We consider $M = 100$ antennas, 20 dBm transmit power per UE in both UL and DL, and the Gaussian local scattering model with ASD $\sigma_\varphi = 10°$. Figure 7.7 shows the DL sum SE as a function of the number of active UEs, for two different channel coherence block lengths: $\tau_c = 200$ and $\tau_c = 400$. In each case, $\tau_p = fK$ samples are used for pilots and the remaining samples are used for DL data transmission. For each number of antennas, each coherence block length,

and each of MR, RZF, and M-MMSE precoding, we use the pilot reuse factor $f \in \{1, 2\}$ and bounding technique that maximizes the sum SE.

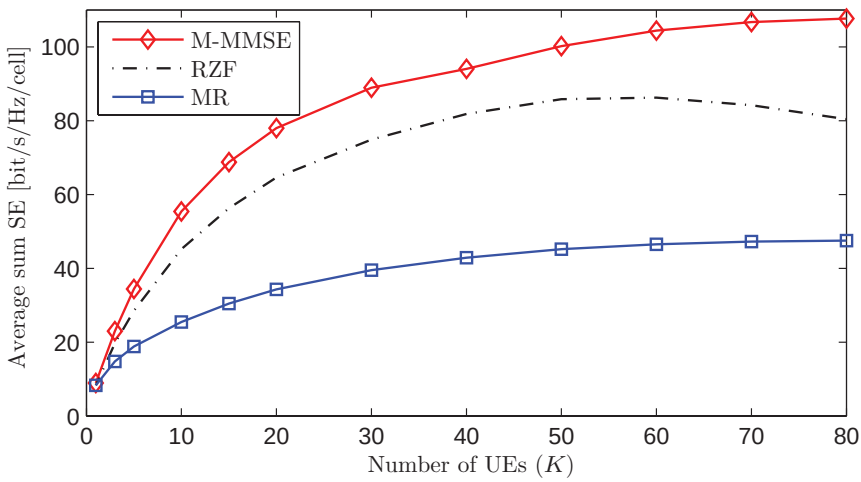
The simulation results match well with the schematic behavior described in Figure 7.6. The sum SE grows steeply with K for the first 20 UEs. It then keeps increasing linearly with K when continuing to increase the number of UEs, but with a smaller slope. Hence, the SE per UE decays while the sum SE grows. The pilot reuse factor $f = 2$ is beneficial when K is small, while $f = 1$ gives the largest sum SE as K increases (since it becomes important to reduce the pilot overhead). The location of the maximum value depends on the length of the channel coherence block and precoding scheme. With $\tau_c = 200$, the maximum sum SE is achieved at around $K = 50$ for all precoding schemes. All SE values increase when using $\tau_c = 400$ instead since the relative size of the pilot overhead is reduced. The maximum sum SE is now achieved at $K = 60$ for RZF, while MR and M-MMSE can handle 80 UEs without reaching the maximum point.

The numerical comparison between different precoding schemes in Section 4.3 on p. 162 focused on $K = 10$. Figure 7.7 shows that the differences between the precoding schemes grow as more UEs are added. M-MMSE and RZF can achieve more than twice the SE of MR. RZF is competitive with M-MMSE for $K \leq 40$, but the performance gap is substantial for larger K.

In summary, Massive MIMO leads to a paradigm shift when it comes to scheduling, which becomes the last resort to deal with peak traffic loads, instead of the main method for resource allocation—as in conventional networks. The maximum sum SE in this example is achieved when scheduling tens of UEs and having an antenna-UE ratio of around $M/K = 100/50 = 2$. The SE per UE is not particularly large at the sum SE maximizing point. With $K = 50$, the average SE per UE is 1.7–2.0 bit/s/Hz with M-MMSE and 0.8–0.9 bit/s/Hz with MR. This can be achieved in practice using 4-QAM modulation and different channel codes. The average throughput per UE is still substantial because there is no time/frequency scheduling, so every UE enjoys the full bandwidth. Recall that, in the running example, we consider 20 MHz bandwidth, which with M-MMSE leads to an average



(a) Coherence block length: $\tau_c = 200$.



(b) Coherence block length: $\tau_c = 400$.

Figure 7.7: Average DL sum SE as a function of the number of UEs, for $M = 100$ and different precoding schemes. We consider the local scattering model with Gaussian angular distribution and ASD $\sigma_\varphi = 10°$. The pilot reuse factor is optimized for each point on the curves.

of 34–40 Mbit/s for each of the 50 UEs and a total of 1.7–2.0 Gbit/s/cell. The latter number can be compared with the 64 Mbit/s/cell that is achieved by a contemporary LTE system for an equivalent bandwidth (see Remark 4.1 on p. 138). Hence, the simulated Massive MIMO setup provides more than an order-of-magnitude higher cell throughput.

7.2.3 Impact of Traffic Load Variations

As mentioned in the beginning of Section 7.2, the number of pilots τ_p is typically constant in practice, while the number of UEs that have data to transmit/receive can change rapidly due to user behavior and the bursty nature of packet transmission. The number of active UEs in an arbitrary coherence block can be treated as a random variable and Poisson distributions are commonly used to model such traffic variations. For example, $K^{\text{active}} \sim \text{Po}(K)$ is a random integer with mean K and standard deviation $\sqrt{K}$. If we use time/frequency scheduling to only serve τ_p UEs when $K^{\text{active}} > \tau_p$, we can use $\min(K^{\text{active}}, \tau_p)$ as a way to randomly generate the number of active UEs in an arbitrary coherence block. This distribution is shown in Figure 7.8 for $K \in \{1, 10, 20, 40\}$ and $\tau_p = 40$. The figure shows that the load distribution has a Gaussian-like shape and the variations grow with K, so in a cell with more UEs, there will also be larger traffic variations. Note that when $K = \tau_p$, it happens frequently that $K^{\text{active}} > \tau_p$ and then scheduling is often needed to handle the traffic load variations.

Both the load variations and the concepts of multiplexing/saturation regimes, illustrated in Figure 7.6, are important when dimensioning the number of pilots in practice. When the traffic load is high, τ_p can be selected based on what is needed to reach the maximum sum SE point. When the traffic load is low, τ_p can be selected based on the distribution in Figure 7.8 to balance between having a low probability of pilot shortage with $K^{\text{active}} > \tau_p$ and keeping the pilot overhead low.

As long as there are sufficiently many pilots, each UE should connect to the BS that provides the best channel conditions [82], which can be measured as having the largest trace of the spatial correlation matrix. However, when there is a pilot shortage in a cell, which calls for time-

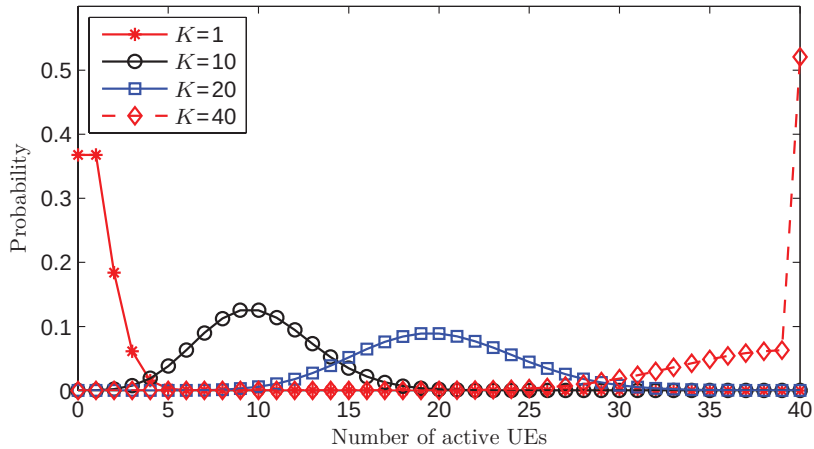


Figure 7.8: Distribution of the number of active UEs in a cell, modeled as $\min(K^{\text{active}}, \tau_p)$ where $K^{\text{active}} \sim \text{Po}(K)$ and $\tau_p = 40$. Note that the probability of $K^{\text{active}} > \tau_p$ is substantial for $K = 40$, which leads to the large probability of having 40 active UEs in that case.

frequency scheduling, it can happen that a few UEs get higher SE by connecting to neighboring cells, which currently have lower traffic load [35]. This type of load balancing is important in conventional networks, which mainly rely on time-frequency scheduling, but is less critical in Massive MIMO. For example, if $K^{\text{active}} > \tau_p$, then each UE can be scheduled in a fraction τ_p/K^{active} of all coherence blocks. If there are 10 more UEs than pilots, the fraction becomes 0.09 in a network with $\tau_p = 1$ and 0.67 in a Massive MIMO network with $\tau_p = 20$.

We will now illustrate how the sum SE of a cell is affected by having a varying number of UEs per cell. To this end, we continue the running example that was defined in Section 4.1.3 on p. 134. We consider $M = 100$ antennas, 20 dBm transmit power per UE in both UL and DL, and spatially correlated channels based on the Gaussian local scattering model with ASD $\sigma_\varphi = 10°$. We denote the average number of UEs per cell as K and consider two scenarios. In the first scenario, the number of UEs is exactly K in every coherence block. In the second scenario, the number of UEs is independently randomized in each cell as $\min(K^{\text{active}}, \tau_p)$ where $K^{\text{active}} \sim \text{Po}(K)$. The number of pilot sequences

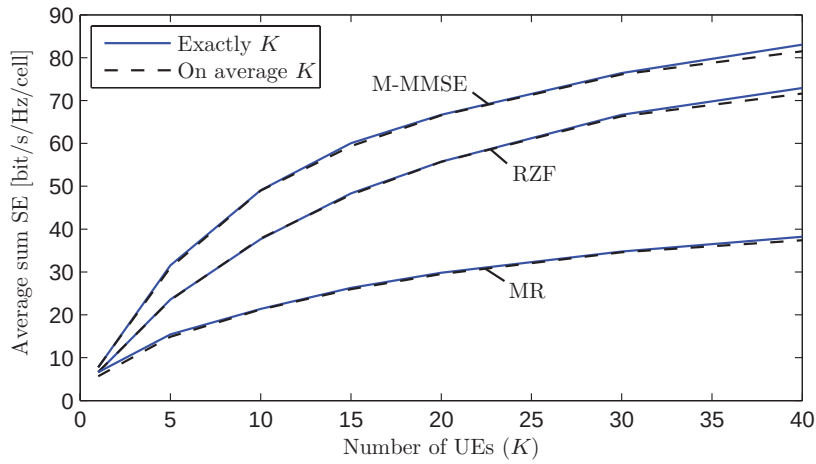


Figure 7.9: Average DL sum SE as a function of the average number of UEs for $M = 100$, $\tau_p = 40$, and different precoding schemes. In each coherence block, the number of UEs in a cell is either exactly equal to K or computed as $\min(K^{\text{active}}, \tau_p)$, where $K^{\text{active}} \sim \text{Po}(K)$. We consider the local scattering model with Gaussian angular distribution and ASD $\sigma_\varphi = 10°$.

is $\tau_p = 40$ and each BS independently selects a random subset of pilots so that its own UEs use orthogonal pilots, but the pilot contamination across cells is random.

The average DL sum SE is shown in Figure 7.9 for different K, using M-MMSE, RZF, or MR precoding. These schemes behave as expected from previous examples. The key observation is that the two scenarios give basically the same sum SE in all cases. This is explained by the fact that the sum SE grows almost linearly with the number of UEs, thus constantly serving 10 UEs gives roughly the same average sum SE as switching between serving 8 and 12 UEs with equal probability. The largest deviation between the curves occurs when K is close to τ_p, in which case the sum SE is slightly smaller when having a random number of UEs. This loss is due to the scheduling because $\mathbb{E}\{\min(K^{\text{active}}, \tau_p)\} < K$ in these cases, although $\mathbb{E}\{K^{\text{active}}\} = K$.

In summary, the number of active UEs varies between coherence blocks in practice, due to the bursty traffic demand. Substantial changes can occur over a few tens of coherence times, but we need to select a

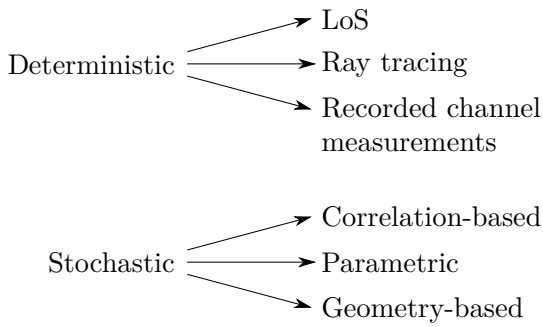


Figure 7.10: Taxonomy of wireless channel models.

fixed pilot length to accommodate different loads in the cells at different times. When quantifying the sum SE for network planning, it is not necessary to randomize the number of UEs per cell, since the results are almost the same as when we have a fixed number of UEs that equals the average number of UEs.

7.3 Channel Modeling

Realistic performance assessment of Massive MIMO systems requires the use of a channel model that reflects the main characteristics of large antenna arrays. Such a model must account at least for the array geometry, the correlation between the channel responses of different antennas, and the physical location and orientation of BSs and UEs. Our goal here is not to provide an introduction to channel modeling for MIMO systems, but rather to convey some insights from simple and analytically tractable models, which are commonly used in the research literature. For further details, the interested reader is referred to classical textbooks and tutorial papers on channel modeling for MIMO systems, such as [255, 225, 12, 88].

The taxonomy of channel models for wireless communications is provided in Figure 7.10. In a nutshell, channel models are either deterministic or stochastic. Deterministic models depend on a given environment with fixed locations of transmitters, receivers, scatterers, reflectors, etc. Examples of such models are ray tracing based on 3D-building models and recorded channel measurements. Also, the previously introduced

LoS model for a horizontal ULA in (1.23) is an example of a very simple deterministic channel model. We will provide an extension of this model to three dimensions and arbitrary array geometries in Section 7.3.1. A drawback of deterministic models is that they are only valid for a specific scenario and, consequently, do not allow for far-reaching conclusions. Moreover, the results cannot be easily reproduced by others as there are very few openly accessible databases of channel measurements and 3D-building models. However, deterministic models can provide very accurate performance predictions for their specific scenarios.

Stochastic channel models are independent of a particular environment and can be used to generate an essentially unlimited amount of channel realizations with the desired statistical properties. These models can be roughly separated into correlation-based, parametric, and geometry-based channel models. The correlated Rayleigh fading channel model, as introduced in (2.1), is an example of a correlation-based model. The channel responses are all Gaussian distributed with zero mean and entirely defined through the correlation matrices. However, it is a frequency flat-fading channel model since all multipath components are assumed to arrive with the same delay (or to be irresolvable with the sampling frequency of the system). For most of the analysis in this monograph, this is sufficient as we focus only on communication over a coherence block in which the channel is assumed to be constant (see Definition 2.2 on p. 66). In contrast, parametric channel models define stochastic distributions of the number of multipath clusters and the delay, power, angle of arrival (AoA), and angle of departure (AoD) of the individual multipath components. Examples include the Saleh-Valenzuela model [283] and extensions thereof [338, 85]. Since parametric models are independent of the geometry of the propagation environment, they can generally not be used for system-level simulations with time-evolution of the channel, caused by movements of the transmitters or receivers. Lastly, stochastic geometry-based models define a distribution of the physical location of scatterers around the transmitters and receivers. Once the locations of all scatterers are chosen, individual propagation paths are modeled in a quasi-deterministic manner. Such models are frequently adopted by standardization bodies such as the

3GPP or the Institute of Electrical and Electronics Engineers (IEEE) since they are easy to simulate, agree very well with measurements, and enable time-evolution. Geometry-based models can be seen as a balance between the two extremes of purely stochastic and purely deterministic channel modeling. We will discuss the 3GPP 3D MIMO channel model [1] as an example in Section 7.3.3.

In (1.23), we introduced a deterministic LoS channel model for a horizontal ULA in a two-dimensional setting. The only parameters of this model are the azimuth angle of the incoming wave and the channel gain. It is important to remember that this model is based on a plane-wave assumption, that is only valid if the UE is located within the far-field of the antenna arrays.[3] We will now extend this channel model to three dimensions and arbitrary antenna array geometries.

Antenna arrays come in arbitrary shapes and sizes depending on the use case, carrier frequency, and the area to be covered. The most widely used arrays in cellular communications are either linear or planar, but also cylindrical arrays find applications in ground-based military communications. Figure 7.11 shows some common antenna array architectures, namely the horizontal and vertical ULA as well as the planar rectangular and cylindrical array. While a horizontal (vertical) ULA is only capable of separating UEs in the azimuth (elevation) domain, a planar or cylindrical array is capable of separating UEs in both azimuth and elevation domains. This aspect becomes very important in metropolitan areas with high-rising buildings, where UEs are located on different floors. For this reason, it is important that a channel model for Massive MIMO systems captures the 3D nature of the propagation environment. We will discuss the impact of the array geometry in detail in Section 7.4.

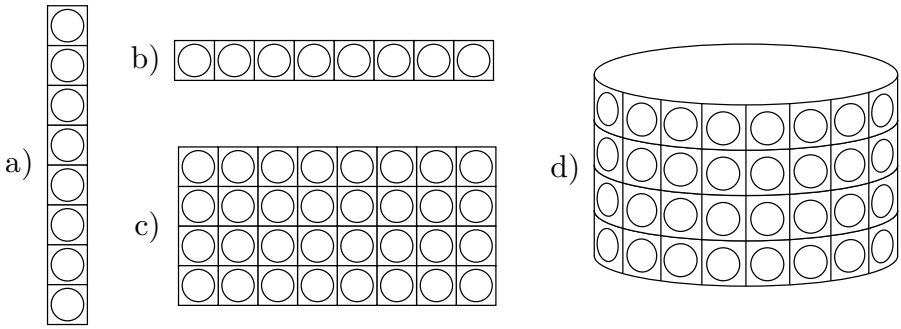


Figure 7.11: Examples of different antenna array geometries: a) linear vertical; b) linear horizontal; c) planar; d) cylindrical. Each circle represents one antenna.

7.3.1 3D LoS Model with Arbitrary Array Geometry

To proceed, we need to define the wave vector $\mathbf{k}(\varphi, \theta) \in \mathbb{R}^3$ of a plane wave with wavelength λ that impinges on the antenna array under the azimuth angle $\varphi \in [-\pi, \pi)$ and the elevation angle $\theta \in [-\pi/2, \pi/2)$:

$$\mathbf{k}(\varphi, \theta) = \frac{2\pi}{\lambda} \begin{pmatrix} \cos(\theta)\cos(\varphi) \\ \cos(\theta)\sin(\varphi) \\ \sin(\theta) \end{pmatrix}. \tag{7.12}$$

The wave vector $\mathbf{k}(\varphi, \theta)$ describes the phase variation of a plane wave with respect to the three Cartesian coordinates (see Figure 7.12). Thus, the wave observed at location $\mathbf{u} \in \mathbb{R}^3$ experiences a phase shift of $\mathbf{k}(\varphi, \theta)^{\mathrm{T}}\mathbf{u}$ with respect to the origin. Consequently, the LoS channel response $\mathbf{h} \in \mathbb{C}^M$ of an antenna array with M antennas, respectively placed at the locations $\mathbf{u}_m \in \mathbb{R}^3$, $m = 1, \ldots, M$, is given by

$$\mathbf{h} = \sqrt{\beta} \underbrace{\left[e^{\mathrm{j}\mathbf{k}(\varphi,\theta)^{\mathrm{T}}\mathbf{u}_1}, \ldots, e^{\mathrm{j}\mathbf{k}(\varphi,\theta)^{\mathrm{T}}\mathbf{u}_M} \right]^{\mathrm{T}}}_{\triangleq \mathbf{a}(\varphi,\theta)} \tag{7.13}$$

where β describes the macroscopic large-scale fading and is assumed to be the same for all antennas (i.e., the array aperture is small as

[3] In the far-field of an antenna in free space, the power intensity of the EM radiation is inversely proportional to the squared distance. The far-field region generally begins at a few wavelengths away from an antenna, while the far-field region of an array begins at a distance that grows with the square of the array aperture.

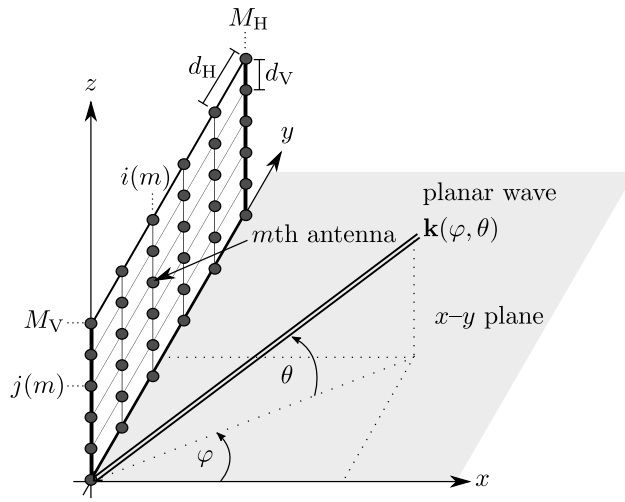


Figure 7.12: Planar rectangular antenna array with an impinging plane wave from elevation angle θ and azimuth angle φ. The $M = M_{\mathrm{V}}M_{\mathrm{H}}$ antennas are horizontally and vertically equally spaced with spacing d_{H} and d_{V}, respectively. The mth antenna has the horizontal index $i(m)$ and the vertical index $j(m)$.

compared to the propagation distance). The vector $\mathbf{a}(\varphi,\theta) \in \mathbb{C}^M$ is the so called *array response* or steering vector. For a horizontal ULA along the y-axis with antenna spacing d_{H} (in multiples of the wavelength) and waves arriving only from directions within the x-y plane (i.e., $\theta = 0$ and $\mathbf{u}_m = [0, \lambda(m-1)d_{\mathrm{H}}, 0]^{\mathrm{T}}$), it is easy to see that (7.13) coincides with (1.23).

Figure 7.12 shows a planar array in the y-z-plane consisting of M_{V} horizontal rows with M_{H} antennas each. The antennas are uniformly spaced with horizontal and vertical spacing d_{H} and d_{V} (measured in multiples of the wavelength), respectively. The antennas are consecutively indexed row-by-row by $m \in [1, M]$, $M = M_{\mathrm{V}}M_{\mathrm{H}}$, so that the location of the mth antenna can be described as

$$\mathbf{u}_m = \begin{bmatrix} 0 \\ i(m)d_{\mathrm{H}}\lambda \\ j(m)d_{\mathrm{V}}\lambda \end{bmatrix} \tag{7.14}$$

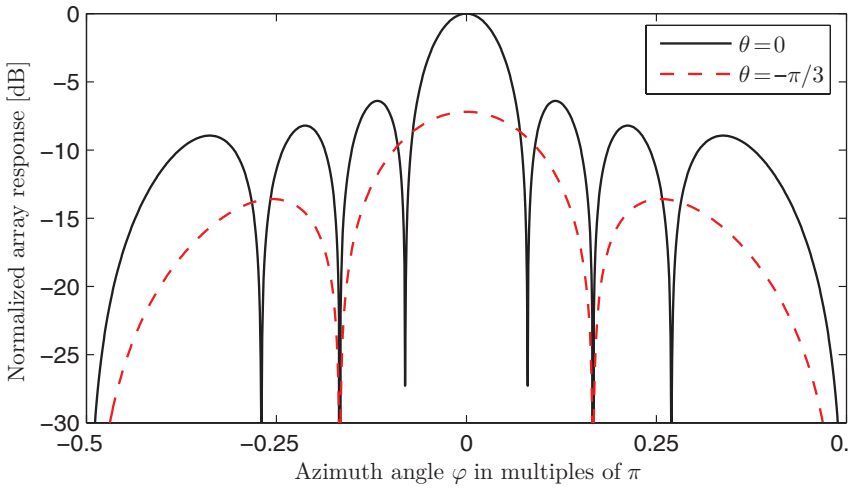


Figure 7.13: Normalized array response $|\frac{1}{M}\mathbf{a}(0,0)^{\mathrm{T}}\mathbf{a}(\varphi,\theta)|$ as a function of the azimuth angle φ. The array geometry is given by $M_{\mathrm{V}} = 4$, $M_{\mathrm{H}} = 8$ and $d_{\mathrm{V}} = d_{\mathrm{H}} = 0.5$.

where

$$i(m) = \mathrm{mod}(m-1, M_{\mathrm{H}}) \tag{7.15}$$

$$j(m) = \lfloor (m-1)/M_{\mathrm{H}} \rfloor \tag{7.16}$$

are the horizontal and vertical index of antenna m, respectively. Figure 7.13 shows the normalized array response $|\frac{1}{M}\mathbf{a}(0,0)^{\mathrm{T}}\mathbf{a}(\varphi,\theta)|$ of the planar array defined in (7.14) with $M_{\mathrm{V}} = 4$, $M_{\mathrm{H}} = 8$, and $d_{\mathrm{V}} = d_{\mathrm{H}} = 0.5$ as function of φ for different elevation angles θ. The figure shows how much an interfering signal arriving from azimuth angle φ is attenuated when receiving a signal from $\varphi = \theta = 0$. The resolution of the array is better when the main lobe is narrower because a smaller angular difference between the desired and interfering signals is sufficient to get a certain attenuation of the interfering signal. We observe that the horizontal resolution of the array is significantly reduced for lower elevation angles (e.g., $\theta = -\pi/3$).

7.3.2 3D Local Scattering Model with Arbitrary Array Geometry

The local scattering model was introduced in Section 2.2 on p. 82 for a horizontal ULA in a two-dimensional scenario. It provides an easy

way to compute the correlation matrix $\mathbf{R} \in \mathbb{C}^{M \times M}$ of a channel vector $\mathbf{h} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}_M, \mathbf{R})$ as a function of the distribution of the AoA of the incoming waves. With the definition of the array response in (7.13), we can now extend the local scattering model in (2.23) to three dimensions and arbitrary array geometries. If we redefine (2.20) as $\mathbf{a}_n = g_n \mathbf{a}(\varphi_n, \theta_n)$ and follow the same subsequent steps as in Section 2.2, we obtain a new expression for the elements of $\mathbf{R}$:

$$[\mathbf{R}]_{m,l} = \beta \iint e^{\mathrm{j}\mathbf{k}(\varphi,\theta)^{\mathrm{T}}(\mathbf{u}_m - \mathbf{u}_l)} f(\varphi, \theta) d\varphi d\theta \tag{7.17}$$

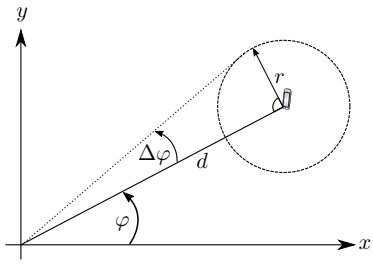
where $f(\varphi, \theta)$ is the joint PDF of the azimuth and elevation angle and the integration is over all angles. For the case of the planar array as shown in Figure 7.12, this expression simplifies to

$$[\mathbf{R}]_{m,l} = \beta \iint \underbrace{e^{\mathrm{j}2\pi d_{\mathrm{V}}[j(m)-j(l)]\sin(\theta)}}_{\text{Vertical correlation}} \underbrace{e^{\mathrm{j}2\pi d_{\mathrm{H}}[i(m)-i(l)]\cos(\theta)\sin(\varphi)}}_{\text{Horizontal correlation}} f(\varphi, \theta) d\varphi d\theta. \tag{7.18}$$

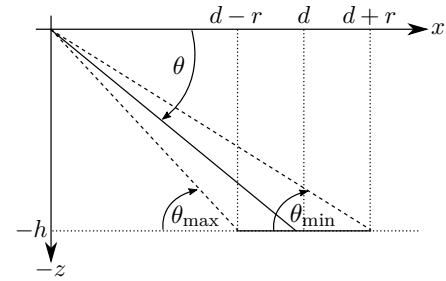
The terms $d_{\mathrm{V}}[j(m) - j(l)]$ and $d_{\mathrm{H}}[i(m) - i(l)]$ represent the vertical and horizontal distance between antennas m and l, respectively. Thus, the two complex exponential terms in (7.18) can be interpreted as the vertical and horizontal correlation of the array. Consider the set of antenna pairs (m, l) such that either $i(l) = i(m)$ or $j(l) = j(m)$; that is, the columns and rows of the antenna array. For these pairs, the correlation matrix has the following values:

$$[\mathbf{R}]_{m,l} = \begin{cases} \beta \int e^{\mathrm{j}2\pi d_{\mathrm{V}}[j(m)-j(l)]\sin(\theta)} f(\theta) d\theta & i(l) = i(m) \\ \beta \iint e^{\mathrm{j}2\pi d_{\mathrm{H}}[i(m)-i(l)]\cos(\theta)\sin(\varphi)} f(\varphi, \theta) d\varphi d\theta & j(l) = j(m). \end{cases} \tag{7.19}$$

Interestingly, while the correlation of the vertical columns coincides with the ULA in (2.23), this is not the case for the correlation of the horizontal rows. The larger the absolute value of the elevation angle θ, the smaller is the effective horizontal antenna spacing $d_{\mathrm{H}}(i(m) - i(l))\cos(\theta)$ of the array. Thus, the correlation between the antennas of the horizontal rows depends on the elevation angle. The same effect has already been observed in Figure 7.13 for a pure LoS scenario.



(a) A UE is located at horizontal distance d and azimuth angle φ from an antenna array in the origin. A ring of local scatterers of radius r around the UE creates a horizontal angular spread of $\Delta\varphi$.



(b) The same ring of local scatterers in the horizontal plane creates also a vertical angular spread of the incoming waves of $\Delta\theta = (\theta_{\max} - \theta_{\min})/2$. The mean elevation angle is $\theta = (\theta_{\max} + \theta_{\min})/2$.

Figure 7.14: Derivation of the angular spread of the 3D one-ring model.

The local scattering model for the planar array is entirely defined by the joint PDF of θ and φ. We will now provide a way to define this distribution for a somewhat realistic system, which takes into account the height at which the antenna array is installed and the distance between the UE and array [7]. As shown in Figure 7.14, we consider a UE being located at azimuth angle φ, horizontal distance d, and height $z = -h$ below a planar antenna array. We assume that the UE is surrounded by a horizontal ring of local scatterers of radius r.[4] As shown in Figure 7.14a, the radius r defines the horizontal *angular spread* $\Delta\varphi = \tan^{-1}(r/d)$. The same scatterers give also rise to a vertical angular spread which can be computed as follows. As can be seen from Figure 7.14b, the maximum elevation angle $\theta_{\max} = \tan^{-1}(h/(d-r))$ is achieved by a scatterer at distance $d - r$. Similarly, the minimum elevation angle $\theta_{\min} = \tan^{-1}(h/(d+r))$ is achieved by a scatterer at distance $d + r$. Assuming that the elevation angles are uniformly distributed in the interval $[\theta_{\min}, \theta_{\max}]$, the mean elevation angle is computed as $\theta = (\theta_{\max} + \theta_{\min})/2$ and the vertical spread is $\Delta\theta =$

[4]The scatterer ring is only a means to define reasonable values of the angular spread, but has no other physical meaning.

$(\theta_{\max} - \theta_{\min})/2$.[5] Thus, the spatial correlation matrix is finally given as

$$[\mathbf{R}]_{m,l} = \frac{\beta}{4\Delta\varphi\Delta\theta} \int_{\theta-\Delta\theta}^{\theta+\Delta\theta} \int_{\varphi-\Delta\varphi}^{\varphi+\Delta\varphi} e^{\mathrm{j}\mathbf{k}(\varphi,\theta)^{\mathrm{T}}(\mathbf{u}_m - \mathbf{u}_l)} d\varphi d\theta. \tag{7.20}$$

Note that the angular spread with this model is rather small. For example, for a BS at a height of $h = 25\,\mathrm{m}$, a UE located at a distance of $d = 200\,\mathrm{m}$ with a scattering radius of $r = 50\,\mathrm{m}$ sees a horizontal spread of $\Delta\varphi = 14^\circ$ and a vertical spread of less than $\Delta\theta = 2^\circ$. To avoid angular distributions with small finite support, we can alternatively use the Gaussian [4, 373, 313, 363] or Laplacian distribution [225, Section 7.4.2], [161] rather than the uniform distribution in (7.20) (cf. Section 2.6 on p. 82). Recall that Gaussian distributions were used in the running example of previous sections. The mean angles and angular spreads derived above can be used as values for the ASD and mean of these distributions. In the remainder of this monograph, we will sometimes refer to the model in (7.20) as the 3D one-ring model or simply the one-ring model.

Dominant Eigenspace and Chordal Distance

As anticipated in Section 2.6 on p. 82 for the local scattering model, correlation matrices generally have a mix of many weak and a few strong eigendirections. To further exemplify this, Figure 7.15 shows the eigenvalues in decreasing order when using the 3D one-ring model in (7.20) with an 8×8 antenna array and different scatter radii. We can clearly see that the smaller the scatter radius, the more energy is packed into a few strong eigendirections of the channel; 99% of the energy is contained in 5 out of the 64 eigenvalues when $r = 50\,\mathrm{m}$ and 20 eigenvalues when $r = 200\,\mathrm{m}$. Strictly speaking, the correlation matrix might still have full rank, but we can define the following notion of p-dominant eigenspace to capture the eigenspace that contains most of the energy.

Definition 7.1 (p-Dominant eigenspace)**.** Let $\mathbf{X} \in \mathbb{C}^{M \times M}$ be a Hermitian matrix with eigenvalue decomposition $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathrm{H}}$, where $\mathbf{U} =$

[5] This is a crude approximation. A ring of uniformly distributed scatterers would not lead to uniformly distributed azimuth and elevation angles.

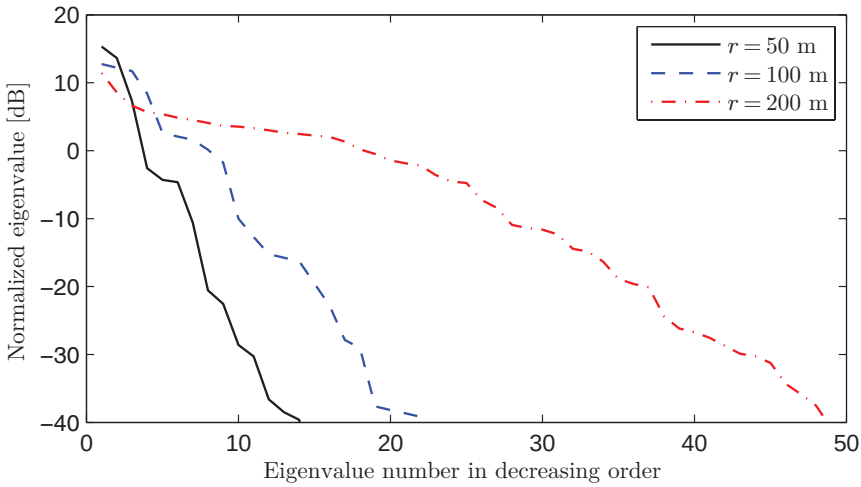


Figure 7.15: Eigenvalues of the spatial correlation matrix $\mathbf{R}$ when using the 3D one-ring model in (7.20) for an 8×8 antenna array and different scatter-radii r. The other model parameters are $d = 200\,\mathrm{m}$, $\varphi = 23.5^\circ$ and $h = 23.5\,\mathrm{m}$.

$[\mathbf{u}_1 \dots \mathbf{u}_M] \in \mathbb{C}^{M\times M}$ is unitary and $\mathbf{D} = \mathrm{diag}(\lambda_1, \dots, \lambda_M) \in \mathbb{R}^{M\times M}$ consists of the non-negative eigenvalues of $\mathbf{X}$ in non-increasing order (i.e., $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_M \geq 0$). The *p-dominant eigenspace* $\mathrm{eig}_p(\mathbf{X}) \in \mathbb{C}^{M\times p}$ is defined as $\mathrm{eig}_p(\mathbf{X}) = [\mathbf{u}_1 \dots \mathbf{u}_p]$.

The p-dominant eigenspace of a Hermitian matrix is the (tall) unitary matrix composed of the p eigenvectors belonging to its p largest eigenvalues. In order to define a metric for measuring orthogonality between two eigenspaces, we consider the chordal distance (see, e.g., [180, 235]) which is defined for arbitrary matrices as follows.[6]

Definition 7.2 (Chordal distance)**.** The *chordal distance* $\mathrm{d_C}(\mathbf{X}, \mathbf{Y})$ between two matrices $\mathbf{X}$ and $\mathbf{Y}$ is defined as

$$\mathrm{d_C}(\mathbf{X}, \mathbf{Y}) = \|\mathbf{X}\mathbf{X}^{\mathrm{H}} - \mathbf{Y}\mathbf{Y}^{\mathrm{H}}\|_F^2. \tag{7.21}$$

For two (tall) unitary matrices $\mathbf{U}_1, \mathbf{U}_2 \in \mathbb{C}^{M\times p}$, the chordal distance

[6] Other distance or orthogonality metrics between correlation matrices can be also defined. We refer the interested reader to [88, Sec. 3.1.1].

takes the form

$$\begin{aligned} \mathrm{d_C}(\mathbf{U}_1, \mathbf{U}_2) &= \|\mathbf{U}_1\mathbf{U}_1^{\mathrm{H}} - \mathbf{U}_2\mathbf{U}_2^{\mathrm{H}}\|_F^2 \\ &= \mathrm{tr}\left((\mathbf{U}_1\mathbf{U}_1^{\mathrm{H}} - \mathbf{U}_2\mathbf{U}_2^{\mathrm{H}})\,(\mathbf{U}_1\mathbf{U}_1^{\mathrm{H}} - \mathbf{U}_2\mathbf{U}_2^{\mathrm{H}})^{\mathrm{H}}\right) \\ &= \mathrm{tr}\left(\mathbf{U}_1\mathbf{U}_1^{\mathrm{H}} + \mathbf{U}_2\mathbf{U}_2^{\mathrm{H}} - 2\mathbf{U}_1\mathbf{U}_1^{\mathrm{H}}\mathbf{U}_2\mathbf{U}_2^{\mathrm{H}}\right) \\ &= 2p - 2\sum_{i=1}^{p}\sum_{j=1}^{p}|\mathbf{u}_{1i}^{\mathrm{H}}\mathbf{u}_{2j}|^2 \end{aligned} \tag{7.22}$$

where $\mathbf{u}_{ki}$ denotes the ith column vector of matrix $\mathbf{U}_k$ for $k = 1, 2$. The chordal distance can be interpreted as the number of dimensions of the subspace that can be reached by a linear combination of the column vectors of only one of the two matrices. For example, if $\mathbf{U}_1 = \mathbf{U}_2$, we have $\mathrm{d_C}(\mathbf{U}_1, \mathbf{U}_2) = 0$. Although each matrix individually spans p dimensions, all of them can be reached through a linear combination of the column vectors of $\mathbf{U}_1$ or $\mathbf{U}_2$. On the other hand, for $\mathbf{U}_1^{\mathrm{H}}\mathbf{U}_2 = \mathbf{0}_{p\times p}$, we have $\mathrm{d_C}(\mathbf{U}_1, \mathbf{U}_2) = 2p$ because each matrix spans a p-dimensional space which cannot be reached through a linear combination of the column vectors of the other matrix.

Figure 7.16 shows the chordal distance for different values of M between the 6-dominant eigenspaces of the correlation matrices of a UE at azimuth 0° and distance 200 m and that of a UE at the same distance but at azimuth angle φ. The 3D one-ring model in (7.20) is used with scatter radius $r = 50$ m. We can see that the chordal distance increases with M, due to the increased spatial resolution of the array, and with φ since the spatial correlation matrices become increasingly different.

7.3.3 The 3GPP 3D MIMO Channel Model

The previously presented LoS and local scattering channel models are mathematically convenient and very easy to simulate while still capturing essential characteristics of the wireless channel between BSs equipped with large antenna arrays and single-antenna UEs. In particular, we have seen the dependence of the channel (correlation) matrix on the antenna array geometry as well as on the physical location and orientation of the BS and UEs. Both channel models are *spatially consistent*; that is, the channel statistics for a given location are always the

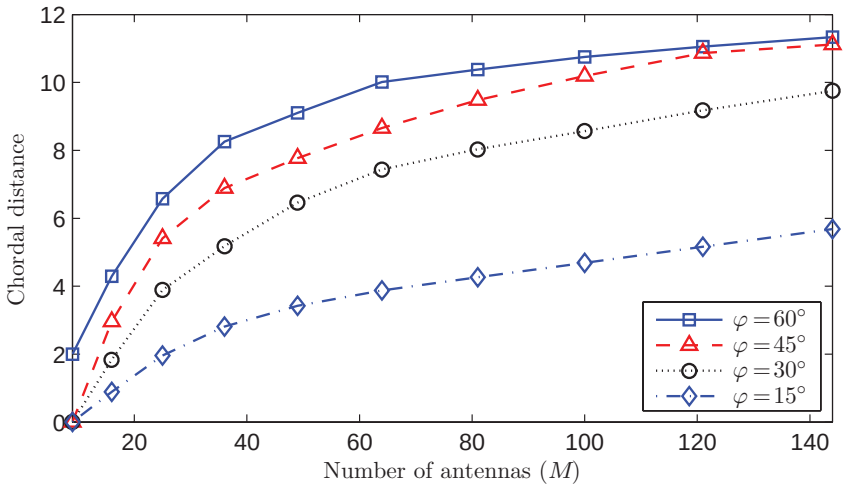


Figure 7.16: Chordal distance between the 6-dominating eigenspaces of the correlation matrices of a UE at azimuth 0° and distance $d = 200\,\text{m}$ and of a UE at the same distance but at azimuth angle φ. The antenna array is assumed to be quadratic with $d_\text{H} = d_\text{V} = 0.5$ and installed at height $h = 23.5\,\text{m}$. The correlation matrices are generated by the 3D one-ring model in (7.20) with scatter radius $r = 50\,\text{m}$.

same and do not depend on the simulation run. However, these models suffer from several shortcomings, which render them inappropriate for large-scale system-level simulations with the goal of quantifying the real-world performance of MIMO systems. In particular, it is neither realistic to assume pure LoS conditions nor to consider a single local cluster of scatterers around the UE. For this reason, the 3GPP has defined a 3D stochastic geometry-based channel model for MIMO systems [1], which explicitly accounts for multiple spatial clusters of scatterers and a mix of NLoS and LoS propagation paths. This so-called 3GPP 3D MIMO channel model is an extension of previously standardized channel models, such as Winner II [347], which assumes that all scatterers, reflectors, UEs and BSs are located in a two-dimensional plane. This extension was required as it was impossible with such models to simulate MIMO systems exploiting the elevation dimension.

The 3GPP 3D MIMO channel model is characterized by a deterministic system layout (i.e., BS and UE locations, antenna orientations, field

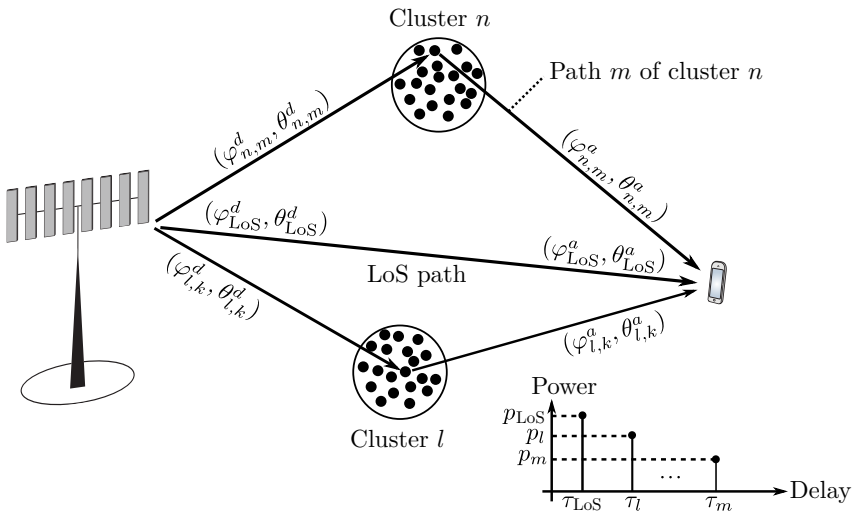


Figure 7.17: The 3GPP 3D MIMO channel model [1] is a stochastic geometry-based model. For each BS-UE link, C scattering clusters with random angles of arrival/departure are generated. Each cluster is associated with a time delay τ_l and a power p_l, for $l = 1, \ldots, C$. The distributions of angles, delays, and powers depend on the chosen scenario. Each cluster generates 20 multipath components, which are assumed to be resolvable in the angular domain, but not in the time domain.

patterns[7], and carrier frequency) and a set of random parameters (i.e., delay spread, delay values, angular spread, shadow fading, and cluster powers) which are generated from statistical distributions obtained from extensive channel measurements. The model is valid for carrier frequencies in the range 2–6 GHz and bandwidths of up to 100 MHz.

The modeling approach is as follows. For each pair of UE and BS, C scattering clusters with random AoAs, AoDs, delays, and power levels are generated. With a scenario-dependent[8] probability, also a LoS path is present. This is schematically shown in Figure 7.17. The value of C depends on the scenario and is in the range of 12–20. Each cluster is assumed to have 20 resolvable multipath components in the angular

[7]The field or radiation pattern of an antenna describes the angular dependence of the strength of the transmitted or received radio waves.

[8]The 3GPP 3D MIMO channel model can be used to simulate different propagation scenarios (e.g., coverage tier, hotspot tier, LoS, NLoS, indoor, outdoor, outdoor-to-indoor, urban, and suburban), which are detailed in [1].

domain with random angular offsets from the cluster AoA/AoD. The angular spread of the multipath components of each cluster is rather low, within 1°–6°, while the angular spread of the clusters themselves is large, within 20°–90°. The multipath components of each cluster are assumed to be unresolvable in the time domain (i.e., they all arrive with the same cluster delay). The 3GPP 3D MIMO channel model considers a multitude of different scenarios with different BS and UE heights and arbitrary antenna characteristics. In particular, high-rise buildings with UEs on multiple floors can be simulated to gauge the gains of elevation precoding. Dual-polarized antennas (see Section 7.4.4), as well as time-evolution of the channel due to movement of UEs, can be also considered. An important aspect of this channel model is that it is not spatially consistent; that is, the location of scattering clusters are random and uncorrelated between different UEs. In other words, two UEs at almost the same location do not share any scatterers. Thus, it cannot be used for simulations requiring spatial consistency. In this case, one must resort to ray tracing, recorded channel measurements, or simpler channel models as presented earlier. There are also scatterer-centric channel models, such as COST2100 (e.g., [88, Sec. 4.4.5]), where a global set of scatterers is shared by all UEs. Although this type of channel model is spatially consistent, there is currently no widely accepted model to be used with large antenna arrays (see [123, 160]). Two open-source implementations of the 3GPP 3D MIMO model are currently available: the standalone QuaDRiGa channel model [159] developed by the Fraunhofer Heinrich Hertz Institute and the LTE Advanced system simulator from the Vienna University of Technology [5]. The QuaDRiGa model will be used for the case study in Section 7.7.

7.3.4 Observations from Channel Measurements

Several channel measurement campaigns with large antenna arrays have been conducted during the last years. Among the first were [120, 150] that provided the first verifications of favorable propagation (see Definition 2.5 on p. 80) in practice.[9] In [120], a cylindrical array of

[9]The authors of [300] built the first 64-antenna Massive MIMO prototype and confirmed the expected tremendous performance. However, their publication does

128 antennas was used for indoor-to-outdoor measurements, while the authors of [150] conducted outdoor measurements with a virtual planar array of 112 antennas. Both papers quantify the level of favorable propagation (see Section 2.5.2 on p. 80) by computing the correlation metric

$$\delta_{\text{corr}} = \mathbb{E}\left\{ \frac{|\mathbf{h}_1^{\text{H}}\mathbf{h}_2|^2}{\|\mathbf{h}_1\|^2\|\mathbf{h}_2\|^2} \right\} \tag{7.23}$$

between the channels of two randomly chosen measurement locations. This metric is similar to (2.19) and essentially describes the variance of the inner products of the normalized channels. The two papers conclude that close to ideal results (i.e., i.i.d. Rayleigh fading channels for which the average correlation can be shown to be $\delta_{\text{corr}} = 1/M$) can be achieved in practice. However, the larger the number of antennas, the higher was the observed gap to the ideal case, giving rise to the conclusion that some saturation effect appears and that the marginal gain of additional antennas rapidly diminishes. Similar measurements were conducted in [124] and different antenna architectures (horizontal, planar, vertical) were compared. Figure 7.18 shows the average correlation over randomly picked UE locations within a cell sector. The measurement results are reproduced from [124][10] and simulation results are based on the 3GPP 3D MIMO channel model as implemented in the QuaDRiGa software [159]. We can clearly see that a horizontal array provides a decorrelation similar to i.i.d. channels, which implies that the UEs' channels will be nearly orthogonal. In contrast, the planar and vertical arrays exhibit higher δ_{corr} due to the lower vertical angular spread between UEs; that is, it is easier to separate UEs in the azimuth domain than in the elevation domain. If we would instead consider a scenario with UEs located at different floors of high-rise buildings, the results would be more favorable towards planar (and vertical) arrays as the elevation angular spreads are then higher. The differences between the results for simulated and measured channels can be partially attributed to a reduced number of measurement locations and the lack of spatial consistency of the 3GPP 3D MIMO channel model.

not report specifically on channel measurements.

[10] We would like to thank Marc Gauger for recomputing some of the results.

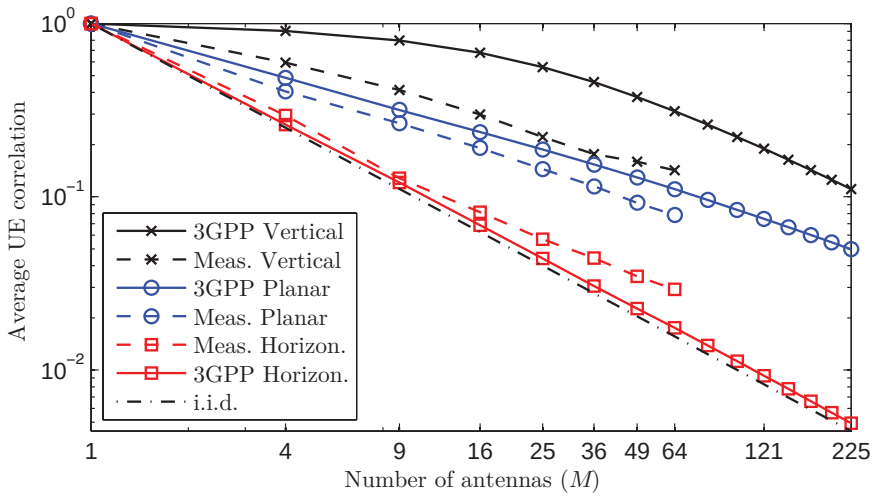


Figure 7.18: Average UE correlation δ_{corr} in (7.23) between the channel vectors of two UEs at randomly chosen locations within a cell sector as a function of the number of antennas. Horizontal, vertical and quadratic planar antenna geometries are considered. The measurement results are reproduced from [124]. The simulation results are obtained for the urban macro LoS scenario of the 3GPP 3D MIMO channel model.

Several papers investigate the resolvability of closely spaced UEs under LoS conditions. Even though we will see in the next section, from simple analytical models, that low angular separation is detrimental for Massive MIMO, the results of [207, 115] indicate that arrays with a very large aperture can resolve UEs even in close proximity. This has been confirmed for indoor [207] as well as outdoor [115] channels.

The subsequent works [121, 122] reported on refined outdoor channel measurements using a 7.4 m long virtual ULA as well as a more compact cylindrical array with 128 antennas. These works demonstrate that SEs similar to those predicted by closed-form expressions obtained for i.i.d. Rayleigh fading channels can be achieved on measured channels, although the channels have very different characteristics. Moreover, the authors make the interesting observation that substantial variations of the received power over the array exist. In other words, antenna arrays with a large aperture can experience antenna-dependent shadow fading. We have shown in Section 4.4.1 on p. 184 that this phenomenon leads

to linearly independent correlation matrices, which is a key property in the asymptotic performance analysis. The antenna-dependent shadow fading was observed for the ULA as well as for the compact cylindrical array. However, the explanation for this phenomenon is different in these cases. Different parts of a ULA "see" different parts of the propagation environment such that obstacles might only block certain parts of the array. A similar effect arises in distributed antenna systems, where most antennas have independent propagation paths to a UE. On the other hand, the antennas of a cylindrical array point in different directions, where different scattering clusters or obstacles are relevant. The authors of [122] also observed that shadowing over the array can give rise to the effect that some antennas contribute more to the overall channel than others. Thus, some antennas might be dynamically turned on/off to save energy and/or processing power. This phenomenon is further discussed in the following example.

Large-Scale Fading Variations over the Antenna Array

Consider the scenario in Figure 7.19 where two obstacles block two different parts of the antenna array in two different cell regions A and B. In region A, the first M' antennas of the array exhibit strong shadowing with attenuation $\beta \in [0, 1]$, while the remaining $M - M'$ antennas have an unobstructed path to the UEs with attenuation $(M-\beta M')/(M-M')$. This value is chosen such that the average per-antenna energy of the channel is independent of β and equal to one. In region B, the situation is the opposite, which means that the last M' antennas are obstructed. We neglect other types of spatial channel correlation in this example to focus exclusively on the effects of antenna-dependent shadowing. The channel vectors of UEs in region A and B are distributed as $\mathbf{h}_A \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}_M, \mathbf{R}_A)$ and $\mathbf{h}_B \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}_M, \mathbf{R}_B)$, respectively, where[11]

$$\mathbf{R}_A = \begin{bmatrix} \beta \mathbf{I}_{M'} & \mathbf{0} \\ \mathbf{0} & \frac{M-\beta M'}{M-M'} \mathbf{I}_{M-M'} \end{bmatrix} \tag{7.24}$$

$$\mathbf{R}_B = \begin{bmatrix} \frac{M-\beta M'}{M-M'} \mathbf{I}_{M-M'} & \mathbf{0} \\ \mathbf{0} & \beta \mathbf{I}_{M'} \end{bmatrix}. \tag{7.25}$$

[11] For brevity, we did not write out the dimensions of the all-zero matrices.

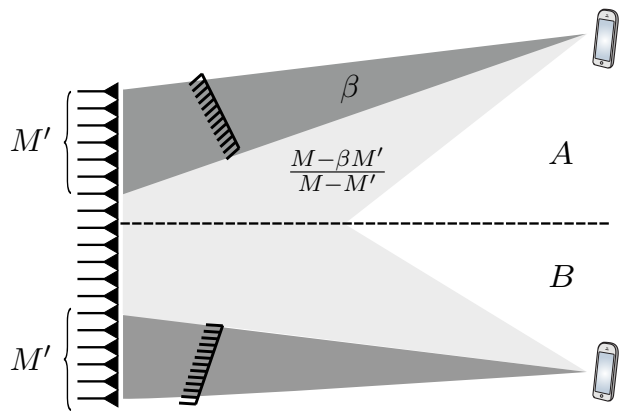


Figure 7.19: Obstacles which only partially block propagation paths towards a UE can lead to large-scale fading variations over the antenna array.

The effect of shadowing in this example is as follows. Two UEs located in the same region see a channel with an effectively reduced number of antennas. For instance, with $\beta = 0$, only $M - M'$ antennas are visible. Figure 7.20 illustrates the effect with $M = 64$, $M' = 30$, and different values of β. As can be seen from Figure 7.20, shadowing increases the average correlation δ_{corr} as defined in (7.23) between the UEs' channels in the same region case. On the other hand, for two UEs in different regions, shadowing decreases the correlation since two parts of the antenna array receive more energy from one UE than from the other. Thus, antenna-dependent shadowing can have a positive or negative effect depending on which UEs are simultaneously scheduled in a cell.

One can also consider shadowing over the antenna array as a particular form of spatial channel correlation which could be exploited in a manner similar to the previously discussed JSDM in Remark 7.3 on p. 319. In this case, the dominating eigenspace of a group would be simply the subset of antennas of the array over which most energy is received. In the example in Figure 7.19, the UEs in region A and B would be respectively served by the lower and upper part of the antenna array so that the present obstacles would naturally reduce the interference between the groups.

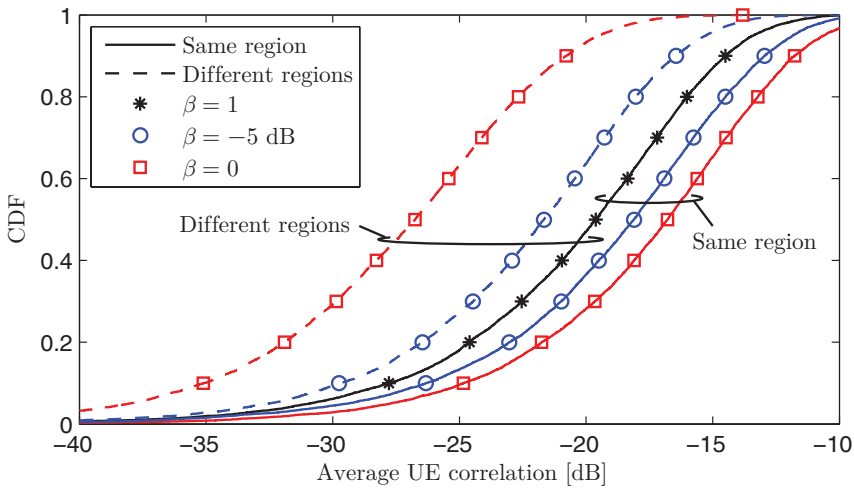


Figure 7.20: Average UE correlation δ_{corr} in (7.23) of the channels of two UEs in either the same $(A - A)$ or different $(A - B)$ regions of Figure 7.19. Depending on the scenario, antenna-dependent shadowing can either increase or decrease the correlation.

7.4 Array Deployment

Until now, we have characterized antenna arrays only indirectly through their spatial channel correlation. We will now shift focus to the arrangement of the individual antennas; that is, the array geometry. In this section, we will explore the effects of different antenna array geometries and have an in-depth look at the effects of antenna spacing and polarization. For an introduction to the design of BS antennas and antenna arrays for cellular communications, we refer to [79].

The most important factors of an antenna array are the antenna spacing and its total size (relative to the wavelength), which is known as the aperture. The size determines the array's directivity; that is, its ability to focus the radiated energy towards certain directions, while the number of antennas determines the radiated/received energy. Some of these aspects will be discussed in more detail in Section 7.4.2. Each individual antenna consists of one or more radiating elements[12] that have

[12] A radiating element can of course also receive energy.

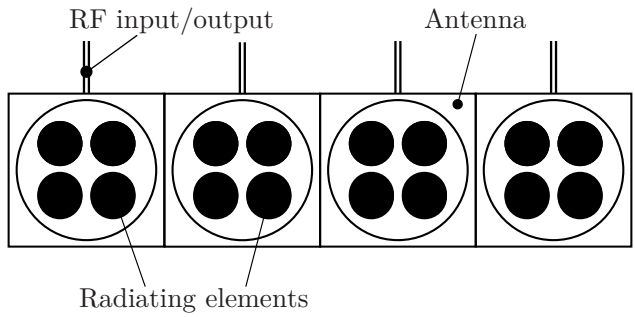


Figure 7.21: Antenna array consisting of four antennas each of which is composed of four radiating elements. The radiating elements of an antenna share the same RF input and output.

a fixed size that depends on the wavelength λ of the carrier frequency. For example, a half-wave dipole, is in essence, a piece of wire of length $\lambda/2$. For a carrier frequency of 2.6 GHz, such a dipole has a length of 5.8 cm. Since the size of a dipole cannot simply be made larger, multiple dipoles or other radiating elements need to be connected together if the captured energy of an antenna shall be increased. The following definition, which is also visualized in Figure 7.21, makes the relation between the terms radiating element, antenna, and antenna array clear.

Definition 7.3 (Radiating element, antenna, antenna array)**.** An *antenna* consists of one or more *radiating elements* (e.g., dipoles) which are fed by the same RF signal. An *antenna array* is composed of multiple antennas with individual RF chains.

In legacy mobile communication systems (e.g., GSM, UMTS), a BS with a single antenna and single RF chain needs to provide coverage for an entire cell sector with a horizontal width of 120° and a radius of up to several kilometers. At the same time, it must not radiate energy to neighboring cell sectors. Such BS antennas have typically an azimuth beamwidth of 65° and an elevation beamwidth of 3°–15° [79, Sec. 2.2].[13] The resulting coverage gap between the 65° sectors is generally filled by

[13]The *beamwidth* is defined as the angular distance between the half-power points of the main lobe of the antenna's radiation pattern. It is also referred to as the half-power beamwidth or 3-dB beamwidth. The beamwidth is generally different in the azimuth and elevation directions.

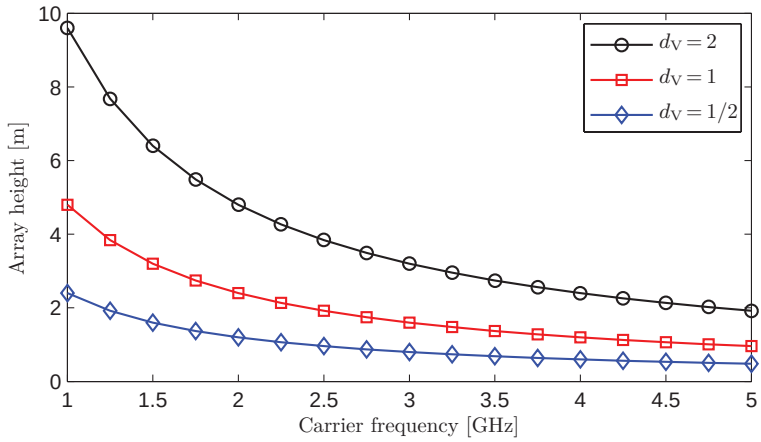


Figure 7.22: Height of a planar antenna array as a function of the carrier frequency for $M_V = 16$ antennas per vertical column and different vertical antenna spacings $d_V = [\frac{1}{2}, 1, 2]$.

neighboring BSs. As we will see later, to achieve such a focused radiation of energy in the elevation domain, the antenna must be composed of multiple vertically stacked radiating elements. Common antenna heights are 8–16 wavelengths [79, Sec. 2.2.1.3]. With a typical element spacing of $0.8\,\lambda$, a single BS antenna consists of 10–20 radiating elements. This number is doubled since dual-polarized co-located antennas are the norm for cellular communications (see Section 7.4.4). Thus, a legacy BS covering three cell sectors is already equipped with 60–120 radiating elements but has only three RF chains, one per sector. In contrast, a Massive MIMO system has a similar number of radiating elements but an individual RF chain for each of them. The practical challenge of Massive MIMO is therefore not to accommodate a large number of radiating elements but to process their individual RF signals.

7.4.1 Physical Array Size and Antenna Spacing

Consider the planar antenna array shown in Figure 7.12. The horizontal width and vertical height of this array are respectively given by $M_H d_H \lambda$ and $M_V d_V \lambda$, where d_H and d_V are the horizontal and vertical antenna spacing (in multiples of the wavelength). In Figure 7.22, the array height

is exemplified as a function of the carrier frequency for $M_{\mathrm{V}} = 16$ and different values of d_{V}. Clearly, the higher the carrier frequency, the smaller is the array form factor. This makes arrays with a large number of antennas especially attractive at mmWave frequencies. This aspect will be discussed in more detail in Section 7.5. As Figure 7.22 shows, antenna arrays can easily become bulky for frequencies below 5 GHz. For example, a 16×16 planar array with $\lambda/2$ spacing at a carrier frequency of 2.5 GHz has the physical dimensions of $1\,\mathrm{m} \times 1\,\mathrm{m}$. Thus, in order to optimally design large antenna arrays, it is important to understand the role of the antenna spacing as well as the array height and width.

7.4.2 Physical Array Size and Antenna Spacing

To get some simple insight into the array design problem, we will revisit the analysis of the two-UE LoS uplink channel of (1.25) in a 3D setting with a planar array. To this end, we focus on the LoS channel model in (7.13) and assume that the two UEs are respectively located at the azimuth and elevation angles (φ_1, θ_1) and (φ_2, θ_2). Our goal is to study how the correlation of the array responses $|\frac{1}{M}\mathbf{a}(\varphi_2, \theta_2)^{\mathrm{H}}\mathbf{a}(\varphi_1, \theta_1)|$ behaves as a function of the array geometry and the angular separation of the UEs.[14] For ease of notation, we introduce the quantities

$$\Omega = \sin(\theta_1) - \sin(\theta_2) \tag{7.26}$$

$$\Psi = \cos(\theta_1)\sin(\varphi_1) - \cos(\theta_2)\sin(\varphi_2). \tag{7.27}$$

Since the azimuth and elevation angles are confined within the interval $[-\pi/2, \pi/2]$, it follows that $\Omega, \Psi \in [-2, 2]$. The distributions of these quantities are in general not uniform as they depend on the distribution of the UEs in the cell sector and on the height at which the antenna array is installed. This has an impact on the ranges of values of Ω and Ψ that are of practical importance. Using the definition of $\mathbf{a}(\varphi, \theta)$ in

[14] A detailed discussion for the case of a ULA can be found in [314, Sec. 7.2.4].

(7.13) and (7.14), it is easy to show that

$$\begin{aligned}
&\left|\frac{1}{M}\mathbf{a}(\varphi_2,\theta_2)^{\mathrm{H}}\mathbf{a}(\varphi_1,\theta_1)\right| \\
&= \left|\frac{1}{M}\sum_{m=1}^{M} e^{\mathrm{j}2\pi(d_{\mathrm{V}}j(m)\Omega+d_{\mathrm{H}}i(m)\Psi)}\right| \\
&= \underbrace{\left|\frac{1}{M_{\mathrm{V}}}\sum_{k=0}^{M_V-1} e^{\mathrm{j}2\pi d_{\mathrm{V}}k\Omega}\right|}_{\triangleq S(\Omega)} \underbrace{\left|\frac{1}{M_{\mathrm{H}}}\sum_{l=0}^{M_H-1} e^{\mathrm{j}2\pi d_{\mathrm{H}}l\Psi}\right|}_{\triangleq T(\Psi)}.
\end{aligned} \tag{7.28}$$

Thus, the correlation can be represented as the product of two functions $S(\Omega)$ and $T(\Psi)$. Note that $S(\Omega)$ only depends on the angular separation of the UEs in the elevation domain while $T(\Psi)$ depends on all azimuth and elevation angles. As in the proof of Lemma 1.5 in Appendix C.1.4 on p. 430, we can rely on the identity $\sum_{n=0}^{N-1} q^n = (1-q^N)/(1-q)$ for $q \neq 1$, to simplify $S(\Omega)$ as follows:

$$\begin{aligned}
S(\Omega) &= \left|\frac{1}{M_{\mathrm{V}}}\frac{1-e^{\mathrm{j}2\pi d_{\mathrm{V}}M_{\mathrm{V}}\Omega}}{1-e^{\mathrm{j}2\pi d_{\mathrm{V}}\Omega}}\right| \\
&= \left|\frac{e^{\mathrm{j}\pi d_{\mathrm{V}}(M_{\mathrm{V}}-1)\Omega}}{M_{\mathrm{V}}}\frac{e^{-\mathrm{j}\pi d_{\mathrm{V}}M_{\mathrm{V}}\Omega}-e^{\mathrm{j}\pi d_{\mathrm{V}}M_{\mathrm{V}}\Omega}}{e^{-\mathrm{j}\pi d_{\mathrm{V}}\Omega}-e^{\mathrm{j}\pi d_{\mathrm{V}}\Omega}}\right| \\
&= \left|\frac{\sin(\pi L_{\mathrm{V}}\Omega)}{M_{\mathrm{V}}\sin(\pi d_{\mathrm{V}}\Omega)}\right|
\end{aligned} \tag{7.29}$$

where $L_{\mathrm{V}} = M_{\mathrm{V}}d_{\mathrm{V}}$ is the normalized height of the array. It is easy to verify that $S(\Omega) = \sqrt{g(\theta_1,\theta_2)/M_{\mathrm{V}}}$, where $g(\theta_1,\theta_2)$ was defined in Lemma 1.5 on p. 31 (using $d_{\mathrm{H}} = d_{\mathrm{V}}$ and $M = M_{\mathrm{V}}$). Following the same steps and defining the normalized array width $L_{\mathrm{H}} = M_{\mathrm{H}}d_{\mathrm{H}}$, we can express $T(\Psi)$ as

$$T(\Psi) = \left|\frac{\sin(\pi L_{\mathrm{H}}\Psi)}{M_{\mathrm{H}}\sin(\pi d_{\mathrm{H}}\Psi)}\right|. \tag{7.30}$$

We can make the following observations from (7.29) and (7.30):

- $S(\Omega)$ and $T(\Psi)$ are periodic with period $\frac{1}{d_{\mathrm{V}}}$ and $\frac{1}{d_{\mathrm{H}}}$, respectively;
- $S\left(\frac{k}{L_{\mathrm{V}}}\right) = 0$ for $k = 1,\ldots,M_{\mathrm{V}}-1$, and $T\left(\frac{k}{L_{\mathrm{H}}}\right) = 0$ for $k = 1,\ldots,M_{\mathrm{H}}-1$;

- Both functions peak at $\Omega = \Psi = 0$ with $S(0) = T(0) = 1$.[15]

The function $S(\Omega)$ is shown for different values of M_{V} and d_{V} in Figure 7.23. The behavior of $T(\Psi)$ is identical. Both functions have main-lobes around the origin with a width of $\frac{2}{L_{\mathrm{V}}}$ and $\frac{2}{L_{\mathrm{H}}}$, respectively.[16] The maximum values of all other lobes are much smaller. This implies that the array is not able to separate two UEs whenever for some integers k and l we have

$$\left|\Omega - \frac{k}{d_{\mathrm{V}}}\right| \ll \frac{1}{L_{\mathrm{V}}} \text{ and } \left|\Psi - \frac{l}{d_{\mathrm{H}}}\right| \ll \frac{1}{L_{\mathrm{H}}}. \tag{7.31}$$

These conditions simply say that whenever the signals of two UEs arrive from similar azimuth and elevation angles they will interfere strongly with each other. To get a feeling of what "$\ll$" means in (7.31), we can similarly to (1.31) rely on the approximation $\sin(\pi z) \approx \pi z$ for $|z| < 0.2$ to show that $S(\Omega) \approx 1$ whenever $|\Omega - k/d_{\mathrm{V}}| < 0.2/L_{\mathrm{V}}$ (the same holds analogously for $T(\Psi)$). The quantities k, l in the conditions above arise because $S(\Omega)$ and $T(\Psi)$ are periodic and, depending on the antenna spacing, more than one period fits into the interval $[-2, 2]$ on which both functions are defined. The conditions in (7.31) also reveal that the angular resolution of the array in the elevation and azimuth dimension is respectively defined through its normalized height L_{V} and width L_{H}. Note that this resolution is not affected by a change of the number of antennas as long as they are distributed over the same area. For example, doubling the number of antennas while halving their distances does not improve the spatial resolution of the antenna array (although the array gain is increased). Similarly, the width of the main-lobes depends solely on the physical size of the antenna array but not on the number of antennas. For this reason, we require physically large antenna arrays to achieve a high directivity (i.e., narrow main-lobes).

Remark 7.5 (Relation to time-sampling of a band-limited signal)**.** There is an interesting analogy between the time-sampling of a band-limited

[15] This can be most easily seen from the definition of $S(\Omega)$ and $T(\Psi)$ in (7.28).

[16] Since both functions are periodic, they also have such lobes around multiples of $\frac{1}{d_{\mathrm{V}}}$ and $\frac{1}{d_{\mathrm{H}}}$, respectively. These lobes are also called main-lobes, while the smaller lobes in between are called side-lobes.

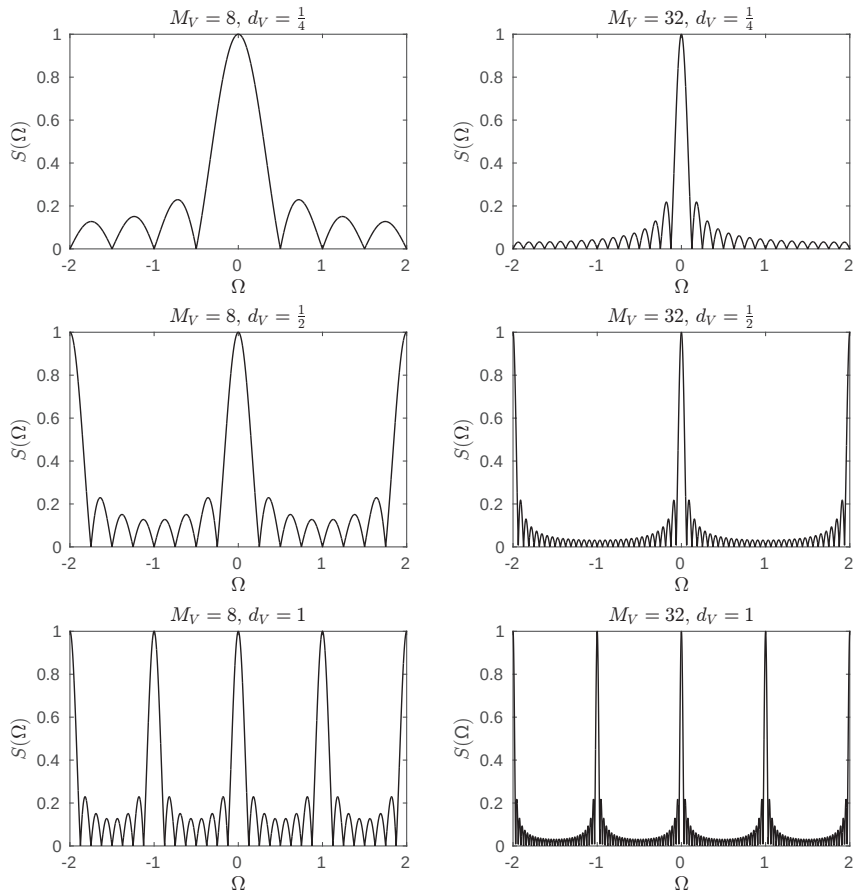


Figure 7.23: The function $S(\Omega)$ in (7.29) for different values of M_V and d_V.

signal and the spatial-sampling of an EM field through an antenna array [268] (see also [314, Sec. 7.3.3]). A passband signal of bandwidth B allows the receiver to resolve two multipath components only if their arrival times are separated by more than $\frac{1}{B}$. Similar to the observation that increasing the number of antennas for a fixed array size does not improve the spatial resolution, increasing the sampling rate does not help to distinguish more multipath components (or to increase the time-resolution). Another analogy can be made with respect to OFDM. A symbol time of T_s creates subcarriers with frequency spacing $1/T_s$. Thus, the frequency resolution is defined by the symbol length and not

by the number of samples per symbol. Increasing the sampling rate will increase the bandwidth but not the frequency resolution.

Even though the beamwidth and the angular resolution are independent of the antenna spacing (or, equivalently, of the number of antennas), this parameter has a very important influence on the characteristics of the antenna array. We have noted earlier that $S(\Omega)$ and $T(\Psi)$ are defined on the interval $[-2, 2]$, that they are periodic with period $\frac{1}{d_\mathrm{V}}$ and $\frac{1}{d_\mathrm{H}}$, respectively, and that they have a main-lobe around the origin. Thus, whenever $d_\mathrm{V} \geq \frac{1}{2}$ or $d_\mathrm{H} \geq \frac{1}{2}$, one or more additional main-lobes appear in the interval of interest. This implies that two UEs whose signals arrive from azimuth and elevation angles such that $\Omega = \frac{k}{d_\mathrm{V}}$ or $\Psi = \frac{k}{d_\mathrm{H}}$, for some integer k, cannot be separated by the array. For this reason, we often consider antenna arrays with *critical spacing* $d_\mathrm{V} = \frac{1}{2}$ and $d_\mathrm{H} = \frac{1}{2}$.[17] Antenna arrays with larger/smaller antenna spacing are called super-/sub-critically spaced. Similar to the fact that undersampling of a signal in the time domain creates aliasing (i.e., higher frequency components are distinguishable from lower frequency components), spatial undersampling creates aliasing in the angular domain since multiple distinct directions cannot be distinguished by the array.

In some cases, it is desirable to have a high angular resolution or directivity, but one cannot afford to have a physically long/high array with a large number of critically spaced antennas. As explained above, super-critical spacing will lead to multiple main-lobes. However, if the additional main-lobes appear at values outside the feasible range of Ω and Ψ it does not matter. For example, in a typical cell sector, the outdoor UEs are located on the ground below the antenna array such that θ is limited to the range $[-\pi/2, 0]$. As a consequence, Ω is limited to the interval $[-1, 1]$ and a spacing of $d_\mathrm{V} = 1$ is sufficient to avoid the negative effect of having multiple main-lobes. Due to this reason, antenna arrays for cellular communications have often super-critical vertical antenna spacing. In some other scenarios, where separability is not an important criterion, multiple main-lobes can even be desired to illuminate better certain areas of a cell-sector.

[17] For critical spacing, the second main-lobe appears only in the pathological case where $\theta_1 = \pi/2$ and $\theta_2 = -\pi/2$ or $\theta_1 = \theta_2 = 0$, $\varphi_1 = \pi/2$ and $\varphi_2 = -\pi/2$.

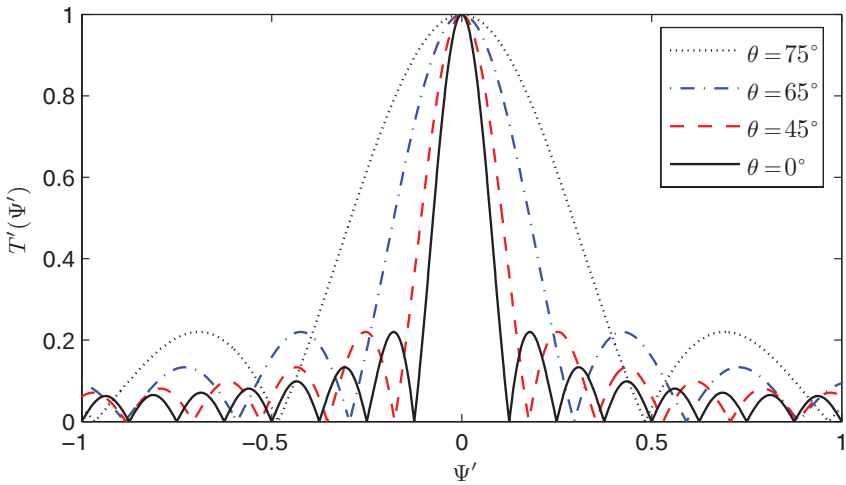


Figure 7.24: The function $T'(\Psi')$ for $M_{\mathrm{H}} = 16$, $d_{\mathrm{H}} = \frac{1}{2}$ and different values of θ.

In Section 7.3.1, we have already observed that the azimuth angular resolution of an array depends on the elevation angle. Using (7.30), this observation can now be mathematically explained. Consider two UEs whose signals arrive from the same elevation angle $\theta_1 = \theta_2 = \theta$ but from different azimuth angles: $\varphi_1 \neq \varphi_2$. This means $\Omega = 0$ and $\Psi = \cos(\theta)(\sin(\varphi_1) - \sin(\varphi_2)) = \cos(\theta)\Psi'$. The function $T'(\Psi') = T(\cos(\theta)\Psi')$ is periodic with period $\frac{1}{\cos(\theta)d_{\mathrm{H}}}$ and has nulls for $\Psi' = k/(\cos(\theta)d_{\mathrm{H}}M_{\mathrm{H}})$. This is equivalent to the behavior of an array with the same number of antennas but with reduced horizontal antenna spacing $|\cos(\theta)|d_{\mathrm{H}} \leq d_{\mathrm{H}}$. In other words, with increasing elevation angle, the spatial resolution in the azimuth domain is reduced. For two UEs with the same elevation angle, the array behaves like a ULA with antenna spacing $\cos(\theta)d_{\mathrm{H}}$. This effect is visualized in Figure 7.24, which shows the function $T'(\Psi')$ for different values of θ. The main-lobe is less wide for smaller values of θ, which improves the resolution.

Although the above discussion is entirely based on a LoS channel model, it also provides important insights into the representation of NLoS channels. As described in Section 7.3.3, the channel response of a NLoS channel is composed of reflections from different scattering clusters which arrive with different delays from different angular directions. While the resolvability of such multipath components in the time domain

depends on the bandwidth, the angular resolvability depends on the array geometry. One can thus decompose the channel response into spatially separable components that essentially determine the degrees of freedom of the wireless channel. For example, a horizontal ULA is able to sample the angular domain with a resolution of $1/L_{\mathrm{H}}$. The number of non-empty "angular bins" determines hence the available degrees of freedom of the channel. In a pure LoS channel, all signals arrive from the same direction and fall consequently in the same angular bin. The angular domain representation of wireless channels was explored in [289] and is discussed in great detail in [314, Sec. 7.3.3-7].

7.4.3 Cell-Free Systems

A radically different way of deploying large antenna arrays, and even cellular networks in general, is by distributing subarrays, connected through optical fibers, over a large geographic area as illustrated in Figure 7.25. This idea is sometimes referred to as *cell-free* Massive MIMO [240, 236], although it is conceptually very close to coordinated multipoint (CoMP) with joint transmission or network MIMO [126, 325, 156, 46] with the difference that the number of deployed antennas per UE is assumed to be large, as in Massive MIMO. Expressed with the parameters of the canonical system model in Definition 2.1 on p. 64, we have a single cell (i.e., $L = 1$) in which a BS consisting of S distributed antenna arrays with M_s antennas, $s = 1, \ldots, S$, respectively, is deployed and serves K UEs. The total number of antennas is $M = \sum_{s=1}^{S} M_s$. The channel $\mathbf{h}_k^s \in \mathbb{C}^{M_s \times 1}$ from UE k to subarray s has the spatial correlation matrix $\mathbf{R}_k^s \in \mathbb{C}^{M_s \times M_s}$, so that the overall channel $\mathbf{h}_k \in \mathbb{C}^{M \times 1}$ of UE k is modeled as $\mathbf{h}_k \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}_M, \mathbf{R}_k)$ where the block-diagonal correlation matrix is

$$\mathbf{R}_k = \operatorname{diag}\left(\mathbf{R}_k^1, \ldots, \mathbf{R}_k^S\right). \tag{7.32}$$

The SE of such a system is then given by the results in Section 4. However, the problems of power control and pilot allocation have a quite different flavor in a setting with distributed arrays and reuse of pilots within the same cell. Thus, the algorithms described in Sections 7.1 and 7.2 are generally not applicable.

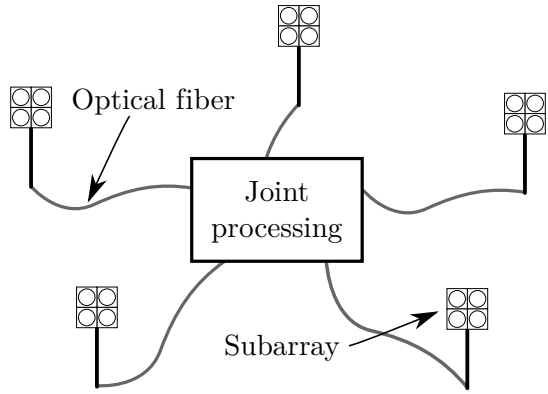


Figure 7.25: A cell-free Massive MIMO system consisting of five spatially distributed subarrays with four antennas each.

The key motivation behind cell-free Massive MIMO is to trade-off channel hardening (as defined in Section 2.5.1 on p. 78) against macro-diversity.[18] Since a UE is likely to have strong channels to only a few subarrays, its overall channel vector is dominated by a few strong components [78]; that is, a few of the matrices $\mathbf{R}_k^s$ have much larger eigenvalues than the others. As a result of this strong spatial channel correlation, the effect of channel hardening is dramatically reduced and SE expressions that rely on hardening can vastly underestimate the achievable performance [78]. However, the UL expression in Theorem 4.1 on p. 123 and the DL expression in Theorem 4.9 on p. 172 are still usable. The increased macro-diversity in cell-free systems can lead to significant power gains and a more uniform distribution of the received signal power, especially in urban scenarios with strong shadow fading.

Cell-free systems employ TDD to exploit channel reciprocity and signal processing relying on only locally available CSI at each subarray, to avoid the costly exchange of CSI and precoding/combining vectors. MR is one suitable scheme due to its distributed nature, but subarrays

[18] Since the subarrays are sufficiently separated, their average channel gain coefficients to a randomly located UE can be viewed as uncorrelated random variables (depending on how the subarrays are distributed). This effect is called "macro-diversity" in contrast to "micro-diversity" which relates to uncorrelated small-scale fading coefficients of sufficiently spaced antennas of the same subarray.

with multiple antennas can also process the signals jointly within the array [61]. Joint power control over all subarrays has been shown to be beneficial [241, 237]. The price to pay for cell-free Massive MIMO systems is a high traffic load on the optical fiber connections because payload data for all UEs must be sent to all subarrays. Given the high deployment cost of optical fiber and the technical challenges related to the synchronization and calibration of a large number of distributed antennas arrays, it is unclear at the time of writing of this monograph if cell-free systems will be used in practice. One step towards reducing the backhaul traffic is to serve each UE only by the few subarrays that have the strongest average channel gains [45]. One alternative but less radical approach is to have multiple cells with distributed subarrays in each of them [53].

We will now demonstrate how the effect of channel hardening is reduced in a cell-free Massive MIMO system. To this end, we consider a single UE located at the center of a cell of fixed radius in which S subarrays with M_S antennas each are distributed uniformly at random. We vary the values of S and M_S in such a way that the total number of antennas $M = SM_S = 256$ remains constant. We ignore spatial channel correlation at the subarrays and focus only on large-scale fading. Thus, the UE's channel $\mathbf{h} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}_M, \mathbf{R})$ has the correlation matrix

$$\mathbf{R} = \text{diag}\left(\beta_1 \mathbf{I}_{M/S}, \ldots, \beta_S \mathbf{I}_{M/S}\right) \tag{7.33}$$

where β_s is the average channel gain of subarray s. The channel gain coefficients are modeled according to (2.3) as $\beta_s = -148.1 - 37.6 \log_{10}(d_s/1\,\text{km}) + F_s$, where d_s is the distance between the UE and subarray s and $F_s \sim \mathcal{N}(0, 8)$ accounts for shadow fading. Recall from (2.17) that the variance of $\|\mathbf{h}\|^2 / \text{tr}(\mathbf{R})$ is a suitable measure of channel hardening, where a smaller variance corresponds to more hardening. Figure 7.26 shows this variance as a function of S for a cell radius of either 100 m or 350 m. We assumed that the subarrays are uniformly distributed within a disc of radius 100 m or 350 m around the UE while keeping a minimum distance of 10 m. The results are calculated over 1000 random drops of subarrays. One can clearly see that the variance increases with the number of subarrays, which means that the channel hardening is reduced. The smaller the density of subarrays (i.e., the

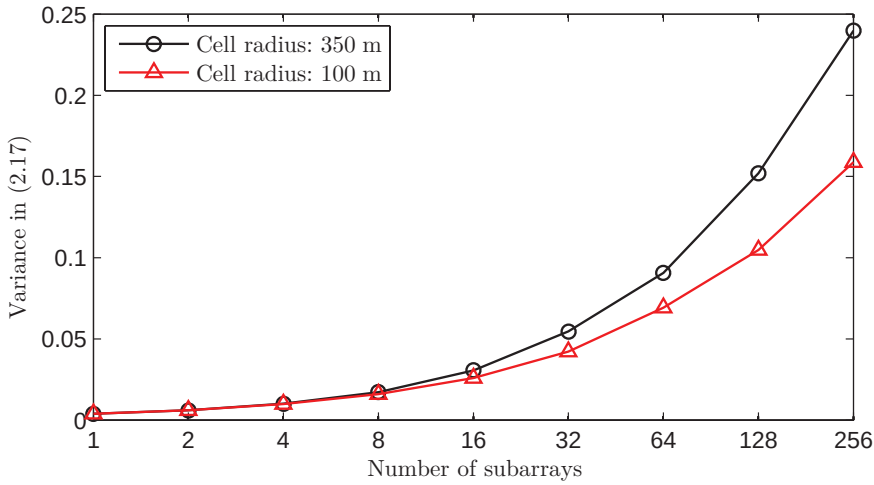


Figure 7.26: Variance in (2.17) of $\|\mathbf{h}\|^2/\mathrm{tr}(\mathbf{R})$, which measures the closeness to channel hardening, as a function of the number of subarrays S for a cell radius of 100 m and 350 m. The variance is computed with respect to random subarray locations and small-scale fading realizations.

larger the cell radius), the more pronounced is this effect since there are fewer dominating subarray links.

Although increasing S reduces the desirable effect of channel hardening, it increases the macro-diversity. This can be seen from Figure 7.27, which shows the CDF of the average channel gain $\beta = \frac{1}{M}\mathrm{tr}(\mathbf{R})$ for different numbers of subarrays. Interestingly, the expected average channel gain $\mathbb{E}\{\beta\}$ is independent of S (where the expectation is taken with respect to the subarray locations as well as the shadow fading realizations) owing to the i.i.d. large-scale fading coefficients of the individual subarrays. However, going from $S = 1$ to $S = 16$ improves the median of β by around 9 dB; increasing this number to $S = 256$ adds another 5 dB. Thus, cell-free systems reduce substantially the fluctuations in average channel gain among the UEs, which is important when the goal is to guarantee a certain service quality to the UEs, uniformly over the coverage area.

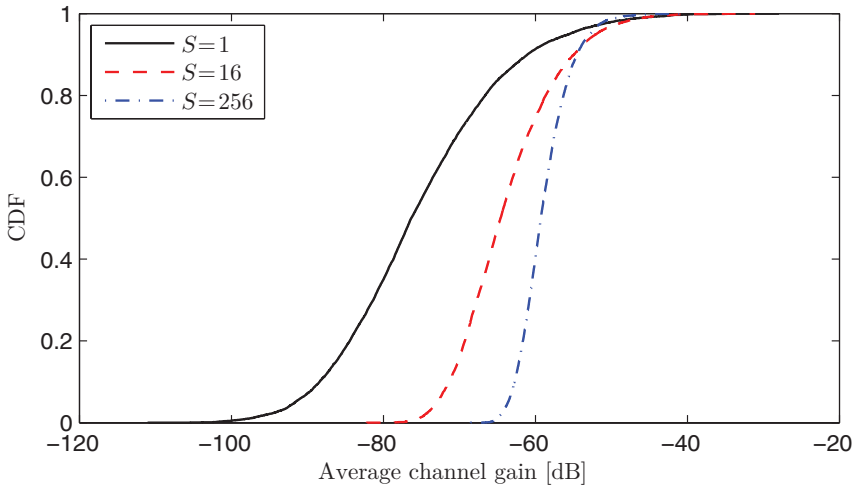


Figure 7.27: CDF of the average channel gain $\beta = \frac{1}{M}\text{tr}(\mathbf{R})$ in a cell-free system for different numbers of subarrays $S \in \{1, 16, 256\}$.

7.4.4 Polarization

One of the aspects related to antenna design we have not been discussed is the polarization of the EM waves radiated by the UEs or BSs. If one tracks at a fixed location the movement of the tip of the electric field vector over time, one obtains a curve called the polarization ellipse. One can then classify the polarization of an EM wave according to the shape of this ellipse which can be either linear, circular, or elliptical. In cellular communications, linearly polarized antennas are most commonly used. The direction of a linear polarization is defined by the tilt angle of the polarization ellipse; for example, 90° (vertical), 0° (horizontal), and ±45° (slant). Linear polarization always come in two orthogonal pairs; for example, horizontal and vertical or ±45° slant. An example of a horizontally and a vertically polarized wave is shown in Figure 7.28. Importantly, any linear polarization can be obtained from a superposition of two orthogonal polarizations. Antennas which radiate (or respond to) EM waves with only one polarization direction are called *uni-polarized*, while antennas that create (or respond to) field components in two orthogonal polarization directions are called *dual-polarized*. Generally, UEs have uni-polarized antennas or antennas with

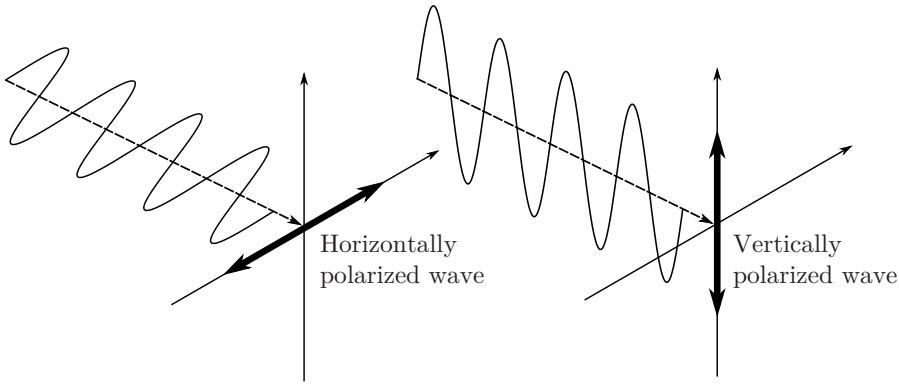


Figure 7.28: Polarization ellipse for a horizontally and a vertically polarized wave.

a strong dominating polarization.[19] However, one needs to keep in mind that the effective polarization depends on the physical orientation of the UE which is generally random (e.g., depending on how a smartphone is held). For example, a UE with a vertically polarized antenna can create (or respond to) an EM wave with vertical and horizontal polarization components if it is rotated by a few degrees. In order to prevent that radiated energy in one of the polarization directions is lost, BSs are almost exclusively equipped with dual-polarized antennas.

Ideally, a horizontally polarized EM wave should not be received by a vertically polarized antenna and vice versa. However, there is generally some form of cross-talk between both polarization directions due to imperfect cross-polar isolation (XPI) of the antennas and imperfect cross-polar discrimination (XPD) of the channel. XPI is a property of the antenna alone and describes how a (supposedly) linearly polarized antenna responds to cross-polar wave components; that is, the components of an EM wave in the orthogonal polarization direction. XPD describes the phenomenon that EM waves can change their polarization when going through a scattering medium. This effect is also called (channel) depolarization. In general, the reflection coefficients for each polarization are different so that the phases of the two orthogonal polarizations undergo different changes for each reflection. Thus, in a sufficiently rich

[19] We use the terms "polarization" and "polarization direction" interchangeably.

scattering propagation environment, the received polarization would be independent of the transmitted polarization. However, in practice, we do not have sufficient scattering to observe full depolarization and some correlation exists [323].

Remark 7.6 (Dual-polarized antenna arrays)**.** When we speak about a dual-polarized antenna array with M antennas in this monograph, we consider an antenna array composed of $M/2$ uni-polarized antennas for each polarization direction. For space reasons, the antennas for both polarization directions are generally co-located. This means that an antenna array with dual-polarized co-located antennas accommodates twice the number of antennas as a uni-polarized array of the same physical dimensions.

Channel Model with Dual-polarized Antennas

We will now present a general channel model for a BS antenna array consisting of M dual-polarized co-located antennas and a UE with a uni-polarized, but randomly oriented, antenna, as shown in Figure 7.29.[20] This model is inspired by [89, 171]. The interested reader is also referred to [88, Sec. 3.3] and [249, 293] for a more detailed discussion of polarization modeling for MIMO systems. Denote by $h_{i,\mathrm{V}}$ and $h_{i,\mathrm{H}}$ the channel coefficients from the UE to the ith vertically and horizontally polarized antenna, respectively. Then, for $i = 1, \ldots, \frac{M}{2}$,

$$\mathbf{h}_i \triangleq \begin{bmatrix} h_{i,\mathrm{V}} \\ h_{i,\mathrm{H}} \end{bmatrix} = \mathbf{F}_{\mathrm{BS}} \mathbf{Z}_i \mathbf{m}_{\mathrm{UE}}(\theta_r) \tag{7.34}$$

where

- $\mathbf{F}_{\mathrm{BS}} \in \mathbb{C}^{2\times 2}$ is the deterministic polarization matrix of the BS's antennas which has orthonormal columns and can be modeled as

$$\mathbf{F}_{\mathrm{BS}} = \frac{1}{\sqrt{1+\chi_a}} \begin{bmatrix} 1 & \sqrt{\chi_a} e^{\mathrm{j}\phi_a} \\ -\sqrt{\chi_a} e^{\mathrm{j}\phi_a} & 1 \end{bmatrix} \tag{7.35}$$

  where χ_a^{-1} is the XPI and ϕ_a a phase offset depending on the antenna characteristics;

[20] We assume isotropic radiating antennas. Thus, only the polarization but not the radiation pattern is affected by rotations.

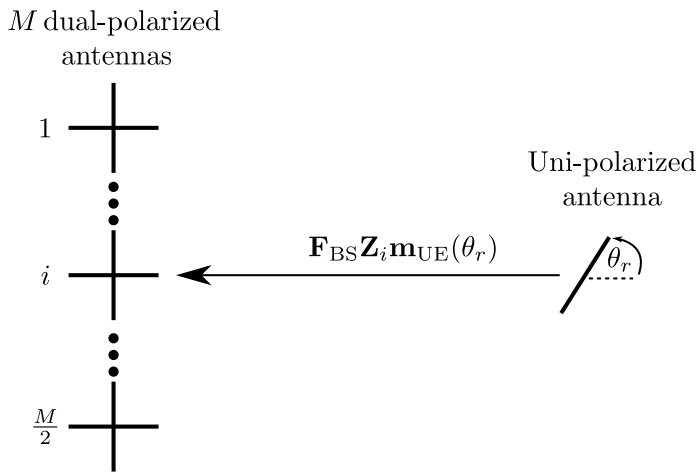


Figure 7.29: A BS antenna array with M dual-polarized co-located antennas and a UE with a uni-polarized, but randomly oriented, antenna. The 2×1 channel from the UE to the ith antenna location is denoted $\mathbf{F}_{\mathrm{BS}}\mathbf{Z}_i\mathbf{m}_{\mathrm{UE}}(\theta_r)$.

- $\mathbf{m}_{\mathrm{UE}}(\theta_r) \in \mathbb{C}^{2\times 1}$ is the unit-norm polarization vector of the UE's antenna, which is modeled as[21]

$$\mathbf{m}_{\mathrm{UE}}(\theta_r) = \frac{1}{\sqrt{1+\chi_b}} \begin{bmatrix} \cos(\theta_r) & -\sin(\theta_r) \\ \sin(\theta_r) & \cos(\theta_r) \end{bmatrix} \begin{bmatrix} 1 \\ \sqrt{\chi_b} e^{\mathrm{j}\phi_b} \end{bmatrix} \tag{7.36}$$

  where $\theta_r \in [0, 2\pi)$ is the rotation angle of the polarization directions, χ_b^{-1} is the XPI, and ϕ_b a phase offset depending on the antenna characteristics;

- $\mathbf{Z}_i \in \mathbb{C}^{2\times 2}$ is the random channel matrix describing the propagation of the vertical and horizontal EM wave components, which is defined as

$$\mathbf{Z}_i = \begin{bmatrix} z_{i,\mathrm{VV}} & z_{i,\mathrm{VH}} \\ z_{i,\mathrm{HV}} & z_{i,\mathrm{HH}} \end{bmatrix}. \tag{7.37}$$

Somewhat surprisingly, the 2×1 channel $\mathbf{h}_i$ from the UE to two co-located dual-polarized antennas depends on four complex random quantities in $\mathbf{Z}_i$ which we will characterize next. The XPD describes the

[21] The matrix in (7.36) is a rotation matrix.

ability of the channel to separate vertical and horizontal polarizations and is defined as the ratio (independent of i)

$$\mathsf{XPD} = \frac{\mathbb{E}\left\{|z_{i,\mathrm{VV}}|^2\right\}}{\mathbb{E}\left\{|z_{i,\mathrm{HV}}|^2\right\}} = \frac{\mathbb{E}\left\{|z_{i,\mathrm{HH}}|^2\right\}}{\mathbb{E}\left\{|z_{i,\mathrm{VH}}|^2\right\}} = \frac{1-q_{\mathrm{XPD}}}{q_{\mathrm{XPD}}} \tag{7.38}$$

that depends on the parameter $q_{\mathrm{XPD}} = 1/(1+\mathsf{XPD})$, which satisfies $0 < q_{\mathrm{XPD}} \leq 1$. In this definition, we have used the following assumptions

$$\begin{aligned} \mathbb{E}\left\{|z_{i,\mathrm{VV}}|^2\right\} &= \mathbb{E}\left\{|z_{i,\mathrm{HH}}|^2\right\} = 1-q_{\mathrm{XPD}} \\ \mathbb{E}\left\{|z_{i,\mathrm{HV}}|^2\right\} &= \mathbb{E}\left\{|z_{i,\mathrm{VH}}|^2\right\} = q_{\mathrm{XPD}} \end{aligned} \tag{7.39}$$

which additionally imply that $\mathbb{E}\left\{\mathrm{tr}\left(\mathbf{Z}_i\mathbf{Z}_i^{\mathrm{H}}\right)\right\} = 2$. This means that the total radiated power is transferred through the channel without any losses, whatever the value of XPD. The higher the XPD (or the smaller q_{XPD}), the smaller is the power that leaks from one polarization to the other. Due to insufficient scattering, it is reasonable to assume that the components of $\mathbf{Z}_i$ are correlated. We adopt here the separable correlation model

$$\mathbf{Z}_i = \left(\mathbf{C}_{r_p}^{\frac{1}{2}}\mathbf{G}_i\mathbf{C}_{t_p}^{\frac{1}{2}}\right) \odot \mathbf{\Sigma} \tag{7.40}$$

where $\odot$ is the Hadamard (or elementwise) product,

$$\mathbf{\Sigma} = \begin{bmatrix} \sqrt{1-q_{\mathrm{XPD}}} & \sqrt{q_{\mathrm{XPD}}} \\ \sqrt{q_{\mathrm{XPD}}} & \sqrt{1-q_{\mathrm{XPD}}} \end{bmatrix} \tag{7.41}$$

is the XPD matrix,

$$\mathbf{C}_{r_p} = \begin{bmatrix} 1 & r_p \\ r_p^\star & 1 \end{bmatrix}, \quad \mathbf{C}_{t_p} = \begin{bmatrix} 1 & t_p \\ t_p^\star & 1 \end{bmatrix} \tag{7.42}$$

are the receive and transmit polarization correlation matrices, respectively, and

$$\mathbf{G}_i = \begin{bmatrix} g_{i,\mathrm{VV}} & g_{i,\mathrm{VH}} \\ g_{i,\mathrm{HV}} & g_{i,\mathrm{HH}} \end{bmatrix} \tag{7.43}$$

contains the small-scale fading channel coefficients (whose distributions also account for the average channel gain β, defined in (2.2)). Since the horizontally and vertically polarized antennas are co-located, there is no

additional spatial correlation on top of the polarization correlation between the elements of $\mathbf{G}_i$. However, the vectors $\mathbf{g}_{xy} = [g_{1,xy}, \ldots, g_{\frac{M}{2},xy}]^{\mathrm{T}}$ for $x, y \in \{\mathrm{V}, \mathrm{H}\}$ consisting of the stacked xy-components of each of the $M/2$ channel matrices $\mathbf{G}_i$ are spatially correlated with the same correlation matrix $\mathbf{R} \in \mathbb{C}^{\frac{M}{2} \times \frac{M}{2}}$; that is, $\mathbf{g}_{\mathrm{xy}} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}_{\frac{M}{2}}, \mathbf{R})$ for $x, y \in \{\mathrm{V}, \mathrm{H}\}$. One can show after some simple algebra that the entire channel matrix $\mathbf{Z} = [\mathbf{Z}_1^{\mathrm{H}}, \ldots, \mathbf{Z}_{\frac{M}{2}}^{\mathrm{H}}]^{\mathrm{H}} \in \mathbb{C}^{M \times 2}$ can be modeled as

$$\mathbf{Z} = \left(\left(\mathbf{R} \otimes \mathbf{C}_{r_p} \right)^{\frac{1}{2}} \mathbf{W} \mathbf{C}_{t_p}^{\frac{1}{2}} \right) \odot \left(\mathbf{1}_{\frac{M}{2}} \otimes \mathbf{\Sigma} \right) \tag{7.44}$$

where $\mathbf{W} \in \mathbb{C}^{M \times 2}$ has i.i.d. $\mathcal{N}_{\mathbb{C}}(0, 1)$ elements and $\otimes$ denotes the Kronecker product. The final channel vector $\mathbf{h} = [\mathbf{h}_1^{\mathrm{H}}, \ldots, \mathbf{h}_{\frac{M}{2}}^{\mathrm{H}}]^{\mathrm{H}} \in \mathbb{C}^{M \times 1}$ is then given as

$$\mathbf{h} = \left(\mathbf{I}_{\frac{M}{2}} \otimes \mathbf{F}_{\mathrm{BS}} \right) \mathbf{Z} \mathbf{m}_{\mathrm{UE}}(\theta_r). \tag{7.45}$$

Measurements have shown that the absolute values of the transmit and receive polarization correlation are rather small (i.e., $|t_p|, |r_p| < 0.2$ [108]) and can be sometimes even fully neglected [25]. The XPD has also been found to be rather strong in cellular systems, lying in the range from 5 dB to 15 dB [25]. We can additionally assume in simulations that the UE's polarization is uniformly randomly distributed: $\theta_r \sim U[0, 2\pi)$.

It is difficult to say if dual-polarized antennas are advantageous for Massive MIMO systems, apart from allowing for more compact arrays. To exemplify this issue, consider a BS with a horizontal ULA of either M critically-spaced uni-polarized or dual-polarized—but not co-located (i.e., neighboring antennas have orthogonal polarizations)—antennas, as shown in Figure 7.30. The array has the same number of antennas and the same length in both cases and hence the same angular resolution (see (7.31)). We assume that there are three UEs in the cell at azimuth angles $\varphi_1 = 30°$, $\varphi_2 = 25°$, and $\varphi_1 = -10°$, respectively. Figure 7.31 shows the average correlation $\mathbb{E}\left\{ |\mathbf{h}_1^{\mathrm{H}} \mathbf{h}_i|^2 / (\|\mathbf{h}_1\| \|\mathbf{h}_i\|)^2 \right\}$ for $i = 2, 3$ of the channels between UE 1 and UE 2 and between UE 1 and UE 3. We use the local scattering model with Gaussian angular distribution and ASD of 5°. The polarization parameters are $r_p = t_p = 0.2$, $\mathsf{XPD} = 7$ dB, and both the BS and the UE have antennas with perfect XPI (i.e.,

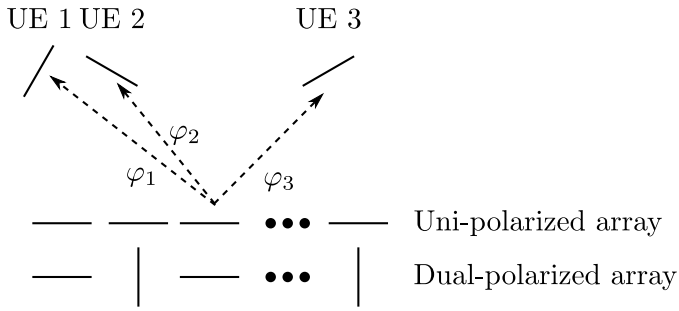


Figure 7.30: A horizontal uni-polarized or dual-polarized (but not co-located) ULA of M critically-spaced antennas communicating with three UEs.

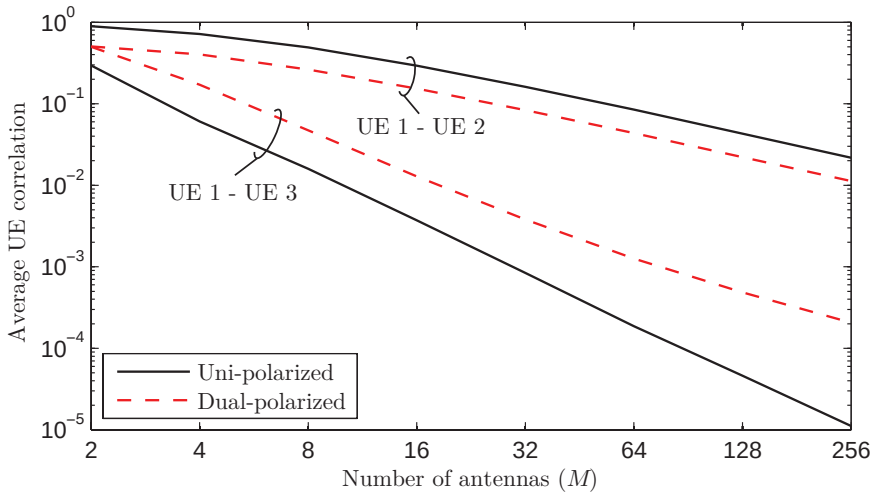


Figure 7.31: Average correlation between the channels of UEs 1 and 2 and between the channels of UEs 1 and 3, as a function of M (in logarithmic scale), for uni- and dual-polarized antenna arrays.

$\chi_a = \chi_b = 0$). The UEs' polarization directions are uniformly randomly distributed.

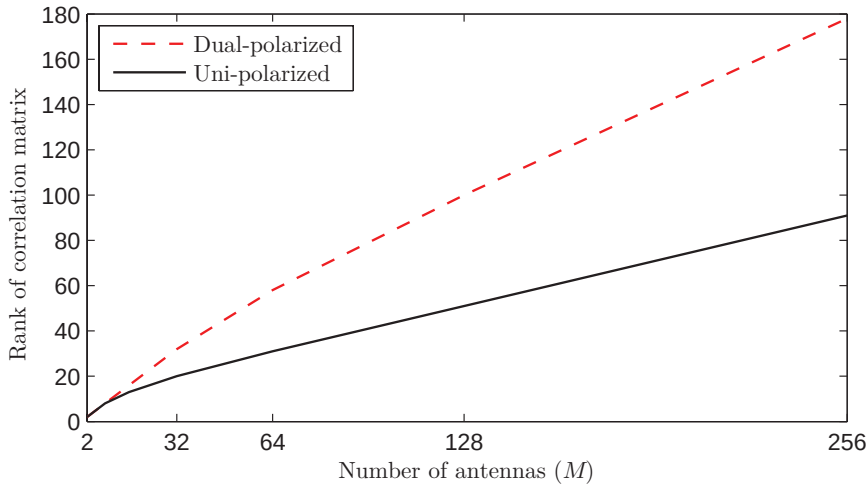
We can observe that the dual-polarized array achieves a better separation of UE 1 and UE 2 than the uni-polarized array. The opposite is true for UE 1 and UE 3. The reason for this behavior is as follows. UEs 1 and 2 are located very closely together and their channels have

similar correlation matrices. We have seen in Figure 7.15 that the dimension p of the dominating eigenspace (which contains almost all energy) can be much smaller than M under spatial channel correlation. Since the UEs essentially share the available degrees of freedom of the same p-dimensional subspace, the average correlation of their channels decays not as $1/M$ but as $1/p$. Now, since the orthogonal polarization components are almost uncorrelated, the dual-polarized array almost doubles the number of degrees of freedom compared to the uni-polarized array. Thus, whenever a system is limited by spatial channel correlation, polarization diversity helps. Figure 7.32a shows the rank of the spatial correlation matrix of UE 1 for the uni- and dual-polarized arrays as a function of M. This demonstrates that the use of a dual-polarized array almost doubles the degrees of freedom. This effect can be also explained mathematically from (7.45). Since $\operatorname{rank}(\mathbf{A} \otimes \mathbf{B}) = \operatorname{rank}(\mathbf{A})\operatorname{rank}(\mathbf{B})$ for two matrices $\mathbf{A}$, $\mathbf{B}$ with suitable dimensions, it follows that

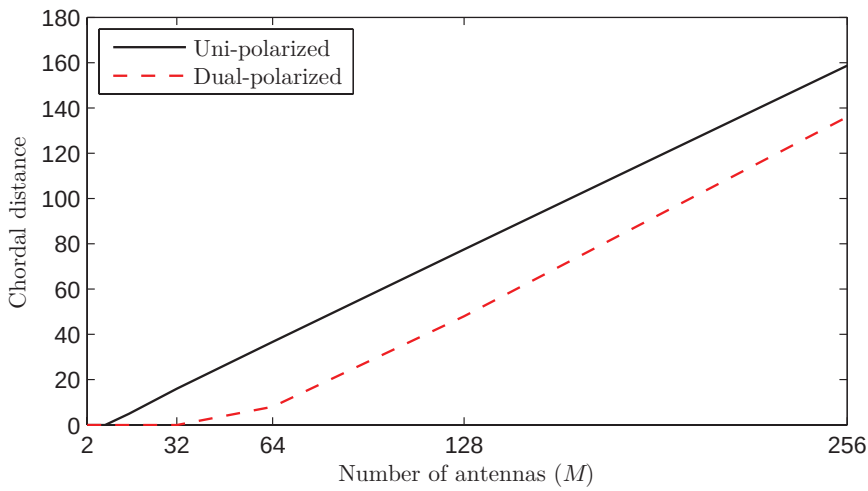
$$\operatorname{rank}\left(\mathbf{R} \otimes \mathbf{C}_{r_p}\right) = \operatorname{rank}\left(\mathbf{R}\right) \operatorname{rank}\left(\mathbf{C}_{r_p}\right) = 2 \operatorname{rank}\left(\mathbf{R}\right). \tag{7.46}$$

On the other hand, UEs 1 and 3 have correlation matrices whose eigenspaces are very different. Thus, even if each of the UEs' spatial correlation matrices had only rank one, their average correlation would be small because the correlation-eigenspaces are almost orthogonal. In this case, dual-polarized arrays are disadvantageous because they increase the dimensions of the subspaces to which the channel vectors are confined. This can be seen from Figure 7.32b, which shows the chordal distance (as defined in (7.21)) of the eigenspaces of the correlation matrices of UE 1 and 3 for the uni- and dual-polarized arrays as a function of M. The chordal distance for the uni-polarized array is much larger than that for the dual-polarized array, indicating that it achieves a better decorrelation of the UEs' channels.

Due to the rather involved distribution of $\mathbf{h}$ in (7.45) with dual-polarized antennas, we have refrained from explicitly considering polarization in the analysis carried out in this monograph. However, since we have considered arbitrary correlation matrices in most sections, the results apply also to correlated Rayleigh fading with dual-polarized antennas. The interested reader is referred to the rather small number of publications dealing with dual-polarized Massive MIMO systems,



(a) Rank of the spatial correlation matrix of UE 1 as a function of M.



(b) Chordal distance between the eigenspaces of the spatial correlation matrices of UE 1 and UE 3 as a function of M.

Figure 7.32: Comparison of spatial correlation matrices with uni- and dual-polarization.

such as [252, 253, 354]. The 3GPP 3D MIMO model [1] and its open-source implementations [159, 5] support the simulation of dual-polarized antenna arrays, which will be used in the case study in Section 7.7.

7.5 Millimeter Wavelength Communications

Most wireless communication systems today make use of the 300 MHz–6 GHz frequency range while the spectrum from 6–300 GHz is comparatively empty. This is mainly due to the very advantageous propagation characteristics at low frequencies which allow the radio waves to penetrate buildings, reflect multiple times, and bend around corners. The mmWave band refers to the frequency range from 30–300 GHz with wavelength ranging from 1–10 mm. MmWaves suffer from high atmospheric absorption, rain and foliage attenuation, strong penetration and reflection losses, and little diffraction, which essentially restrict their use to LoS outdoor-to-outdoor or indoor-to-indoor communications over relatively short distances. The current main use cases are wireless backhaul in the unlicensed 60 GHz band, as a cost-efficient alternative to wired solutions, and WLANs based on the IEEE 802.11ad standard. Nevertheless, recent theoretical considerations and measurement campaigns have provided evidence that outdoor SCs with up to 200 m cell radii are viable if the transmitters and receivers are equipped with sufficiently "large" antenna arrays (in a sense that we will define shortly) to compensate for the otherwise prohibitive propagation losses [259, 275].

To understand why large antenna arrays or—to be more precise—antennas composed of a large number of radiating elements (see Definition 7.3) are needed for mmWave communications, let us have a look at Friis' transmission formula [118] which describes the relation between the received signal power P_r and transmitted signal power P_t for two antennas separated by distance d under ideal conditions and free-space propagation:

$$\frac{P_r}{P_t} = G_r G_t \left(\frac{\lambda}{4\pi d}\right)^2 = \frac{A_r A_t}{(d\lambda)^2} \tag{7.47}$$

where λ is the wavelength, G_t, G_r and A_t, A_r are the gains[22] and the effective areas[23] of the transmitter and receiver antennas, respectively. Note that (7.47) assumes that both antennas are perfectly aligned and neglects that G_t, G_r are generally angle-dependent. The first part of the equation says that for fixed G_t, G_r, the pathloss P_t/P_r is proportional to λ^{-2}, while the second part says that for fixed A_t, A_r, the pathloss is proportional to λ^2. The seeming contradiction between the two results is resolved once we understand that a dipole (or any other radiating element) has a frequency-independent gain but an effective area that shrinks with the carrier frequency. For example, a half-wavelength dipole has an effective area of approximately $0.125\,\lambda^2$, but a fixed gain of 0.5π over an (hypothetical) isotropic antenna [118] with effective area $\lambda^2/(4\pi)$. The effective area of an antenna can therefore only be kept constant as λ decreases if more and more radiating elements are connected together. The number of radiating elements fitting into a given area is proportional to λ^{-2} and so is the resulting antenna gain. The discussion above implies that we can even achieve a net gain in the received signal power proportional to λ^{-2}, given that the antennas at the transmitter and the receiver are composed of a large number of radiating elements that scales at the same speed. However, in order to keep a constant pathloss, it is sufficient to scale the number of elements such that $A_r A_t \sim \lambda^2$. For example, the receiver antenna could have a single radiating element while the number of elements of the transmit antenna scales as λ^{-2} or vice versa. Alternatively, the number of elements at both antennas could be scaled such that their effective areas are linear in λ. For cellular communications, one could hence have UEs with comparatively few radiating elements, while BSs with antennas composed of many radiating elements compensate for the vast majority of the propagation loss.

[22]The antenna gain is the ratio of the maximum power density radiated by the antenna in any direction to that of an ideal isotropic antenna. It includes the antenna's efficiency, i.e., the ratio of the input power to the total radiated power.

[23]The effective area of an antenna is equal to the area, oriented perpendicular to the incoming wave, which would collect the same amount of power as was actually received by the antenna.

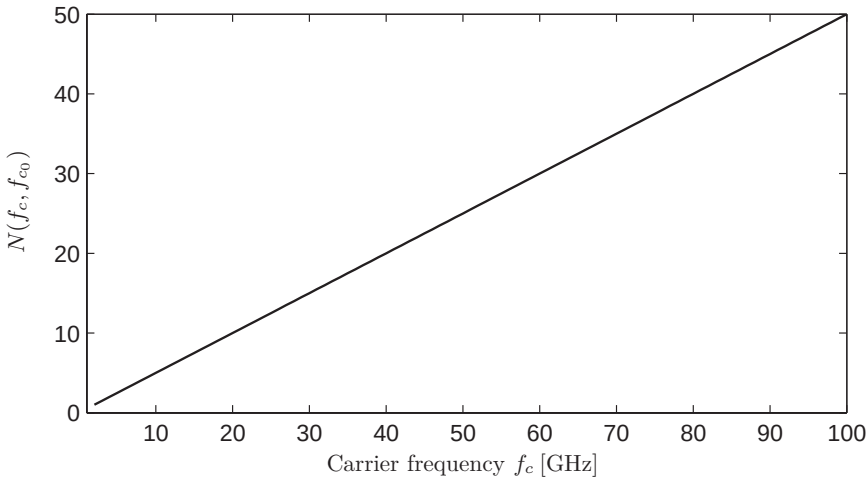


Figure 7.33: The function $N(f_c, f_{c_0})$ versus carrier frequency f_c for a fixed reference carrier frequency $f_{c_0} = 2\,\text{GHz}$. $N(f_c, f_{c_0})$ tells us how many times more half-wavelength dipoles are needed at the transmit and receive antennas at carrier frequency f_c to maintain the same pathloss as observed at f_{c_0}.

In order to get an idea of the required number of radiating elements at mmWave frequencies, denote by $A(\lambda) = 0.125\lambda^2$ the effective area of a half-wavelength dipole. Consider now a communication channel at carrier frequency f_{c_0} (with wavelength λ_0) between two antennas consisting of one (or multiple) such dipoles. According to (7.47), in order to achieve the same pathloss at carrier frequency $f_c > f_{c_0}$ (with wavelength λ), the antennas at the transmitter and the receiver must consist of

$$N(f_c, f_{c_0}) = \frac{\lambda}{\lambda_0}\frac{A(\lambda_0)}{A(\lambda)} = \frac{f_c}{f_{c_0}} \tag{7.48}$$

times more dipoles. As a rule of thumb, we can therefore say that doubling the carrier frequency requires twice the number of radiating elements at both the transmitter and receiver to maintain the same received signal strength. Figure 7.33 shows $N(f_c, f_{c_0})$ as a function of f_c for $f_{c_0} = 2\,\text{GHz}$. However, one must not forget that the main motivation of going to higher frequencies is that the available bandwidth is dramatically increased, but since the total transmit power is generally fixed (due to hardware or regulatory constraints), the SNR

is inversely proportional to the bandwidth. Thus, a communication system operating at $f_c = 60\,\text{GHz}$ with 100 MHz of bandwidth would require transmit and receive antennas consisting of $\sqrt{10} \times 30 \approx 95$ times more radiating elements to maintain the same SNR as a communication system operating at $f_{c_0} = 2\,\text{GHz}$ with 10 MHz of bandwidth.[24] In a cellular setting, this picture might change as the very focused wave propagation at mmWave frequencies creates essentially isolated communication channels. Hence, achieving the same SINR at higher frequencies as at 2 GHz would probably be easier than maintaining the same SNR. The exact comparison is outside the scope of this monograph.

In the discussion above, we considered a single antenna with multiple radiating elements, according to Definition 7.3. But one must not forget that, although an antenna with many radiating elements has a very high gain, which is necessary to overcome the pathloss at mmWave frequencies, the orientation of its radiation pattern is static and cannot be controlled with a single RF input. For this reason, subsets of the radiating elements need to be provided with individual RF inputs to form an antenna array that has a dynamically controllable array response.

Although it is quite encouraging that mmWave cellular communication is rendered possible by large antenna arrays, one must be aware of the following caveats.

First, we illustrated in Section 7.4.2 that the beamwidth of an antenna array in the elevation (azimuth) domain is inversely proportional to its normalized vertical height L_V (horizontal width L_H) measured in multiples of the wavelength. For a fixed effective area, L_V and L_H scale linearly with λ so that the beamwidth is proportional to λ^{-1}. This means that, although large antenna arrays enable communications at mmWaves due to their high gain, they only do so if the narrow transmit and receive beams are well aligned. As a side effect, this implies that mmWave communication suffers strongly from blocking of the strongest (LoS) path since most other paths depart/arrive at angles which are not

[24] The factor 30 is needed to counteract the propagation loss (see Figure 7.33) while the additional factor $\sqrt{10}$ is required to compensate for the increased level of noise due to a ten-fold bandwidth extension.

aligned with the beam directions of the arrays. For these reasons, it is challenging to provide coverage over a large area and to support highly mobile UEs with mmWaves. A quickly growing body of literature deals with the problems of beam training and refinement (see [141, 355] and references therein).

Second, equipping each radiating element of a large antenna with an individual RF chain is impossible with today's technology at mmWave frequencies, because the size, cost, and power consumption of the required hardware is prohibitive for use in UEs [274, 141, 355]. In particular, the PAs and DACs/ADCs are very power consuming at mmWaves (if the bandwidth is increased) and so is the parallel processing of a large number of data streams with billions of samples per second. For this reason, alternative approaches such as analog and hybrid analog-digital beamforming[25] [11] as well as low-resolution DACs/ADCs [222] are subjects of current research.

Third, as discussed in Remark 2.1 on p. 68, λ has a substantial impact on τ_c, which is the size of the coherence block of the wireless channel. Since the channel coherence time T_c is proportional to λ and the coherence bandwidth B_c is inversely proportional to the delay spread T_d, the coherence block $\tau_c = T_c B_c$ satisfies the proportionality

$$\tau_c \sim \frac{\lambda}{T_d}. \tag{7.49}$$

This equation implies that more frequent pilot signals are required in the mmWave bands than in the sub-6 GHz band. However, since mmWave communication systems are foreseen to have rather small cell radii and support only slowly moving UEs, the delay spread is reduced and the coherence time increased which counterbalances the linear scaling in λ to some extent. For example, assuming 60 GHz carrier frequency (i.e., $\lambda = 5\,\text{mm}$), delay spread $T_d = 500\,\text{ns}$ corresponding to a path length difference of 150 m, and $v = 3\,\text{km/h}$ mobility, we obtain with the help of Remark 2.1 a channel with $B_c = 1\,\text{MHz}$ and $T_c = 1.5\,\text{ms}$, resulting in a coherence block of $\tau_c = 1500$ samples.

[25] Analog beamforming introduces phase shifts to the signals going from a single RF chain to the individual radiating elements. Hybrid analog-digital beamforming refers to an antenna array in which each antenna consists of multiple radiating elements whose phases can be individually controlled via analog beamforming.

In the light of the constraints and challenges outlined above, it appears that mmWave communications are best suited for SC hotspot deployments with the goal of providing high throughput to slowly moving UEs. Due to the rather low number of RF chains compared to the number of radiating elements and a relatively small number of UEs per SC (which are likely not to be served simultaneously on the same time-frequency resource), we do not consider mmWave communications as Massive MIMO systems according to Definition 2.1. The interested reader is referred to the textbook [274] and one of the numerous overview papers (e.g., [272, 141]) for a more detailed introduction into the topic.

7.6 Heterogeneous Networks

We have argued in Section 1.3 on p. 20 that a combination of cell size shrinking (i.e., adding more BSs) and increased spatial multiplexing (i.e., adding more antennas to each BS to serve more UEs simultaneously) is needed to satisfy the area throughput requirements of next generation cellular networks. With both techniques, the network is densified, meaning that the number of antennas per unit area is increased. Massive MIMO can be seen as a concentrated form of network densification, while BSs with a small coverage area and only a few antennas, so-called small-cell base stations (SBSs), are a distributed form of network densification. From a pure capacity point of view, there is some theoretical evidence that one should distribute the available antennas as much as possible [104]. This means that if we would have the freedom to distribute M antennas over L BSs, we should choose $L = M$ and spread out the BSs as widely as possible over the coverage area. However, such a network densification quickly reaches its practical limits because with antennas located below the rooftops and a cell radius of less than 50 m, supporting highly mobile UEs and providing seamless coverage over large areas become increasingly difficult. Backhaul provisioning also becomes prohibitively costly. Cell-free Massive MIMO, as described in Section 7.4.3, is one approach to tackle these challenges, but how well it performs under practical conditions remains to be demonstrated. On the other hand, conventional Massive MIMO is particularly suited to provide area coverage and to support high mobility. Thus, a simple

network architecture, integrating the complimentary benefits of Massive MIMO and SBSs, consists of a coverage tier with Massive MIMO BSs overlaid with a hotspot tier of SBSs. In this architecture, the Massive MIMO BSs ensure coverage and serve highly mobile UEs while the SBSs provide high capacity for indoor and outdoor hotspots. There are two main challenges related to such an architecture:

1. How to avoid cross-tier interference if BSs and SBSs share the same spectrum?

2. How to provide backhaul to a large number of SBSs?

It turns out that the significant amount of excess antennas[26] at the Massive MIMO BSs can be used to tackle both of these challenges. This will be discussed in the remainder of this section.

7.6.1 Massive MIMO for Cross-Tier Interference Mitigation

Consider a two-tier network consisting of a canonical Massive MIMO system (according to Definition 2.1 on p. 64) overlaid with a hotspot tier of SBSs. An example of such a network is shown in Figure 7.34. The only differences between SBSs and BSs are that the former has a smaller number of antennas, serve fewer UEs, have less transmit power (and hence a smaller coverage area), and are generally deployed at a lower height. We can, therefore, model such a two-tier network according to Definition 2.1 by choosing appropriate (small) values of M_j and K_j for all indices j that correspond to SBSs. Thus, all results on SE that have been developed in Section 4 can be directly applied. Note, however, that due to the rather small number of antennas, there is little channel hardening, so that the UatF bound (Theorem 4.4 on p. 149) for the UL SE and the hardening bound (Theorem 4.6 on p. 163) for the DL SE are less tight. In this section, we refer to the UEs served by the BSs and SBSs as macro user equipments (MUEs) and small-cell user equipments (SUEs), respectively.

By Definition 2.1, the BSs and SBSs are assumed to operate according to a synchronized TDD protocol and hence share the same spectrum.

[26] We say that a BS with M antennas serving K UEs has $M - K$ excess antennas.

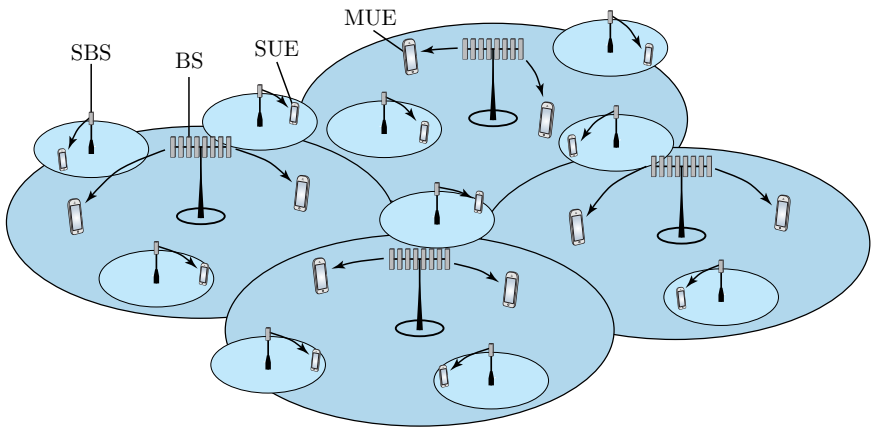


Figure 7.34: A heterogeneous network based on a two-tier deployment with Massive MIMO BSs in the coverage tier and SBSs in the hotspot tier. The BSs serve the MUEs while the SBSs serve the SUEs.

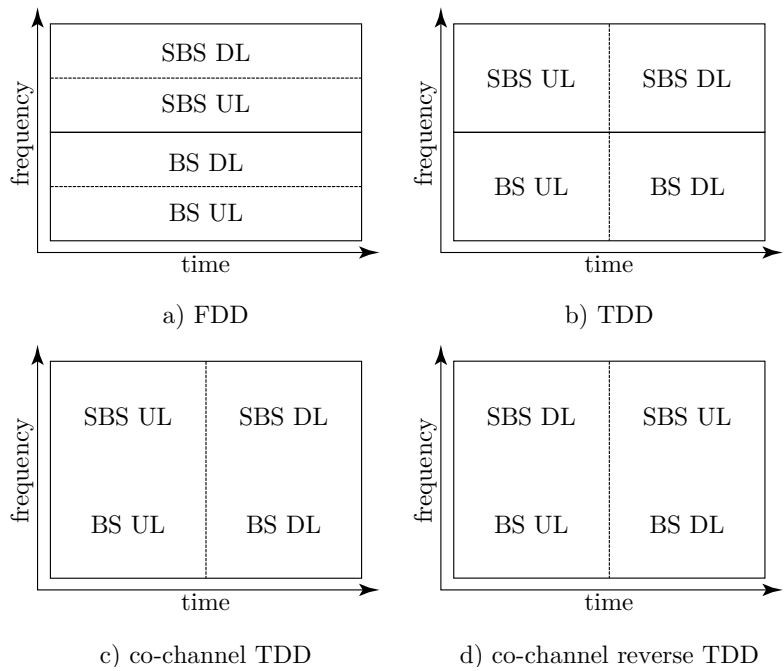


Figure 7.35: Operating principles of different duplexing schemes for heterogeneous networks.

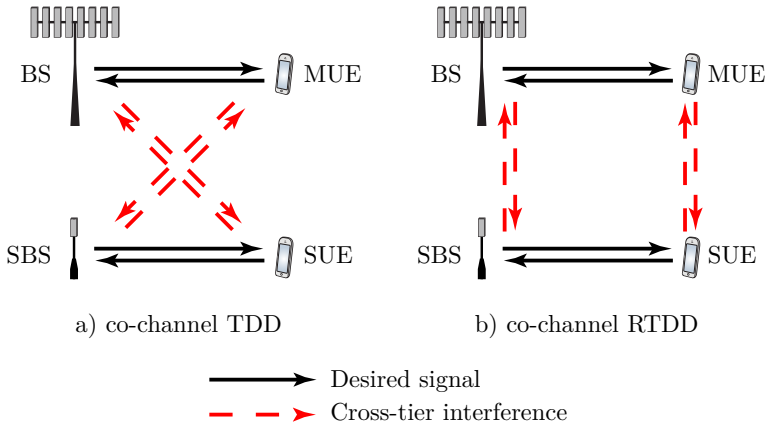


Figure 7.36: Interfering links in co-channel TDD and co-channel RTDD.

We refer to this as co-channel TDD. However, there are several other possible duplexing modes that could be employed to communicate over the available bandwidth: TDD, FDD, and co-channel reverse time-division duplex (RTDD). The operating principles of these different duplexing modes are illustrated in Figure 7.35. In both the FDD and TDD schemes, the BS and SBS tiers operate on non-overlapping frequency bands, while UL and DL transmissions are duplexed in either frequency (FDD) or time (TDD). Thus, transmissions do not interfere across the tiers. Unlike the aforementioned schemes, with co-channel TDD and co-channel RTDD, both tiers share the entire bandwidth. While the UL and DL transmissions are synchronized in both tiers with co-channel TDD, their order is reversed in one of the tiers with co-channel RTDD; that is, the BSs are in DL mode while the SBSs operate in UL mode, and vice versa. The duplexing mode determines which groups of devices interfere with each other. For example, in co-channel TDD, the SUEs interfere with the MUEs in the UL while the BSs interfere with the SBSs in the DL. This fact is shown in Figure 7.36.

A network-wide synchronized TDD protocol (i.e., co-channel TDD or co-channel RTDD) and the resulting channel reciprocity have two important advantages. First, as explained in Section 3.1 on p. 91, the BSs and SBSs can estimate the DL channels from UL pilots sent by the MUEs and SUEs. Second, they can estimate the correlation matrix of

the received signals, which is not only useful for signal detection but also for the design of interference-aware precoding/combining which does not require explicit knowledge of the interfering channels. To see this, recall the M-MMSE combining vector from (4.4):

$$\mathbf{v}_{jk} = p_{jk} \left(\sum_{l=1}^{L} \sum_{i=1}^{K_l} p_{li} \left(\hat{\mathbf{h}}_{li}^{j} (\hat{\mathbf{h}}_{li}^{j})^{\mathrm{H}} + \mathbf{C}_{li}^{j} \right) + \sigma_{\mathrm{UL}}^2 \mathbf{I}_{M_j} \right)^{-1} \hat{\mathbf{h}}_{jk}^{j}. \tag{7.50}$$

The inverse matrix in this expression is $\mathbb{E}\left\{ \mathbf{y}_j \mathbf{y}_j^{\mathrm{H}} | \{\hat{\mathbf{h}}_{li}^{j}\} \right\}$, which is actually an estimate of the conditional correlation matrix $\mathbb{E}\left\{ \mathbf{y}_j \mathbf{y}_j^{\mathrm{H}} | \{\mathbf{h}_{li}^{j}\} \right\}$ of the received signal $\mathbf{y}_j$ at BS j in (2.5). However, rather than computing this matrix based on the channel estimates of all UEs, which requires knowledge of the pilot sequences and spatial correlation matrices (cf. Section 3.1), it can be alternatively estimated from the sample correlation matrix $\hat{\mathbf{Q}}_j$, defined as

$$\hat{\mathbf{Q}}_j = \frac{1}{\tau_u} \sum_{n=1}^{\tau_u} \mathbf{y}_j[n] \mathbf{y}_j^{\mathrm{H}}[n] \tag{7.51}$$

where $\mathbf{y}_j[n]$ denotes the nth sample of the received UL data signal at BS j and τ_u is the total number of UL data symbols within the current coherence block. The approximate receive combining vector $\hat{\mathbf{v}}_{jk}$ based on the sample correlation matrix is then given as

$$\hat{\mathbf{v}}_{jk} = p_{jk} \hat{\mathbf{Q}}_j^{-1} \hat{\mathbf{h}}_{jk}^{j}. \tag{7.52}$$

Note that BS j still needs to estimate the channels $\hat{\mathbf{h}}_{jk}^{j}$ of the served UEs. Moreover, the quality of $\hat{\mathbf{Q}}_j$ depends strongly on τ_u. We have discussed different aspects of correlation matrix estimation in Section 3.3.3 on p. 107. Thanks to the channel reciprocity, (7.52) can be also used as an approximate M-MMSE downlink precoder (cf. (4.37)).

Since the correlation matrix of the received UL signal completely describes the subspace from which the interference was received, the BSs and SBSs can simply precode their transmitted data such that less energy is radiated towards those directions. One can think about this as if every node sacrifices some of its degrees of freedom (or antennas) to reduce the interference it creates. This is an especially attractive

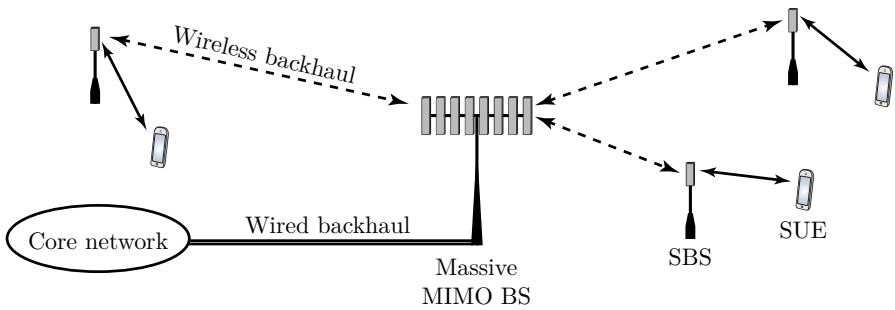


Figure 7.37: A Massive MIMO BS providing wireless backhaul to multiple SBSs.

option for BSs having a large number of excess antennas. Surprisingly, there is a huge gain in SE when also the SBSs are equipped with a few antennas and apply this precoding scheme. For more details and a critical discussion, we refer to [151, 51]. Another possibility to reduce cross-tier interference with the help of large antenna arrays is spatial (angular) blanking of certain cell areas [9, 361]. Similar to the concept of almost-blank subframes in time (as introduced in the LTE Release 10 [14]), this technique creates interference-free spatial zones for SC communications.

7.6.2 Massive MIMO for Wireless Backhaul

With a Massive MIMO infrastructure in place, it is possible to provide point-to-multipoint wireless backhaul from the BSs to a fraction of the SBSs on either the same or a different frequency band. An example is provided in Figure 7.37. In contrast to what is common practice, it would be advantageous to use cellular frequencies ($\leq$ 6 GHz) for backhaul signaling and higher (mmWave) frequencies for the SBSs' data communications. This would have the benefit that LoS backhaul links are not necessary and the inter-cell interference between the SBSs is reduced due to high propagation losses in mmWave bands (cf. Section 7.5). Such a solution would have the following additional advantages:

- As neither standardization nor backward-compatibility is necessary for backhauling, manufacturers can use proprietary solutions and rapidly integrate technological innovations.

- The BS–SBS channels vary very slowly with time, due to the fixed deployment. Thus, complex (cooperative) transmission/detection schemes could be implemented which are prohibitive for BS–UE communications due to practical constraints on channel coherence, latency, and complexity. For example, multiple BSs could jointly provide wireless backhaul to a large number of SBSs in a network MIMO fashion. Also, the utilization of FDD bands might be feasible as the CSI feedback rate is smaller compared to what is necessary for BS–UE communication over fast-fading channels.

- Backhaul can be provided dynamically where it is needed; for example, the backhaul links to empty SBSs can be turned off such that more resources are available for highly loaded cells. This would avoid over-provisioning of backhaul capacity and could also allow for energy savings compared to traditional fiber-optical links whose energy consumption is independent of the traffic.

- With wireless backhaul links, the SBSs only require a power connection to be operational. Thus, they can be installed wherever and whenever needed with a minimum amount of manual configuration (e.g., antenna adjustment and wired power supply). This further reduces the capital and operational expenditures related to their deployment.

A relevant question in this context is how many antennas we would need to satisfy a certain backhaul rate (measured in bit/s) with a given transmit power budget. As the wireless backhaul channels can be seen as quasi-static (or slowly fading), it is not unreasonable to assume that CSI regarding the channels to all neighboring SBSs is available at the BSs and that these can coordinate their transmissions to some extent. Full sharing of UE data among the BSs might be infeasible as this would impose an extremely high traffic load on the wired backhaul network. Therefore, cooperative schemes relying on multicell CSI and only some additional data exchange are preferable.

Consider the canonical Massive MIMO system from Definition 2.1 on p. 64 and think of UEs as SBSs. We make the additional simplifying assumption that all BSs have M antennas and serve S SBSs each.

Moreover, we ignore channel estimation and assume that perfect CSI is available. We will exemplify a power minimization algorithm from [100] that allows us to fix a desired SINR target γ_{js} for each backhaul link in each cell and to find the precoding vectors $\mathbf{w}_{js}$ and transmit powers ρ_{js} that achieve the minimum necessary total transmit power. In other words, our goal is to solve the optimization problem

$$\begin{aligned} \underset{\{\rho_{js},\mathbf{w}_{js}\}}{\text{minimize}} \quad & \sum_{j=1}^{L}\sum_{s=1}^{S}\rho_{js} \\ \text{subject to} \quad & \mathsf{SINR}_{js}^{\mathrm{DL}} \geq \gamma_{js} \quad j=1,\ldots,L, s=1,\ldots,S \\ & \|\mathbf{w}_{js}\| = 1 \quad j=1,\ldots,L, s=1,\ldots,S \end{aligned} \tag{7.53}$$

where $\{\rho_{js}, \mathbf{w}_{js}\}$ denotes the set of transmit powers and precoding vectors and

$$\mathsf{SINR}_{js}^{\mathrm{DL}} = \frac{\rho_{js}\left|\mathbf{w}_{js}^{\mathrm{H}}\mathbf{h}_{js}^{j}\right|^2}{\sum\limits_{l=1}^{L}\sum\limits_{\substack{i=1 \\ (l,i)\neq(j,s)}}^{S}\rho_{li}\left|\mathbf{w}_{li}^{\mathrm{H}}\mathbf{h}_{js}^{l}\right|^2 + \sigma_{\mathrm{DL}}^2} \tag{7.54}$$

is the instantaneous SINR of SBS s in cell j. The solution to (7.53) is provided by the following theorem.

Theorem 7.3 ([100])**.** The solution to (7.53), if it exists, is given by $\rho_{js}^{\star}$ and $\mathbf{w}_{js}^{\star} = \mathbf{v}_{js}^{\star}/\|\mathbf{v}_{js}^{\star}\|$ for $j = 1,\ldots,L$, $s = 1,\ldots,S$, where

$$\mathbf{v}_{js}^{\star} = \left(\sum_{l=1}^{L}\sum_{i=1}^{S}\lambda_{li}^{\star}\mathbf{h}_{li}^{j}\left(\mathbf{h}_{li}^{j}\right)^{\mathrm{H}} + \mathbf{I}_M\right)^{-1}\mathbf{h}_{js}^{j} \tag{7.55}$$

with $\lambda_{js}^{\star}$ being the unique solutions to the set of fixed-point equations

$$\lambda_{js}^{\star} = \frac{\left(1+\frac{1}{\gamma_{js}}\right)^{-1}}{\left(\mathbf{h}_{js}^{j}\right)^{\mathrm{H}}\left(\sum\limits_{l=1}^{L}\sum\limits_{i=1}^{S}\lambda_{li}^{\star}\mathbf{h}_{li}^{j}\left(\mathbf{h}_{li}^{j}\right)^{\mathrm{H}} + \mathbf{I}_M\right)^{-1}\mathbf{h}_{js}^{j}} \tag{7.56}$$

for $j = 1,\ldots,L$ and $s = 1,\ldots,S$, where $\rho_{js}^{\star}$ are the unique solutions to

the set of equations

$$\frac{\rho^{\star}_{js}}{\gamma_{js}} \left| \left(\mathbf{w}^{\star}_{js} \right)^{\mathrm{H}} \mathbf{h}^{j}_{js} \right|^2 - \sum_{\substack{i=1 \\ i \neq s}}^{S} \rho^{\star}_{ji} \left| \left(\mathbf{w}^{\star}_{ji} \right)^{\mathrm{H}} \mathbf{h}^{j}_{js} \right|^2 - \sum_{\substack{l=1 \\ l \neq j}}^{L} \sum_{i=1}^{S} \rho^{\star}_{li} \left| (\mathbf{w}^{\star}_{li})^{\mathrm{H}} \mathbf{h}^{l}_{js} \right|^2 = \sigma^2_{\mathrm{DL}}. \tag{7.57}$$

The solution to (7.56) can be computed by a standard fixed-point algorithm which iteratively updates $\lambda^{\star}_{js}$ starting from some random initial values. Equation (7.57), can be written in a matrix form and solved through matrix inversion. Note, that if (7.53) is infeasible, no solution to (7.56) is found (i.e., the fixed-point algorithm does not converge).

To exemplify the backhauling, we consider an extension of the running example, described in Section 4.1.3 on p. 134, in which $S = 81$ SBSs are distributed on a regular grid within each cell. The channels from the BSs to the SBSs are modeled in the same way as the UE channels; that is, 20 MHz of bandwidth is used, uncorrelated Rayleigh fading is considered, and the shadow fading has $\sigma_{\mathrm{sf}} = 10$. We first fix a maximum transmit power per BS and a desired DL SINR target for each backhaul link. Then using Theorem 7.3, we start with $M = 1$ and find the minimum transmit power necessary to achieve all SINR targets. If an SINR target is infeasible or the necessary transmit power is too high, we increase the number of antennas M until the problem becomes feasible and the transmit power is below the desired level. As the algorithm in (7.56) converges slowly for high SINR targets and it is computationally expensive for large systems, we resort to an asymptotic approximation (assuming very large values of M and S) of this algorithm instead [187, 285]. It only utilizes the average channel gains of all channels and its complexity is independent of the number of antennas. For medium- to large-sized systems (i.e., $M, S > 20$), the difference between the exact and approximate algorithms is generally very small.

In Figure 7.38, we show the minimum number of BS antennas necessary to provide a desired DL backhaul rate to either all, 40, or 20 randomly selected SBSs in each cell with a maximum power budget of

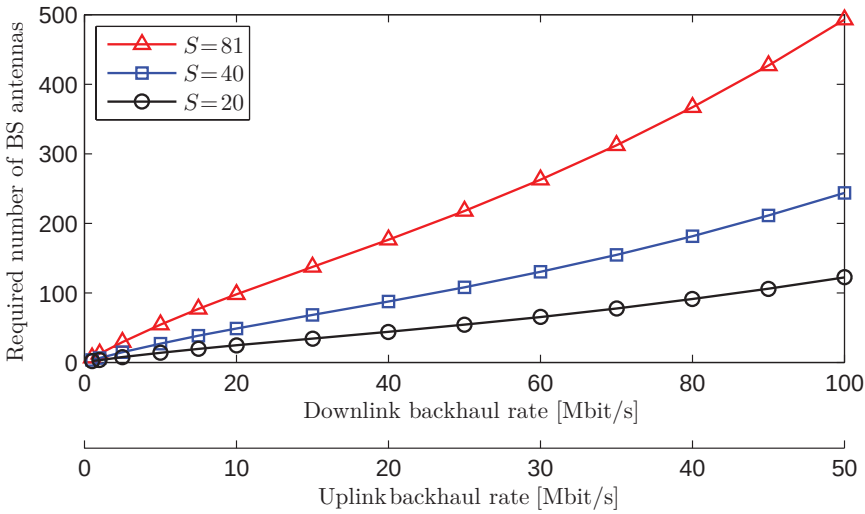


Figure 7.38: Required number of BS antennas M versus the DL/UL backhaul rates for different numbers of randomly selected SBSs $S \in \{20, 40, 81\}$ and a maximum average transmit power of 46 dBm per BS.

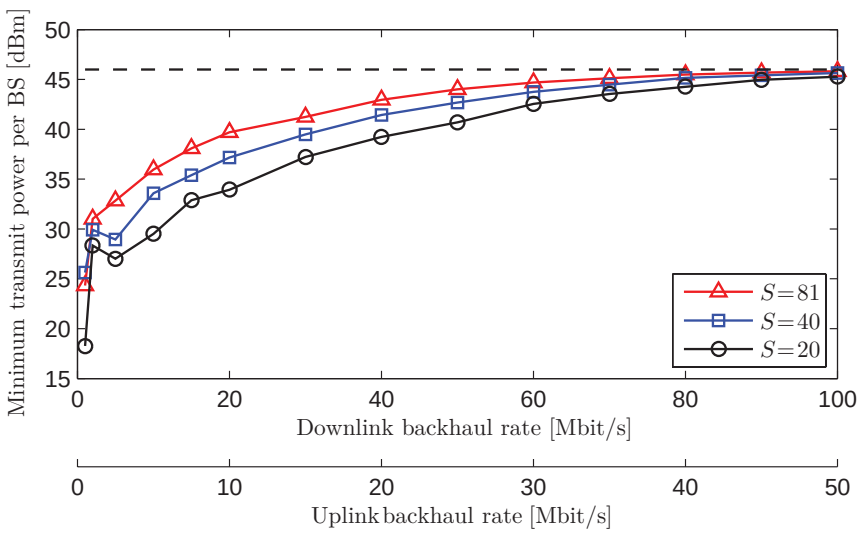


Figure 7.39: Minimum required transmit power per BS versus the DL/UL backhaul rates for different numbers of randomly selected SBSs $S \in \{20, 40, 81\}$. For each target rate, the smallest possible number of antennas is chosen according to Figure 7.38. The dashed line indicates the maximum transmit power of 46 dBm.

46 dBm per BS. Due to the classic UL-DL duality [370, 63, 335, 163], it is possible to achieve the same UL SINR using the precoding vectors $\mathbf{w}_{js}^{\star}$ as receive combining vectors (although the power allocation must be recomputed). By fixing the UL-DL transmission ratio to $\tau_u/\tau_d = 2/3$, the UL rates are 50% smaller than the DL rates. We remark that the UL-DL duality only ensures that the sum of the transmit powers of the BSs and SBSs are equal, but do not respect any individual power constraints. One can see from the figure that the number of antennas increases approximately linearly with the target rate and the number of simultaneously served SBSs. Serving 20 SBSs at a rate of 100 Mbit/s requires $M = 122$ antennas per BS. If the number of SBSs (and hence the aggregated rate) is doubled, 244 antennas are necessary, which becomes 493 if all 81 SBSs are simultaneously provided with backhaul. In the same context, Figure 7.39 shows the average transmit power per BS that is needed to achieve a certain backhaul rate using the smallest possible number of antennas provided in Figure 7.38. As one would expect, the smaller the number of SBSs (and hence the aggregate sum rate which must be delivered by each BS), the smaller is the necessary transmit power. For large target rates, the entire power budget is needed, independently of the number of SBSs. The curves are not entirely smooth since each point on the curves uses a different number of antennas.

In summary, Figures 7.38 and 7.39 provide some evidence that high-speed backhaul provisioning via Massive MIMO BSs is possible up to a very large number of SBSs per cell. In our example, in order to provide backhaul with 100 Mbit/s in the DL and 50 Mbit/s in the UL to 81 SBSs per cell using 20 MHz of bandwidth, which corresponds to an aggregate area throughput of 8.1 Gbit/s/km^2, around 500 antennas and 46 dBm of transmit power per BS are necessary.

7.7 Case Study

We conclude this monograph with a case study that will jointly analyze some of the practical deployment aspects and tradeoffs that have been described earlier in this section. The purpose is to give a baseline for the anticipated throughput of Massive MIMO, using MR or RZF pro-

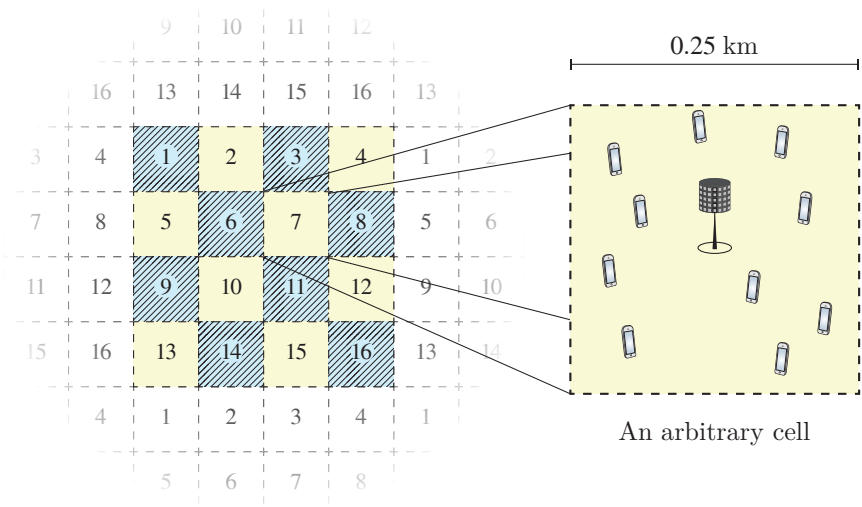


Figure 7.40: The case study considers 16 BSs on a square grid with a wrap-around topology. Note that, due to large-scale fading, the coverage area of each BS will not be the square area around the BS.

cessing, LS channel estimation, and some previously described resource allocation schemes. We stress that higher throughput can potentially be achieved by optimizing the array deployment (based on the propagation environment) and by exploiting MMSE channel estimation and M-MMSE processing.

7.7.1 Network Configuration and Parameters

We consider the network setup illustrated in Figure 7.40. Similar to the running example, which was described in Section 4.1.3 on p. 134, the case study considers $L = 16$ cells that are distributed on a 4×4 square grid with 250 m inter-BS distance. The channels are generated based on the 3GPP 3D UMi NLoS channel model from [1], using the open-source QuaDRiGa implementation developed by the Fraunhofer Heinrich Hertz Institute [159].[27] Within the 250 m $\times$ 250 m geographical area closest to each BS, we distribute 10 UEs uniformly at random (at

[27]The simulation results were generated using QuaDRiGa version 1.4.8-571.

Parameter	Value
Network layout	Square pattern (wrap-around)
Number of cells	$L = 16$
Inter-BS distance	250 m
UE dropping	$K = 10$ UEs in 250 m $\times$ 250 m area around each BS, with 35 m minimum distance
Channel model	3GPP 3D urban microcell (UMi)
BS array configurations	Cylindrical arrays: $10 \times 5 \times 2$ ($M = 100$), $20 \times 5 \times 1$ ($M = 100$), $20 \times 5 \times 2$ ($M = 200$)
BS height	25 m
UE height	1.5 m
Carrier frequency	2 GHz
Bandwidth	$B = 20$ MHz
Number of subcarriers	2000
Subcarrier bandwidth	10 kHz
Maximum UE transmit power	20 dBm
Maximum BS transmit power	30 dBm
Receiver noise power	-94 dBm
Cyclic prefix overhead	5%
Frame dimensions	$B_c = 50$ kHz, $T_c = 4$ ms
Subcarriers per frame	5
Useful samples per frame	$\tau_c = B_c T_c / 1.05 \approx 190$
Pilot reuse factor	$f = 2$
Number of pilot sequences	$\tau_p = 30$
Channel estimation	LS
Combining and precoding	RZF or MR

Table 7.2: Network parameters in the case study.

distances larger than 35 m from the BS). The UEs are at outdoor NLoS locations, 1.5 m above the ground. Each UE is associated with the BS that provides the strongest average channel gain, which results in an

uneven number of UEs per cell due to the shadow fading characteristics of the channel model. Note that one could, potentially, obtain better performance by also taking the spatial channel correlation into account in the UE-BS association.

Each BS is deployed at an elevated location, 25 m above the ground. We consider a cylindrical array that covers an entire cell (i.e., 360°), without the need for cell sectorization. The BS operates at a 2 GHz carrier frequency and we assume an antenna spacing of $\lambda/2$ in the horizontal and vertical direction. We describe the antenna configuration as "horizontal × vertical × polarization", which refers to the number of antennas on each circle, number of circles in the vertical direction, and number of polarizations (see Remark 7.6 on p. 362), respectively. We consider three configurations: $10 \times 5 \times 2$, $20 \times 5 \times 1$, and $20 \times 5 \times 2$. The first and second configurations require $M = 100$ RF chains, while the third one requires $M = 200$ RF chains. The array is 37.5 cm high in all cases, while the diameter is 23.9 cm in the first configuration and 47.7 cm in second and third configurations. Note that by considering dual-polarized co-located antennas, we can effectively double the number of antennas (and RF chains) compared to having uni-polarized antennas, without increasing the array size; see Section 7.4 for further details.

We consider an OFDM system with a bandwidth of $B = 20$ MHz, which is divided into 2000 subcarriers that are each 10 kHz wide. The transmission protocol is based on dividing the time-frequency resources into frames of $B_c = 50$ kHz and $T_c = 4$ ms. All UEs are assumed to have channel coherence blocks that are equal or larger than the frame dimension. Hence, the channel responses are fixed and identical over the five subcarriers in a frame. The cyclic prefix (which combats inter-symbol interference) is assumed to increase the OFDM symbol duration by 5%, thus there are $\tau_c = B_c T_c / 1.05 \approx 190$ useful samples per frame. In each frame, $\tau_p = 30$ samples are used for pilots. We consider a pilot reuse factor of $f = 2$, which results in $\tau_p / f = 15$ pilots per BS. These pilots are distributed uniformly at random to the UEs in the cell, and some remain unused when there are fewer than 15 UEs in the cell. In the unlikely event that more than 15 UEs connect to a BS, a random subset of exactly 15 UEs is scheduled. The remaining 160 samples per

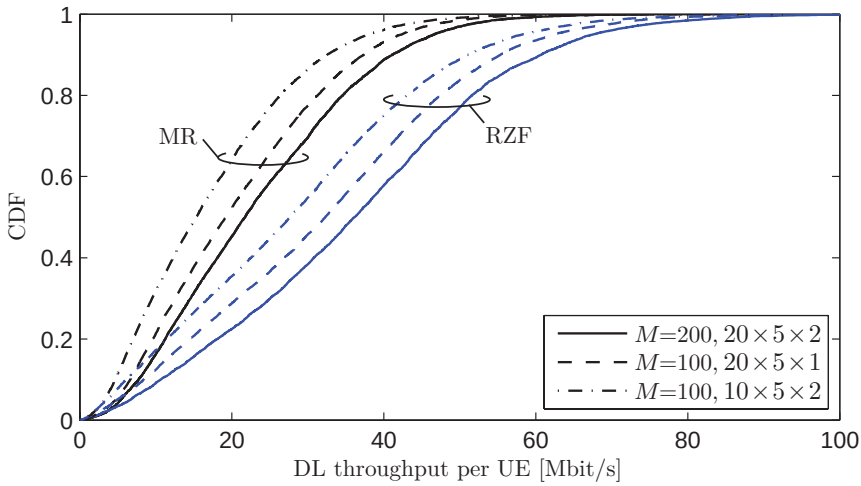
frame are used for payload data, whereof 1/3 is used for UL and 2/3 is used for DL. All active UEs are scheduled over all subcarriers and the maximum UL transmit power is 20 dBm. The maximum DL transmit power per BS is 30 dBm. Note that both numbers are at least an order of magnitude smaller than in current LTE systems (cf. Remark 4.1 on p. 138), but we will anyway achieve higher throughput in Massive MIMO thanks to the array gain and spatial multiplexing. The receiver noise power is -94 dBm.

To showcase the baseline throughput that can be achieved without requiring knowledge of channel statistics, we consider LS channel estimation. Since M-MMSE combining/precoding does not work well under LS estimation (cf. Figures 4.14 and 4.20), we only consider MR and RZF combining/precoding in this case study.

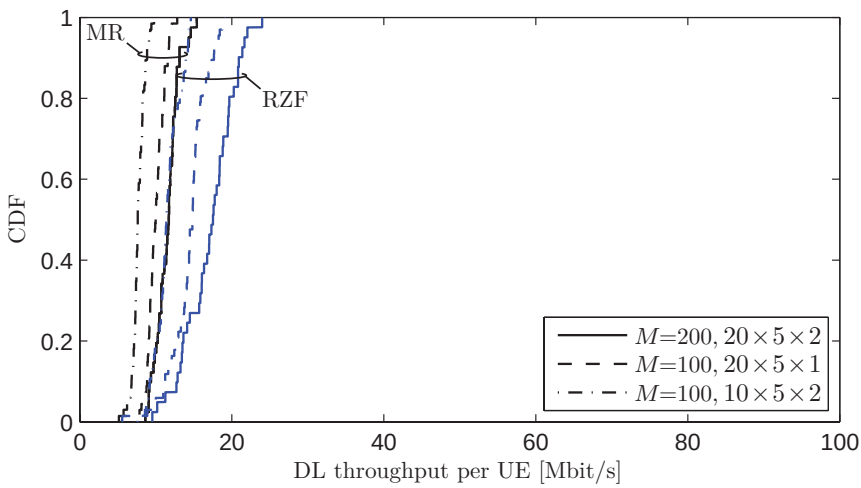
7.7.2 Simulation Results

We consider the DL and UL throughput of the UEs with different power allocation/control schemes and for different random user drops. Figure 7.41a and Figure 7.41b show the CDFs of the DL throughput with max product SINR and max-min fairness power allocation, respectively.[28] These power allocations are obtained using the optimization algorithms described in Section 7.1.1. The first observation is that the choice of antenna configuration has a great impact on the throughput. The dual-polarized configuration $20 \times 5 \times 2$ with $M = 200$ provides higher throughput, for every UE, than the uni-polarized configuration $20 \times 5 \times 1$ with $M = 100$. The uni-polarized configuration is, however, preferable compared with the dual-polarized configuration $10 \times 5 \times 2$, which also has $M = 100$. The conclusion is that, for arrays with a fixed physical size, dual-polarized arrays are beneficial due to the higher array gain that is obtained by having twice the number of antennas. If instead the number of RF chains is the limiting factor in the implementation, a physically larger array with uni-polarization (or dual-polarization but not co-located antennas) is preferable due to the higher spatial resolution.

[28] Equal maximum UL power is assumed for the pilot transmission when evaluated the DL.



(a) Max product SINR power allocation.



(b) Max-min fairness power allocation.

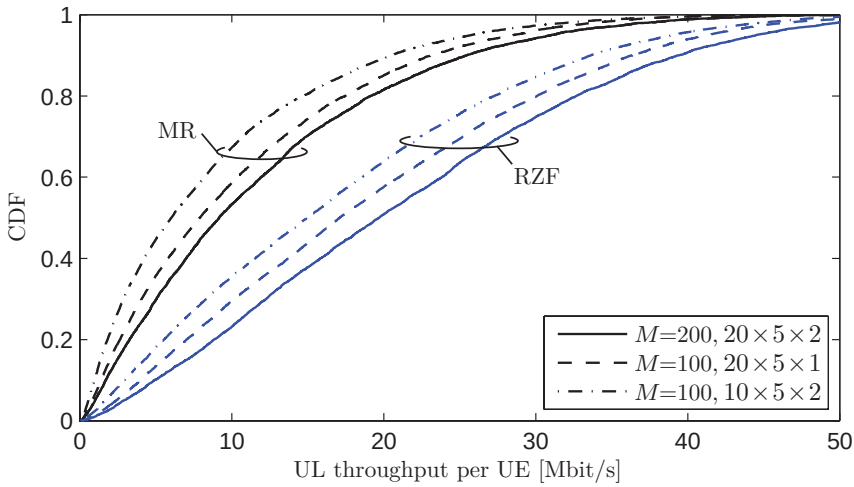
Figure 7.41: CDF of the DL throughput per UE in the case study. We compare different array configurations, transmit precoding schemes, and power allocation schemes.

Scheme	95% likely	Median	5% likely
Max product SINR (MR)	5.3 Mbit/s	21.8 Mbit/s	46.7 Mbit/s
Max product SINR (RZF)	6.7 Mbit/s	36.2 Mbit/s	67.6 Mbit/s
Max-min fairness (MR)	9.1 Mbit/s	11.7 Mbit/s	14.3 Mbit/s
Max-min fairness (RZF)	11.3 Mbit/s	17.5 Mbit/s	21.7 Mbit/s

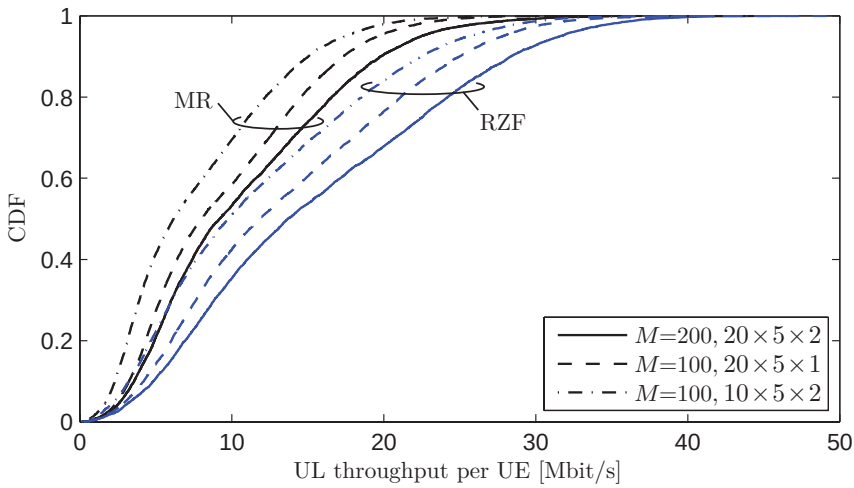
Table 7.3: DL throughput per UE in the case study, with different power allocation and precoding schemes.

We now concentrate on the preferable $20 \times 5 \times 2$ antenna configuration and compare the different power allocation and precoding schemes. With max product SINR power allocation, the average UE throughput is 23.4 Mbit/s with MR and 36.2 Mbit/s with RZF, but there are substantial variations between the UEs. With max-min fairness power allocation, the average throughput is reduced to 11.5 Mbit/s with MR and to 17.1 Mbit/s with RZF, but there is greater fairness in the sense of smaller variations and higher throughput for the UEs with the weakest channel conditions. These observations are further provided in Table 7.3, which shows the 95% likely, median, and 5% likely UE throughput. The throughput that is guaranteed to 95% of the UEs is around 70% higher with max-min fairness, compared with max product SINR. However, the median is 85%–105% higher with max product SINR power allocation, and the UEs with the 5% best channels achieve even higher gains with this scheme.

The CDF of the UL throughput is shown in Figure 7.42. We consider the power control framework described in Section 7.1.2, where the largest received power ratio within a cell is limited to Δ. We consider $\Delta = 20$ dB in Figure 7.42a and $\Delta = 0$ dB in Figure 7.42b. The UL results confirm many of the observations made in the DL. For a given number of RF chains, it is beneficial to have a physically larger uni-polarized array. For a given physical array size, it is instead beneficial to use dual polarization to squeeze in more antennas. There are substantial variations in throughput between the UEs, particularly with $\Delta = 20$ dB, but also with $\Delta = 0$ dB since the heuristic UL power control is performed independently in every cell—in contrast to the DL power allocation,



(a) Heuristic power control policy in (7.11) with $\Delta = 20\,\mathrm{dB}$.



(b) Heuristic power control policy in (7.11) with $\Delta = 0\,\mathrm{dB}$.

Figure 7.42: CDF of the UL throughput per UE in the case study. We compare different array configurations, receive combining schemes, and power control schemes.

Scheme	95% likely	Median	5% likely
$\Delta = 20\,\text{dB}$ (MR)	1.1 Mbit/s	9.7 Mbit/s	33.0 Mbit/s
$\Delta = 20\,\text{dB}$ (RZF)	3.1 Mbit/s	20.9 Mbit/s	47.3 Mbit/s
$\Delta = 0\,\text{dB}$ (MR)	2.8 Mbit/s	9.8 Mbit/s	23.9 Mbit/s
$\Delta = 0\,\text{dB}$ (RZF)	3.5 Mbit/s	14.5 Mbit/s	33.6 Mbit/s

Table 7.4: UL throughput per UE in the case study, with different combining schemes and different values of the maximum received power ratio Δ that is used for power control.

where max-min fairness power allocation attempts to give the same throughput to every UE in the entire network.

Focusing on the preferable $20 \times 5 \times 2$ antenna configuration and $\Delta = 20\,\text{dB}$, the average UE throughput is 12.4 Mbit/s with MR and 22.4 Mbit/s with RZF. In case of $\Delta = 0\,\text{dB}$, the average UE throughput is reduced to 11.4 Mbit/s with MR and 16.4 Mbit/s with RZF, but the 95% likely throughput is higher than with $\Delta = 20\,\text{dB}$. The 95% likely, median, and 5% likely UE throughput are provided in Table 7.4. A smaller Δ improves the baseline throughput that is guaranteed to 95% of the UEs, at the cost of reducing the median throughput and also the throughput of the UEs with the strongest channels. Generally speaking, the throughput differences are larger in the UL than in the DL due to the power control scheme.

In summary, the average DL sum throughput per cell can be as large as 358 Mbit/s over a 20 MHz channel, which corresponds to an area throughput of 5.7 Gbit/s/km^2. The average UL sum throughput per cell can be as large as 209 Mbit/s, which corresponds to an area throughput of 3.3 Gbit/s/km^2. Note that the substantial differences between the throughput in DL and UL are caused by the fact that twice as many samples per frame are used for DL data transmission as for UL data transmission — the average SE is approximately the same in both directions. To put the area throughput values into context, we can compare them with a contemporary LTE system, as described in Remark 4.1 on p. 138. Such a system would deliver 263 Mbit/s/km^2 in the DL and 115 Mbit/s/km^2 in the UL in a corresponding scenario. We conclude that, in this case study, Massive MIMO delivers 20–30 times higher throughput.

7.8 Summary of Key Points in Section 7

- The UEs in a cellular network have conflicting performance goals due to the interference and shared power budgets. Power allocation can be used to find a tradeoff between aggregate throughput and fairness between the UEs. Maximizing the product of the effective SINRs leads to a reasonable tradeoff. Thanks to channel hardening, the SINRs only depend on large-scale fading and the same solution can be used over many channel coherence blocks.
- Despite the large number of UEs, resource allocation is fairly simple since the UEs are separated spatially and, thus, every UE can use the full bandwidth whenever needed. When Massive MIMO is deployed with a high antenna-UE ratio, time-frequency scheduling is only needed to cope with traffic peaks, when there are more UEs than pilots or insufficient spatial resolution. This is a paradigm shift from conventional networks, which constantly rely on time-frequency scheduling to serve the UEs. The assignment of pilots to UEs and traffic load variations have little impact on the average sum SE, but can affect the SE of particular UEs.
- The spatial channel correlation depends mainly on the antenna array geometry (aperture, antenna spacing) and the angular spread, and must be accounted for by any realistic channel model. Channel measurements have confirmed favorable propagation in practice.
- The angular resolution of an antenna array depends on its aperture and not on the number of antennas. Critically spaced antennas are needed to avoid aliasing in the angular domain. Dual-polarized co-located antennas can significantly reduce the size of an array, but the theoretical performance analysis with polarization is difficult. Modern geometry-based

channel models support polarization and can be used in simulations.

- MmWave frequencies suffer from high propagation loss—that can be overcome by large antenna arrays—and are best suited for hotspots and low mobility scenarios. Due to a small number of RF chains compared to the number of radiating elements and a small number of UEs per cell, mmWave communication systems differ fundamentally from the definition of Massive MIMO in Definition 2.1 on p. 64.

- Massive MIMO plays a key role in heterogeneous networks to ensure coverage over large areas and to serve fast-moving UEs. A two-tier network consisting of Massive MIMO BSs and SBSs together with a synchronized TDD protocol allows the BSs to use their excess antennas to reduce intra-tier interference. Massive MIMO is also a promising solution for wireless backhaul provisioning to a large number of SBSs, without the need for LoS links.

- We have provided a case study based on state-of-the-art channel models, optimized resource allocation, and conservative assumptions on the selection of signal processing schemes. It demonstrates that Massive MIMO can deliver tens of Mbit/s per UE in both UL and DL over a channel of 20 MHz bandwidth. With around 10 UEs per cell, this results in an area throughput of several Gbit/s/km^2.